Trapping-signal classification: model report

trapsig 0.1.0 | config fingerprint `7da5ffdaffc35258` | seed 42 | model set **hybrid**

1. Data landscape

CSV twin: `report_data_landscape.csv` carries the same content.

135 files in the pool; 135 used; 0 skipped or failed. Classes: **InVitro_WT_tau** = 56, **InVitro_tau+GSK** = 79. Imbalance ratio 1.41

Event windows for Model B: 868 windows, length 10.0 ms (auto) = 100 samples at 10000.0 Hz. Files with no usable window: 60. Events excluded by the edge rule: 1.

2. Methods actually used

CSV twin: `report_methods.csv` carries the same content.

split unit: file (each file is its own group; no replicate or session metadata)

holdout: none (no replicates to hold out)

tuning: logreg C chosen by inner grouped CV within each outer training fold; the CNN early-stops and temperature-scales on a validation split carved from the outer training fold by file group

imputation: median, fitted within each training fold

window scaling: robust median/IQR fitted on each fold's training windows only

aggregation: window -> file by mean_proba, declared before evaluation

bootstrap: file-level only; a hierarchical bootstrap needs replicate groups

class weighting: none (imbalance below the design threshold of 1.5)

hybrid combination: fixed convex blend, Model A weight 0.5

fusion rule: both

fusion note: fitted per repeat and fold

representation flow: one 1D ConvNet encoder is fitted per outer fold; its global-average-pooled embedding is the single learned representation of the run and is consumed by (1) the softmax head -> cnn1d, (2) the nearest-prototype head -> proto_cnn, (3) the per-fold linear fusion head on handcrafted features + mean embedding -> fusion, (4) the probability blend with the best Model A -> hybrid, and (5) the conditional analyses: few-shot prototypes, semi-supervised pseudo-labels, novelty detection, clustering, and the file-level explanation map. Nothing downstream re-learns a representation, and no strategy is evaluated on files that contributed to the encoder, the prototypes, the fusion head, or the scaler.

matched subset: not applicable

Model B architecture: stem(strided conv) -> residual dilated conv blocks -> GAP -> linear; 31682 parameters; receptive field 231 samples (23.1 ms); selection metric macro_f1; mean temperature 2.3001795325297745; total training time 62.6 s.

3. Results

CSV twin: `report_results.csv` carries the same content.

model	files	balanced acc (mean +/- SD)	aggregated	95% CI	macro F1
dummy_majority	135	0.500 +/- 0.000	0.500	[0.500, 0.500]	0.369
dummy_stratified	135	0.489 +/- 0.052	n/a	n/a	0.488
logreg	135	0.720 +/- 0.022	0.752	[0.677, 0.821]	0.724
rf	135	0.854 +/- 0.024	0.865	[0.801, 0.923]	0.860
hgb	135	0.910 +/- 0.008	0.915	[0.862, 0.962]	0.911
cnn1d	75	0.590 +/- 0.064	0.620	[0.518, 0.726]	0.585
proto_cnn	75	0.648 +/- 0.031	0.720	[0.606, 0.825]	0.644
fusion	75	0.618 +/- 0.066	0.640	[0.544, 0.731]	0.617
hybrid	135	0.864 +/- 0.024	0.868	[0.803, 0.925]	0.869

Gate G7 (does Model B beat or match Model A?): Model A beats Model B. 95% intervals do not overlap and Model A is higher. Compared on the matched subset of 75 files. Model A best = hgb at 0.854; Model B at 0.590. CI overlap: False.

Label-shuffle null, logreg: observed balanced accuracy 0.720 vs null mean 0.490 (95th percentile 0.541), p = 0.048 from 20 permutations. Smallest attainable p = 0.048.

Label-shuffle null, rf: observed balanced accuracy 0.854 vs null mean 0.495 (95th percentile 0.567), p = 0.048 from 20 permutations. Smallest attainable p = 0.048.

Label-shuffle null, hgb: observed balanced accuracy 0.910 vs null mean 0.510 (95th percentile 0.563), p = 0.048 from 20 permutations. Smallest attainable p = 0.048.

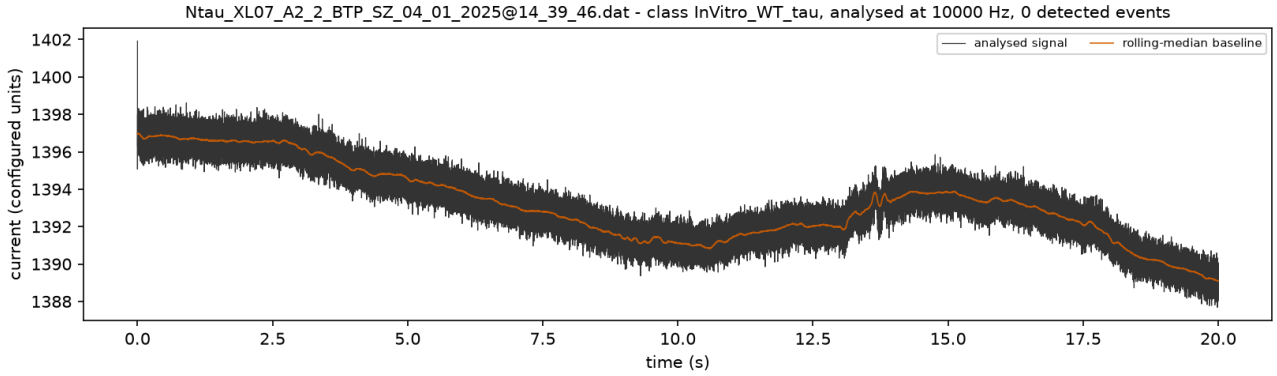
Label-shuffle null, cnn1d: not run - deep.n_permutations is 0; each CNN permutation costs one full repeated cross-validation on CPU. Raise it if a Model B null is required.

Label-shuffle null, hybrid: not run - the ensemble has no free parameters, so its null is implied by the nulls of Model A and Model B; running it would double the cost without adding evidence

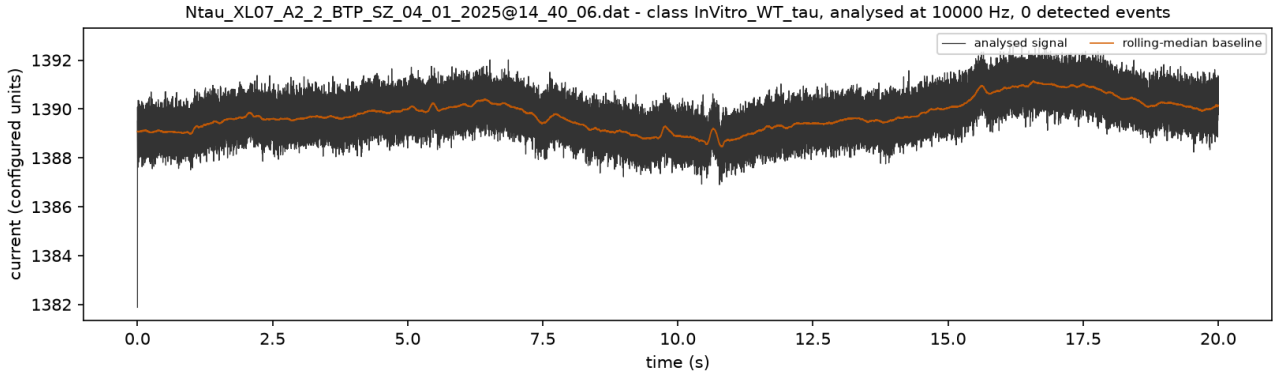
4. Figures

CSV twin: `report_figures.csv` carries the same content.

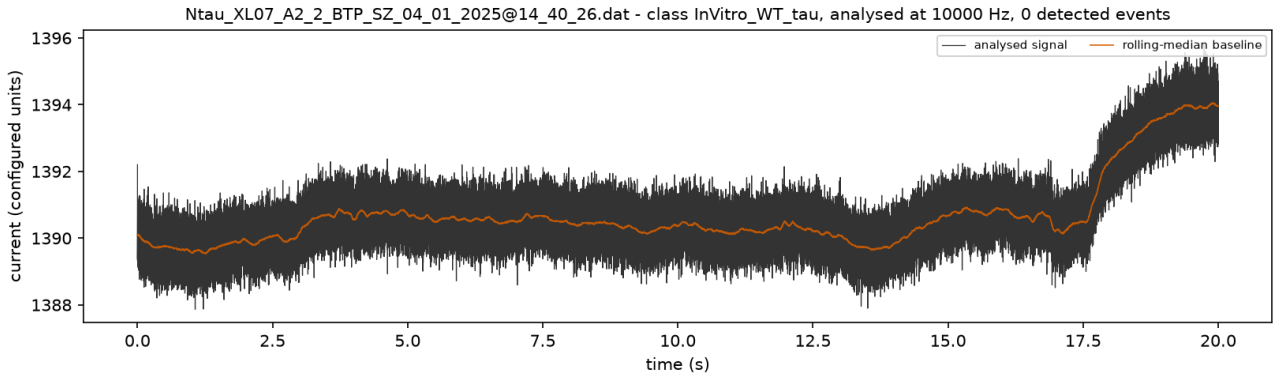
signal c4de5f492bc76186 (class InVitro WT tau)



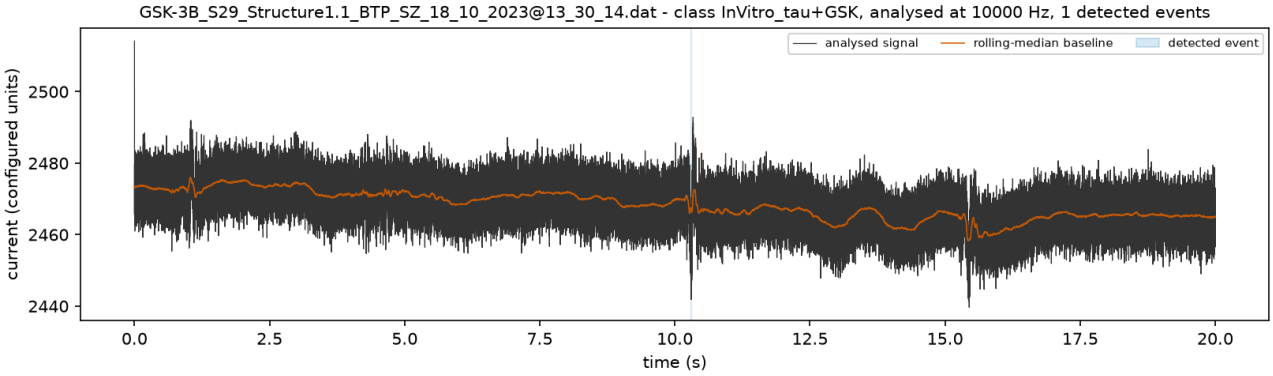
signal ece1386140d9feb3 (class InVitro WT tau)



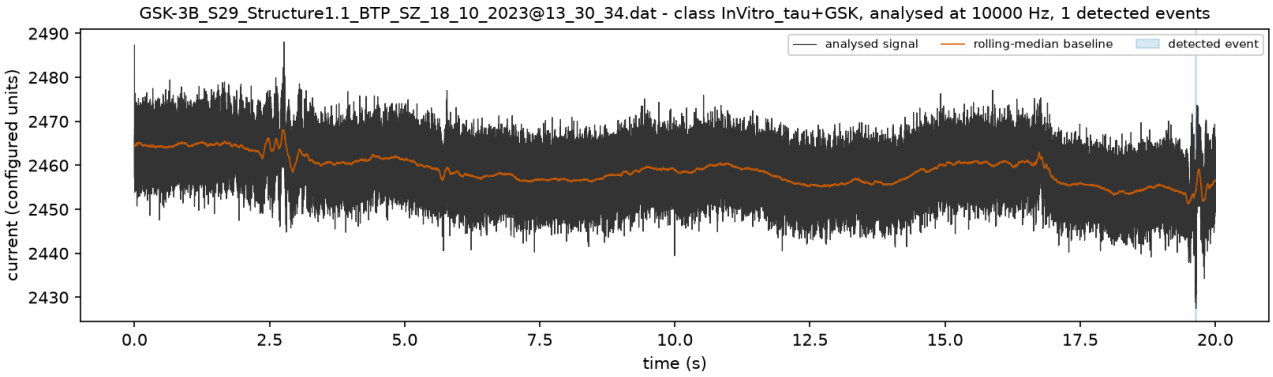
signal be509f6ad0d449aa (class InVitro WT tau)



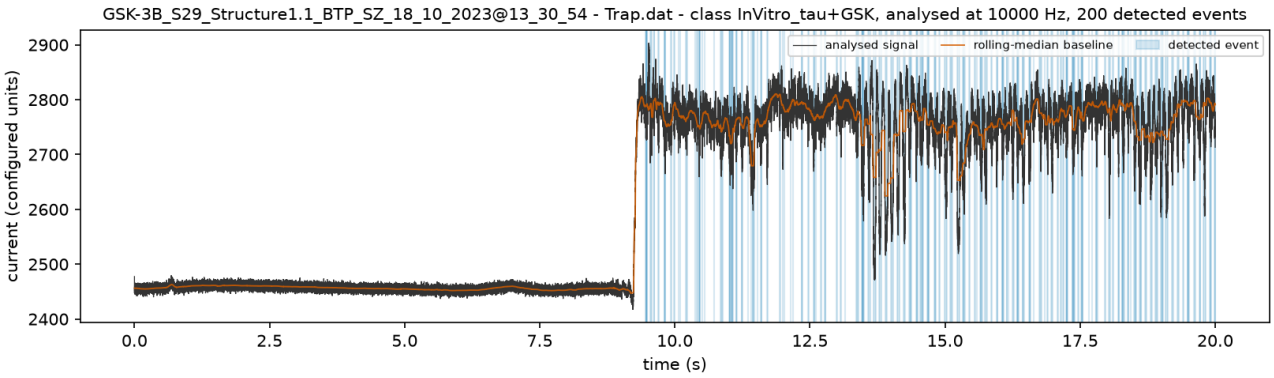
signal 87faede677b3fe14 (class InVitro tau+GSK)



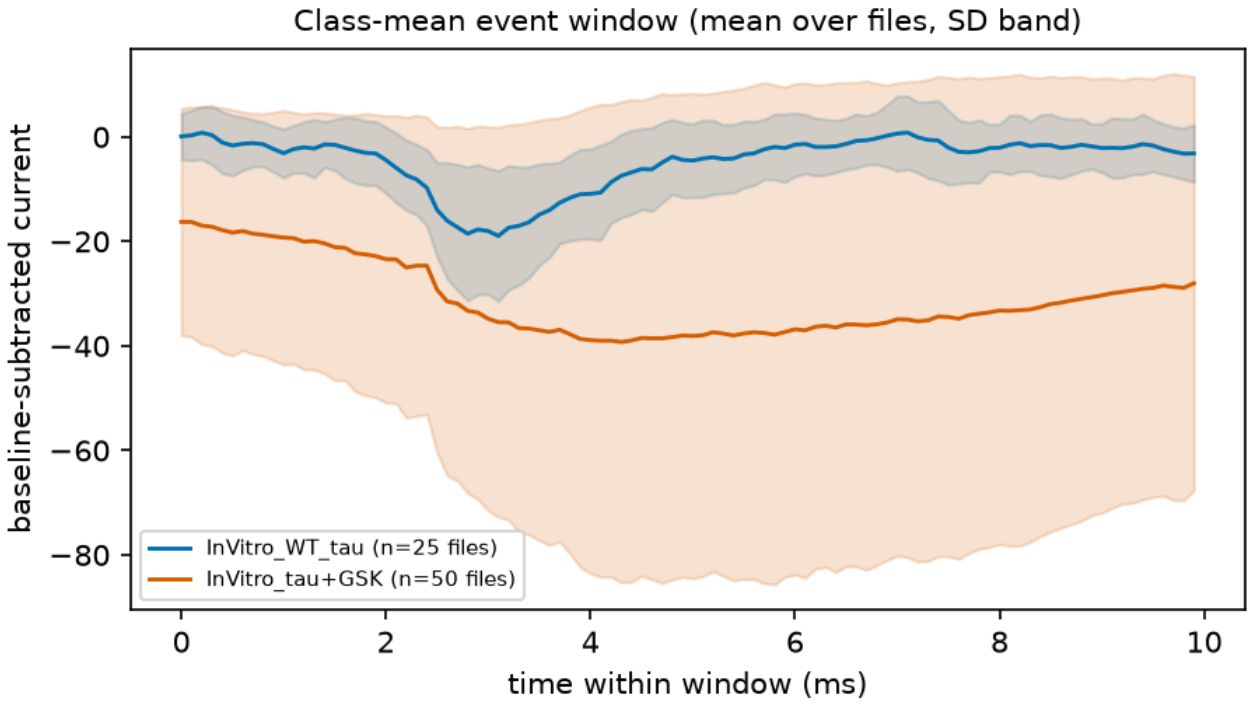
signal fd4cddbd17124fc5 (class InVitro tau+GSK)



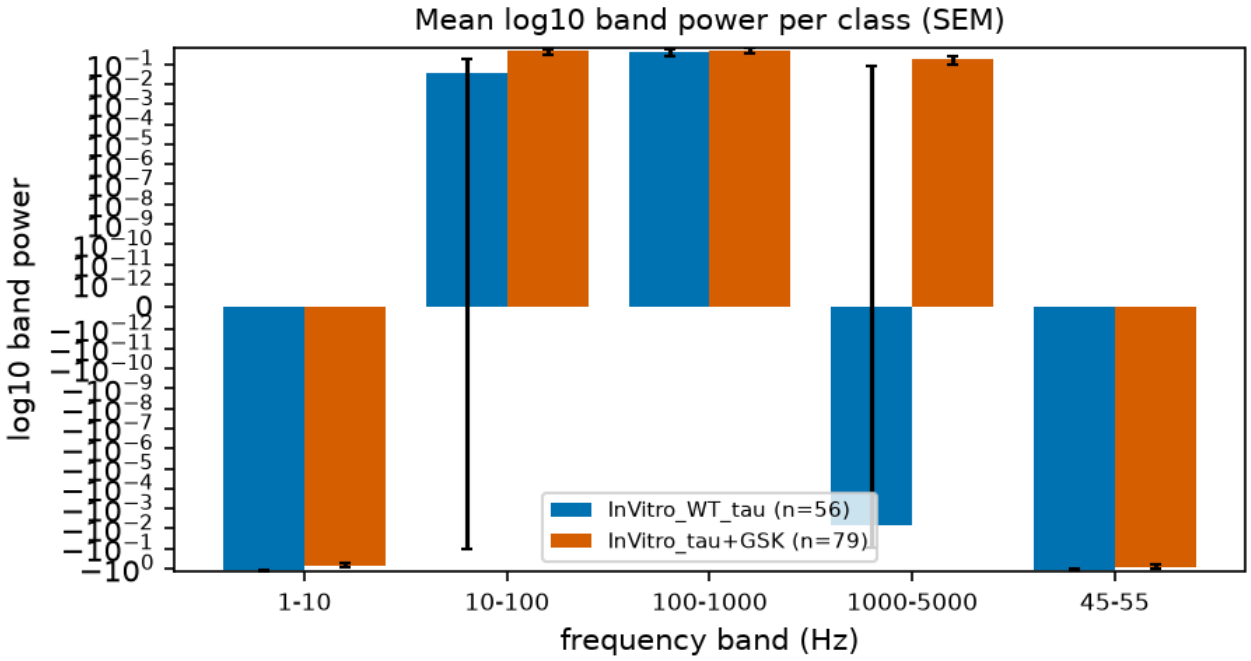
signal 178d697a32a19393 (class InVitro tau+GSK)



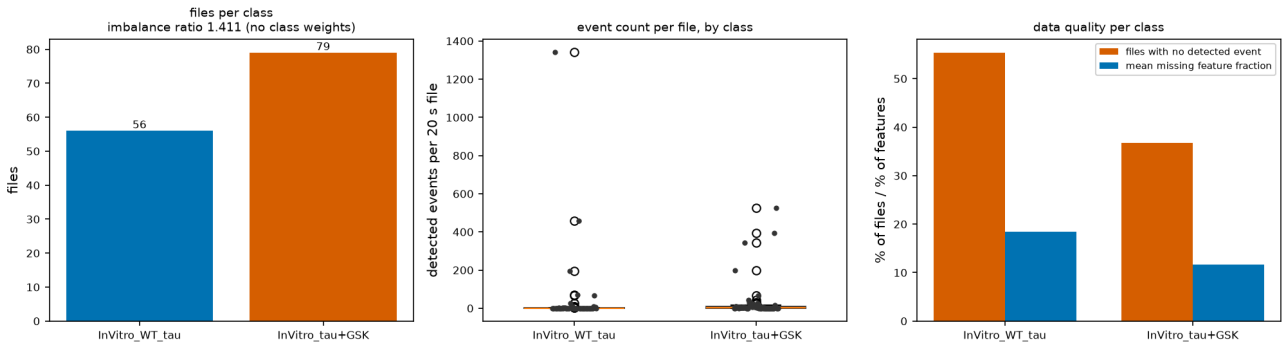
class mean event window



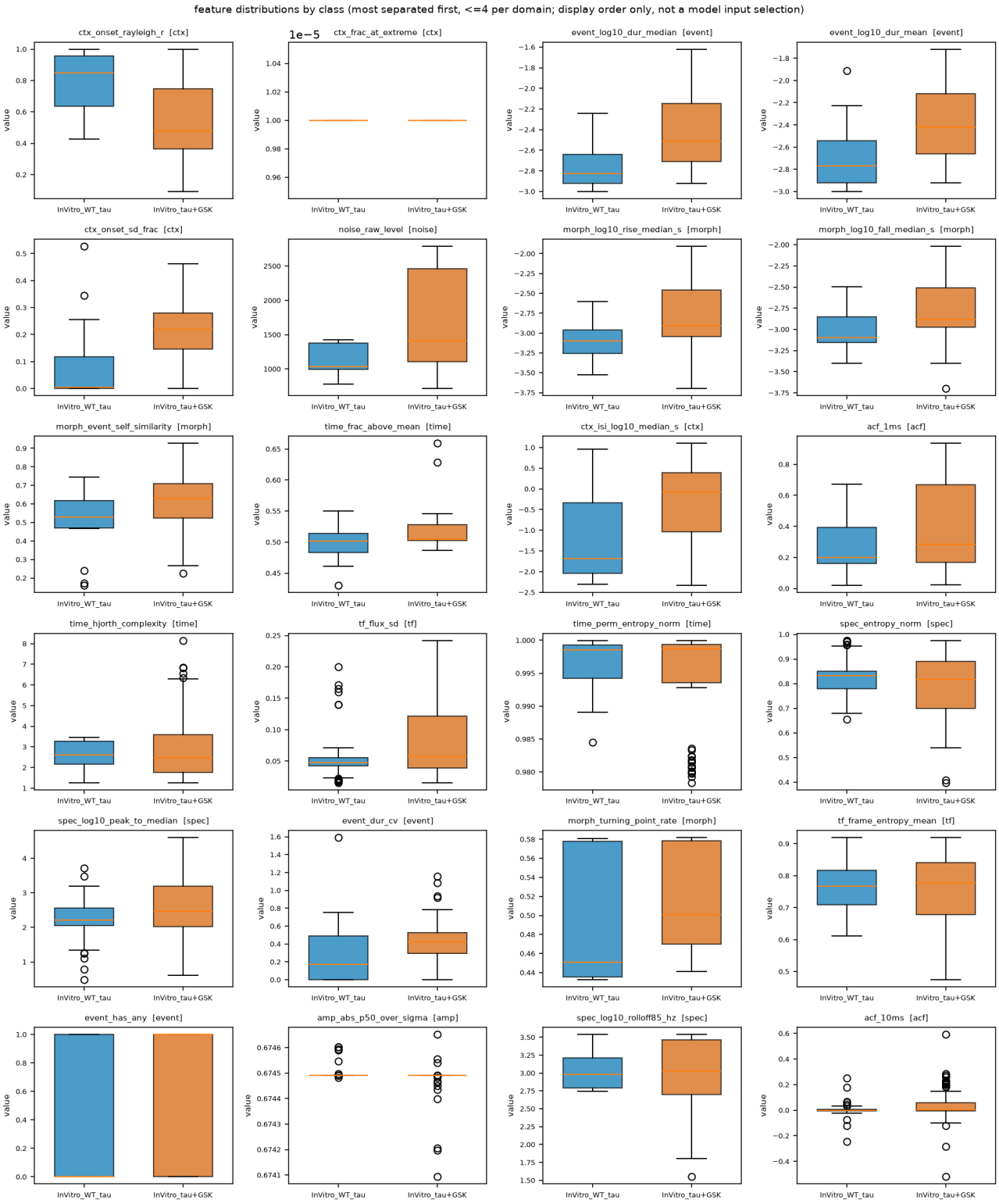
psd by class



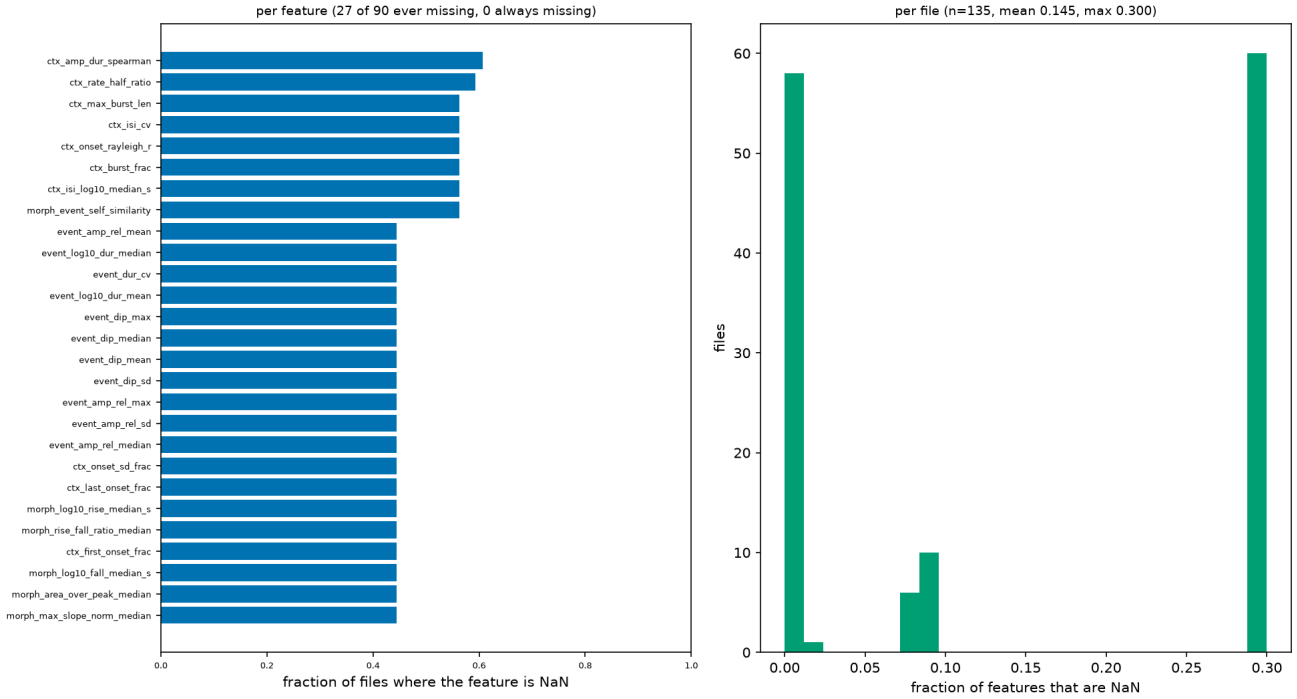
dataset composition



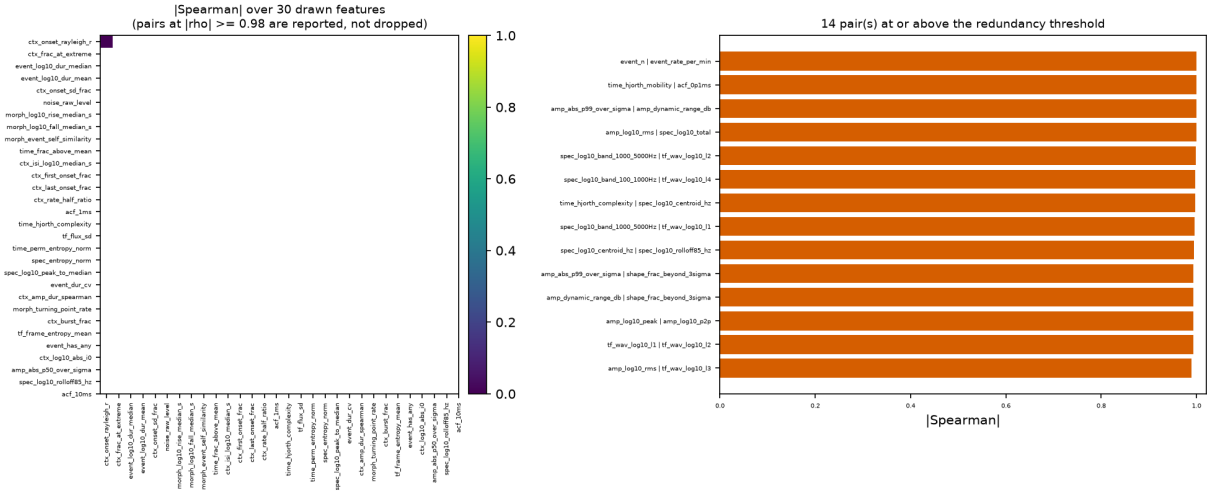
feature distributions



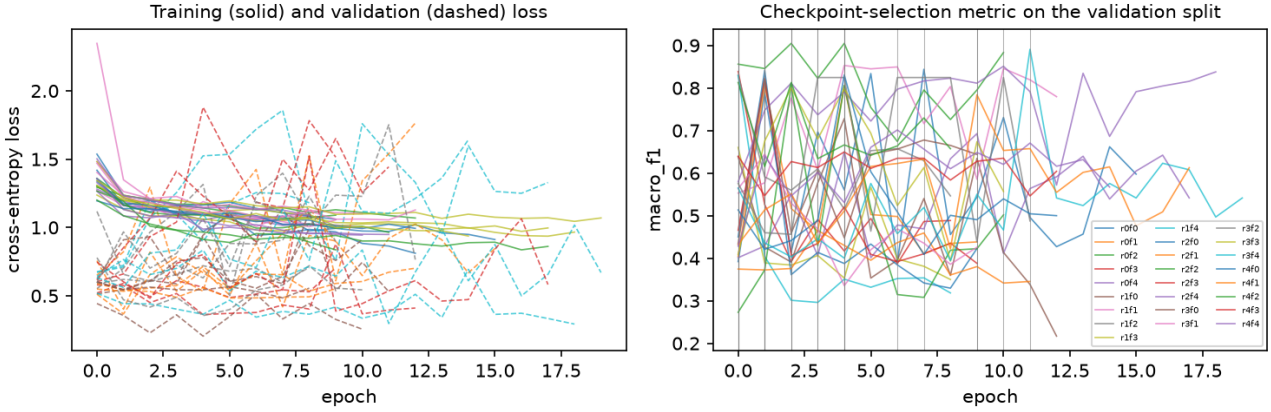
missing values



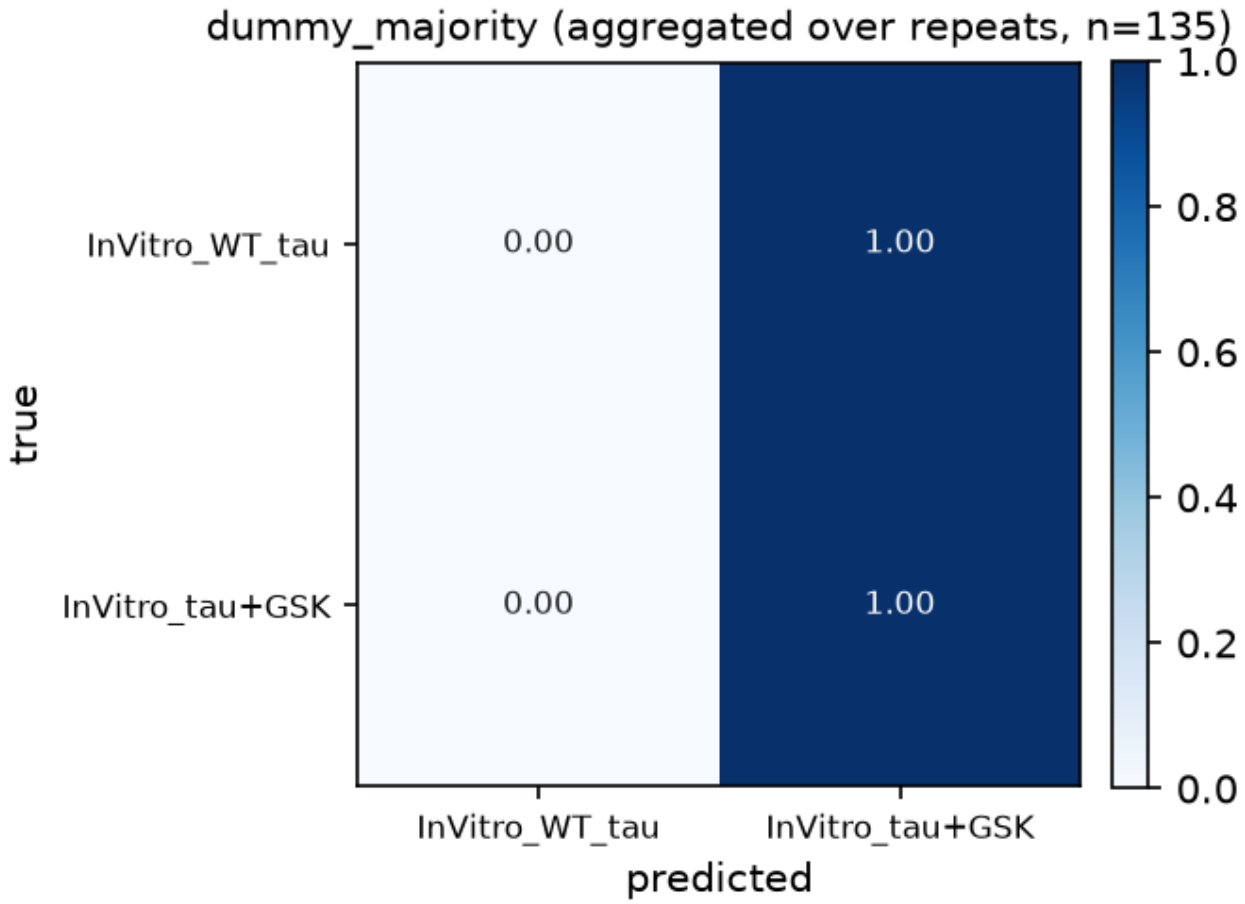
feature correlation



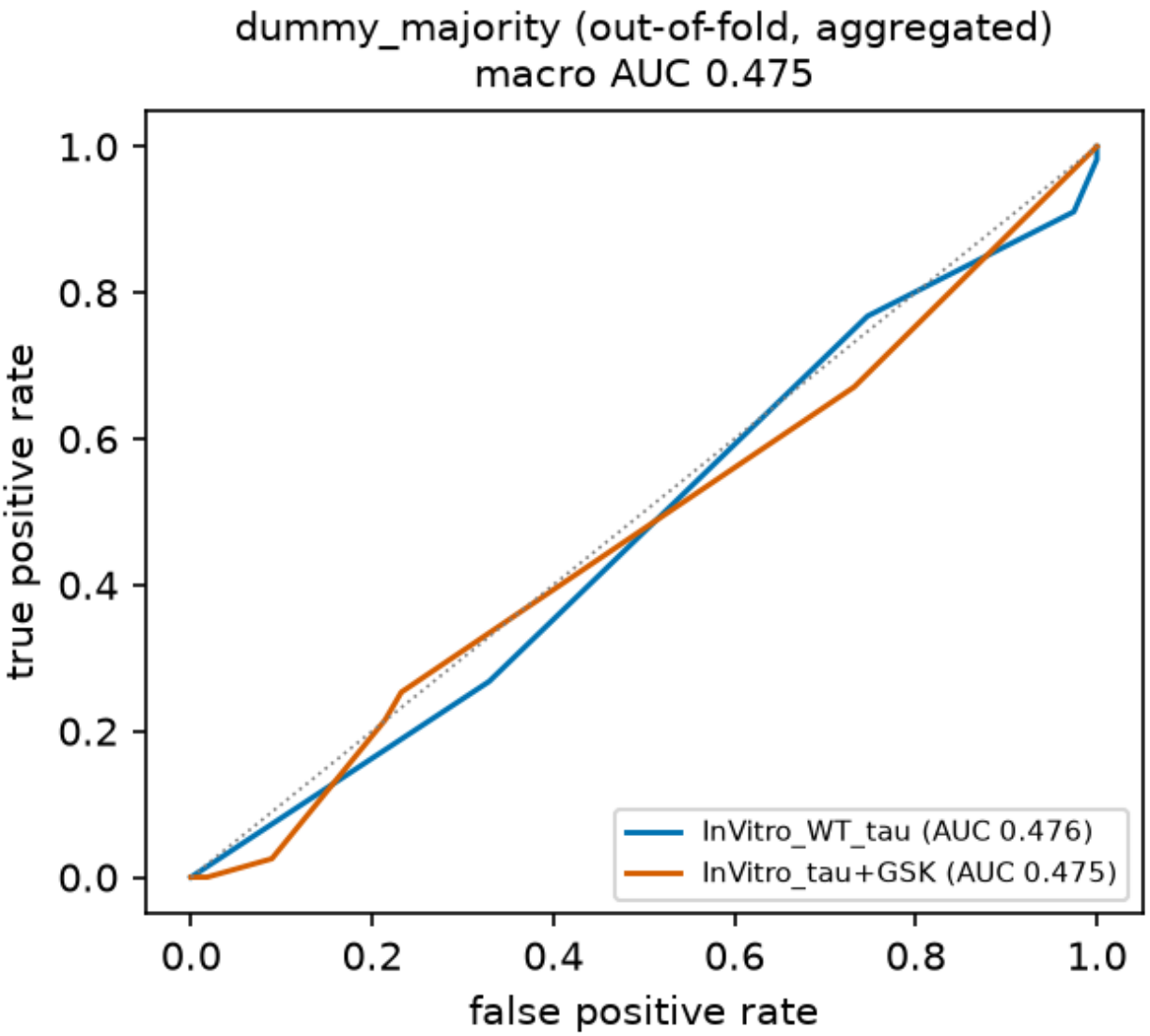
learning curves



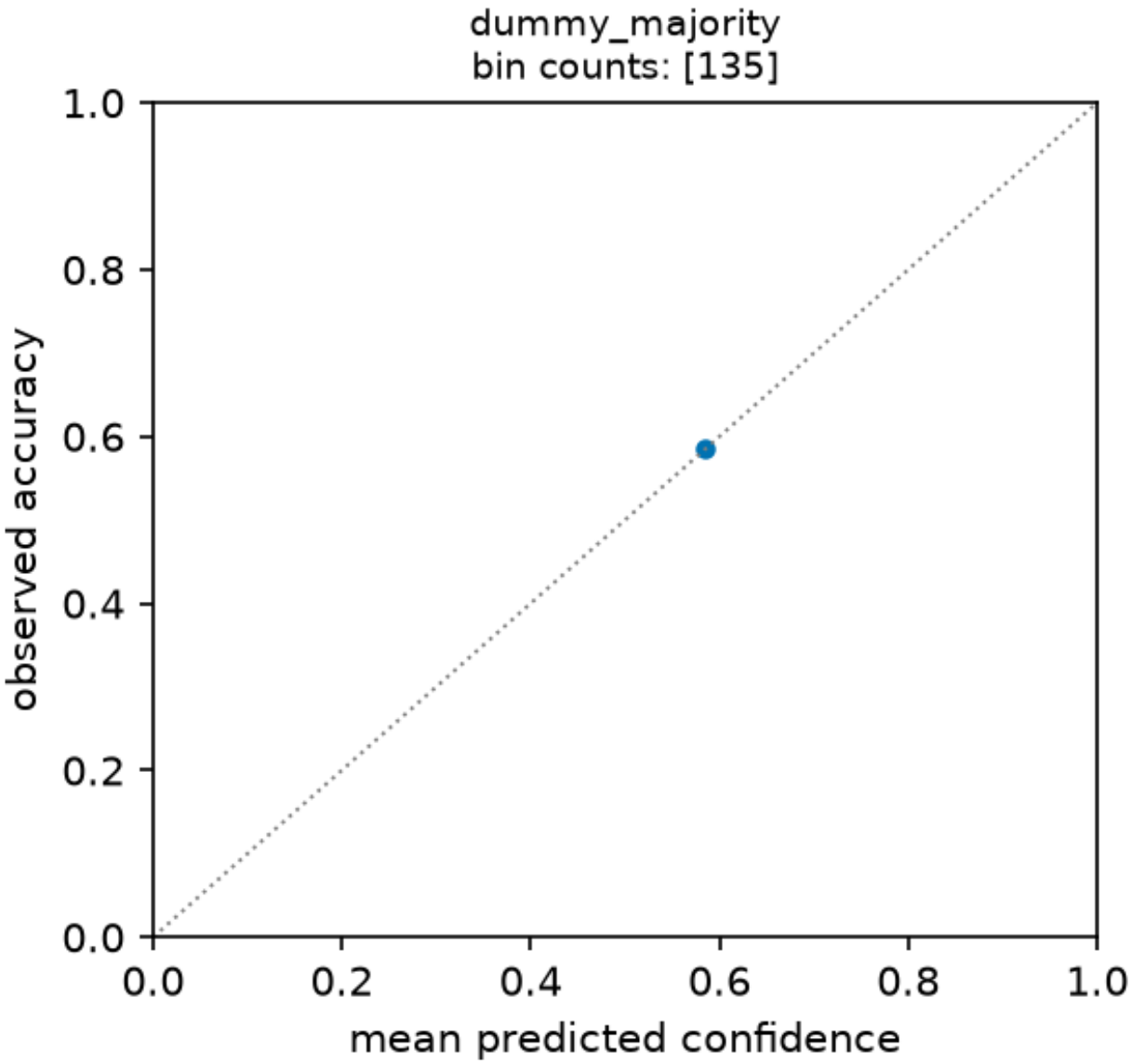
confusion dummy majority



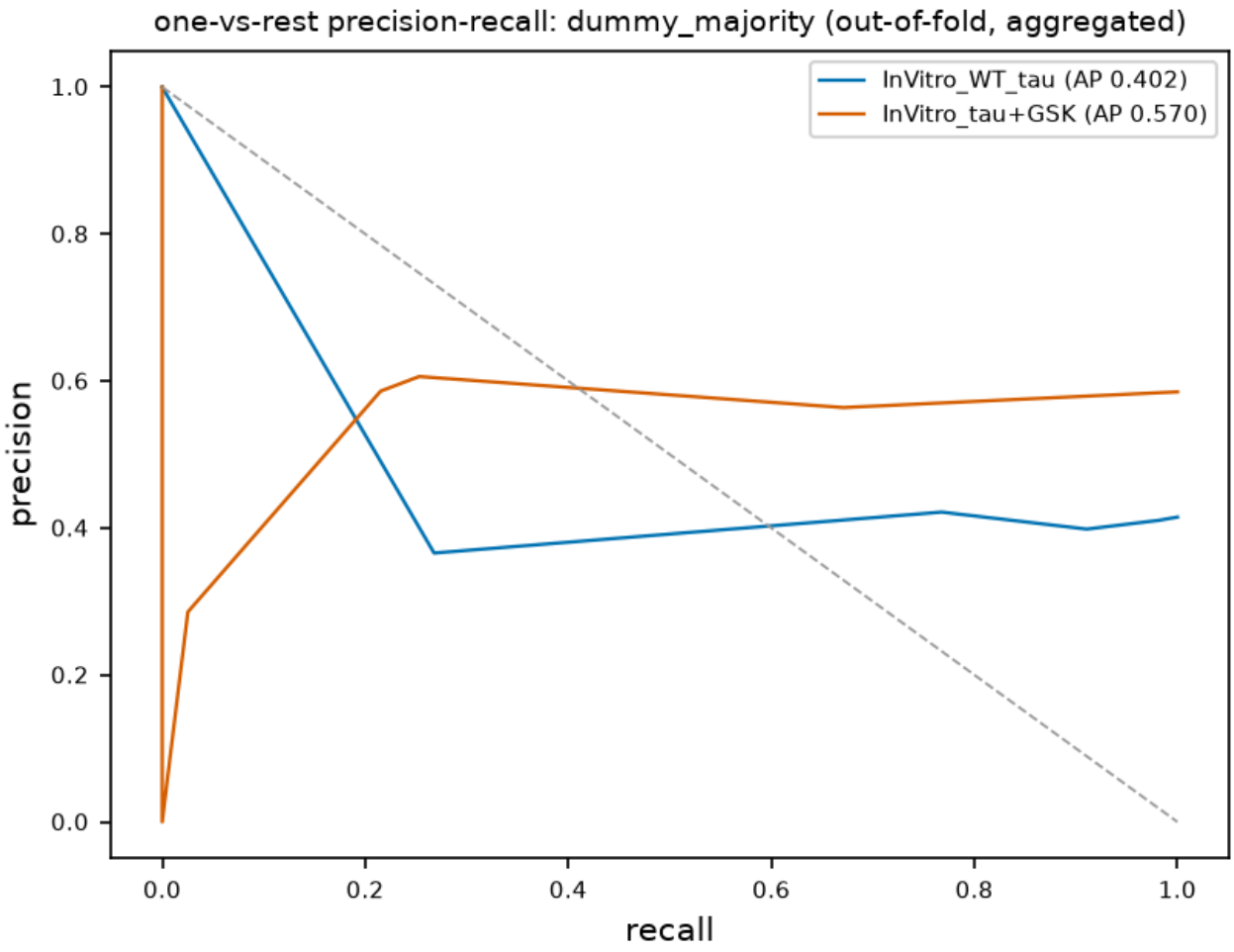
roc dummy majority



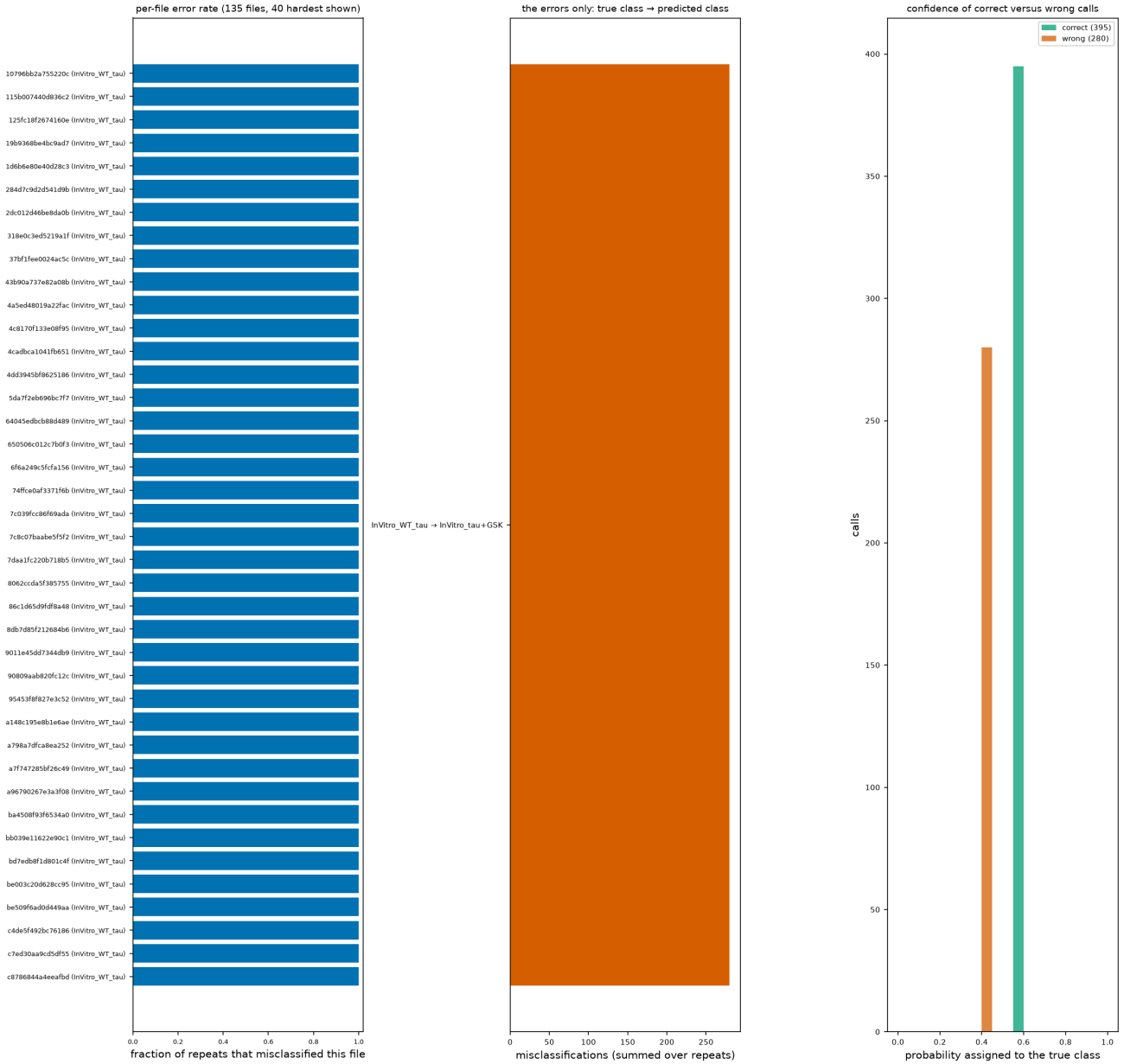
calibration dummy majority



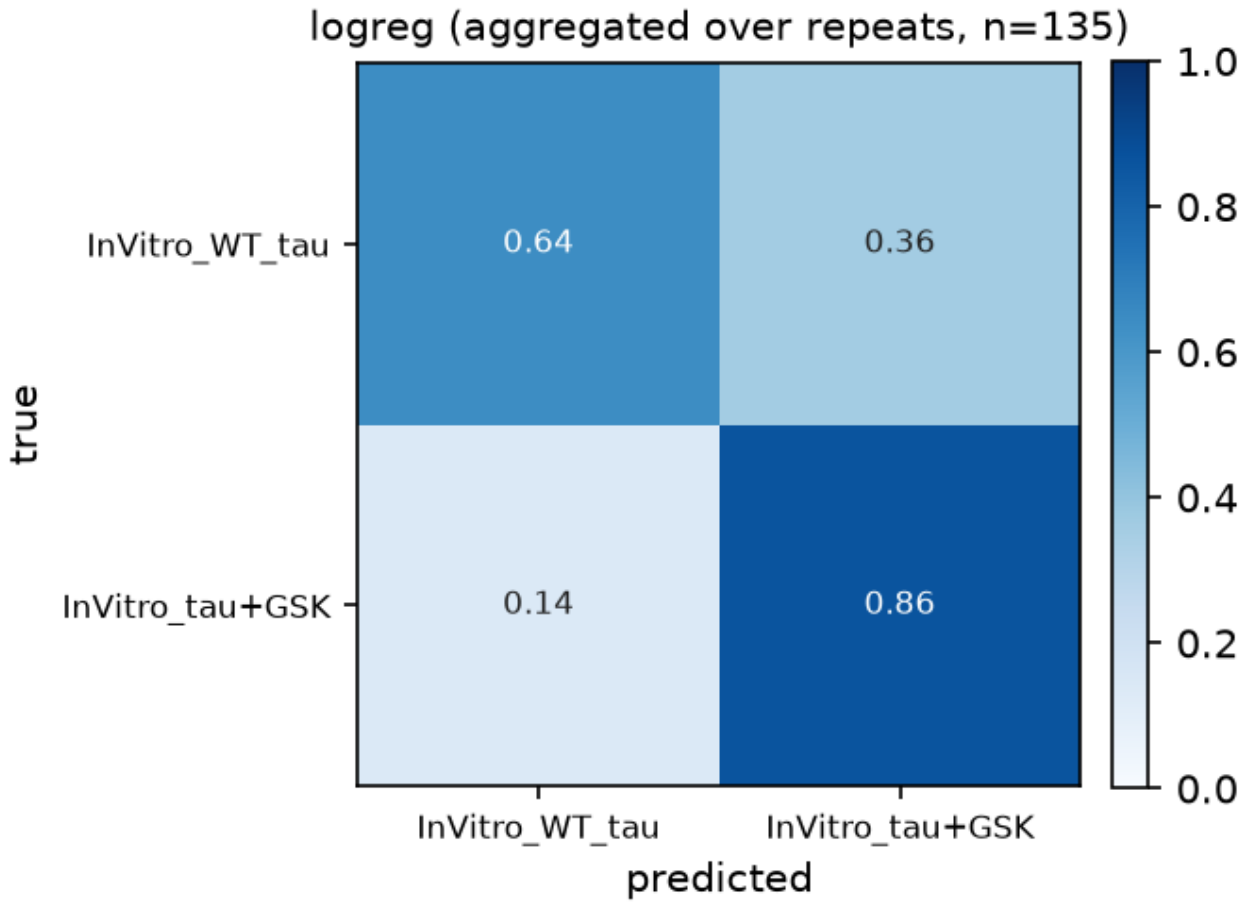
pr dummy majority



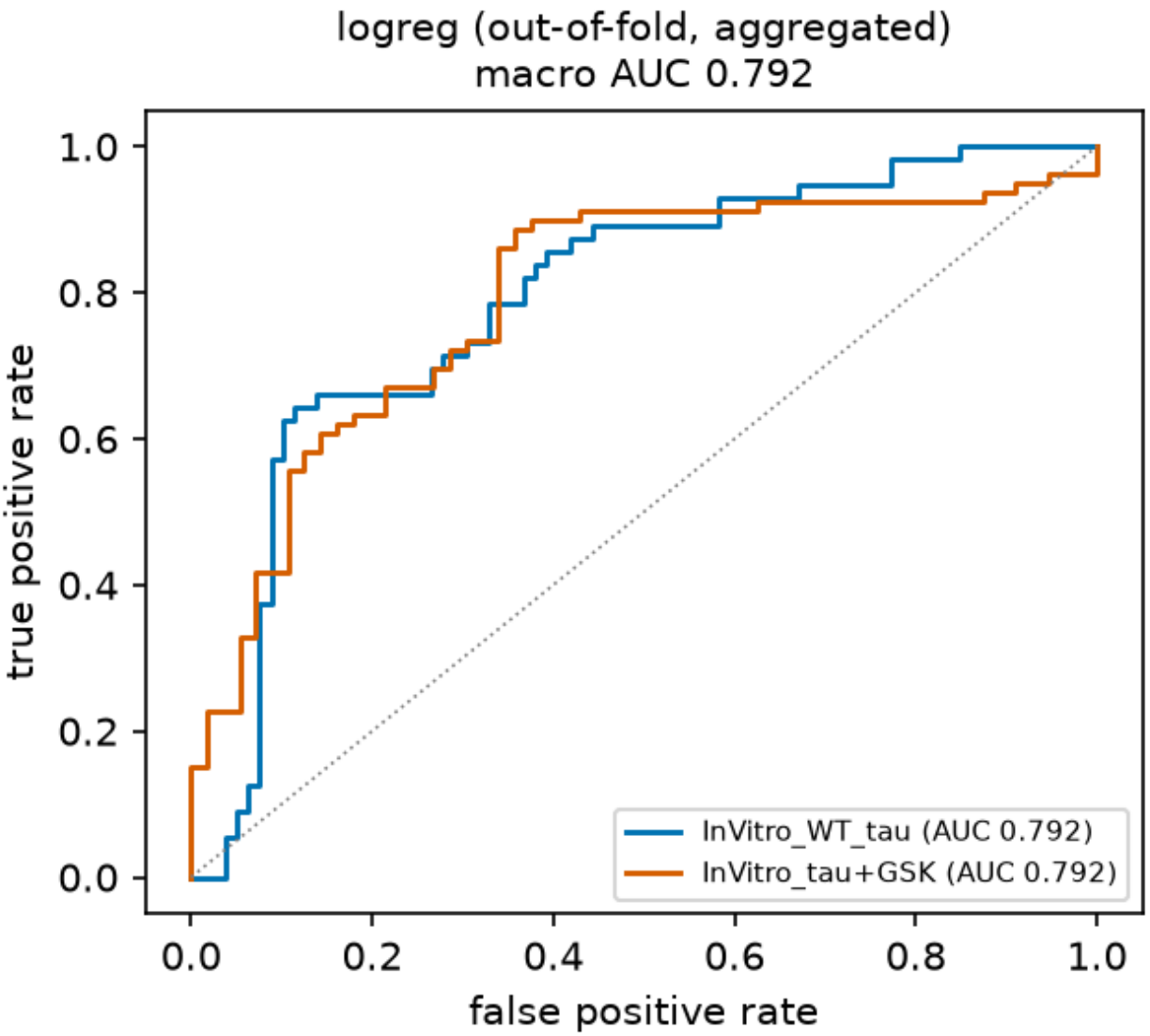
error analysis dummy majority



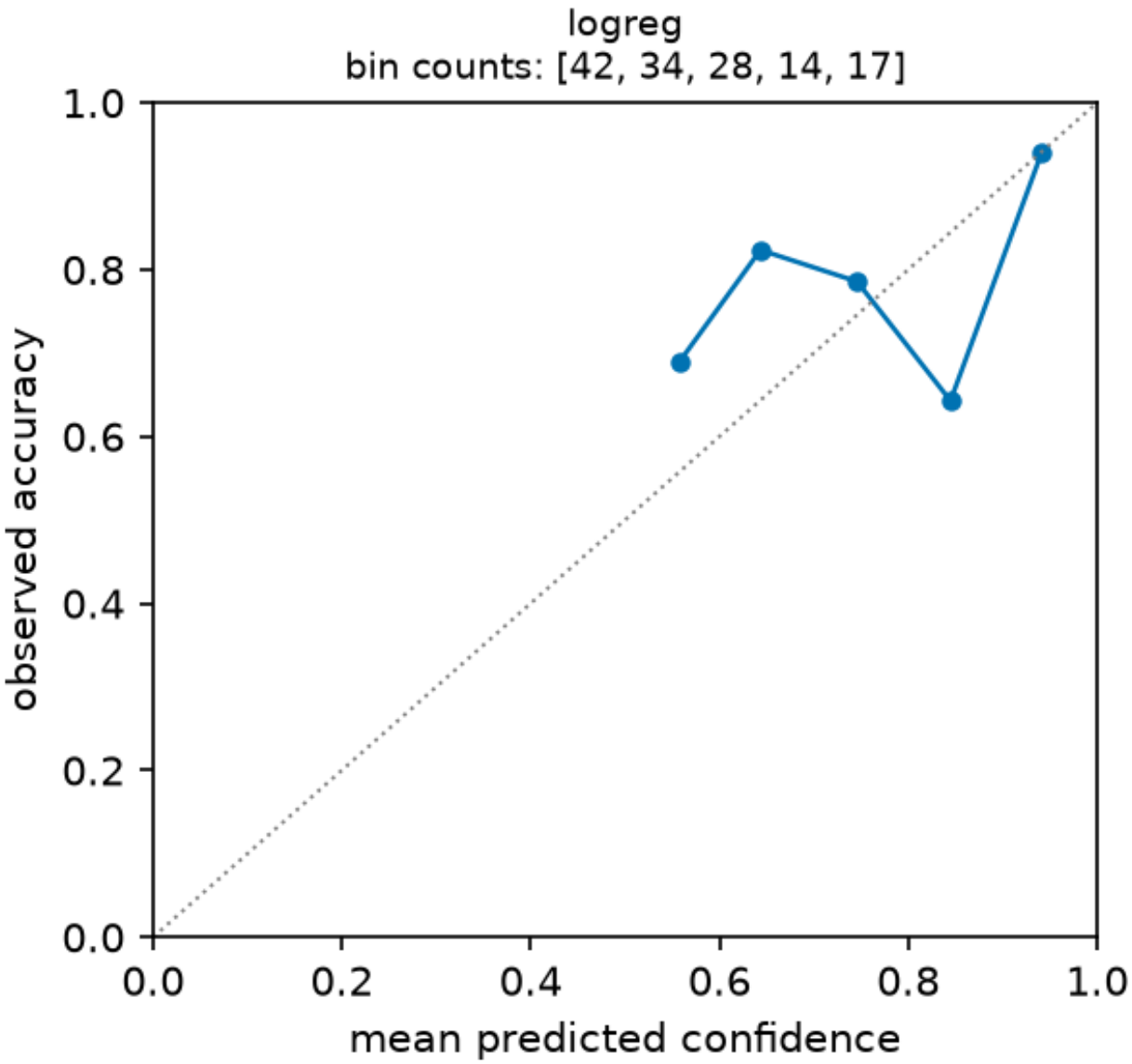
confusion logreg


logreg (aggregated over repeats, n=135)
InVitro_WT_tau
InVitro_tau+GSK
true
0.64
0.36
0.14
0.86
InVitro_WT_tau
InVitro_tau+GSK
predicted
1.0
0.8
0.6
0.4
0.2
0.0

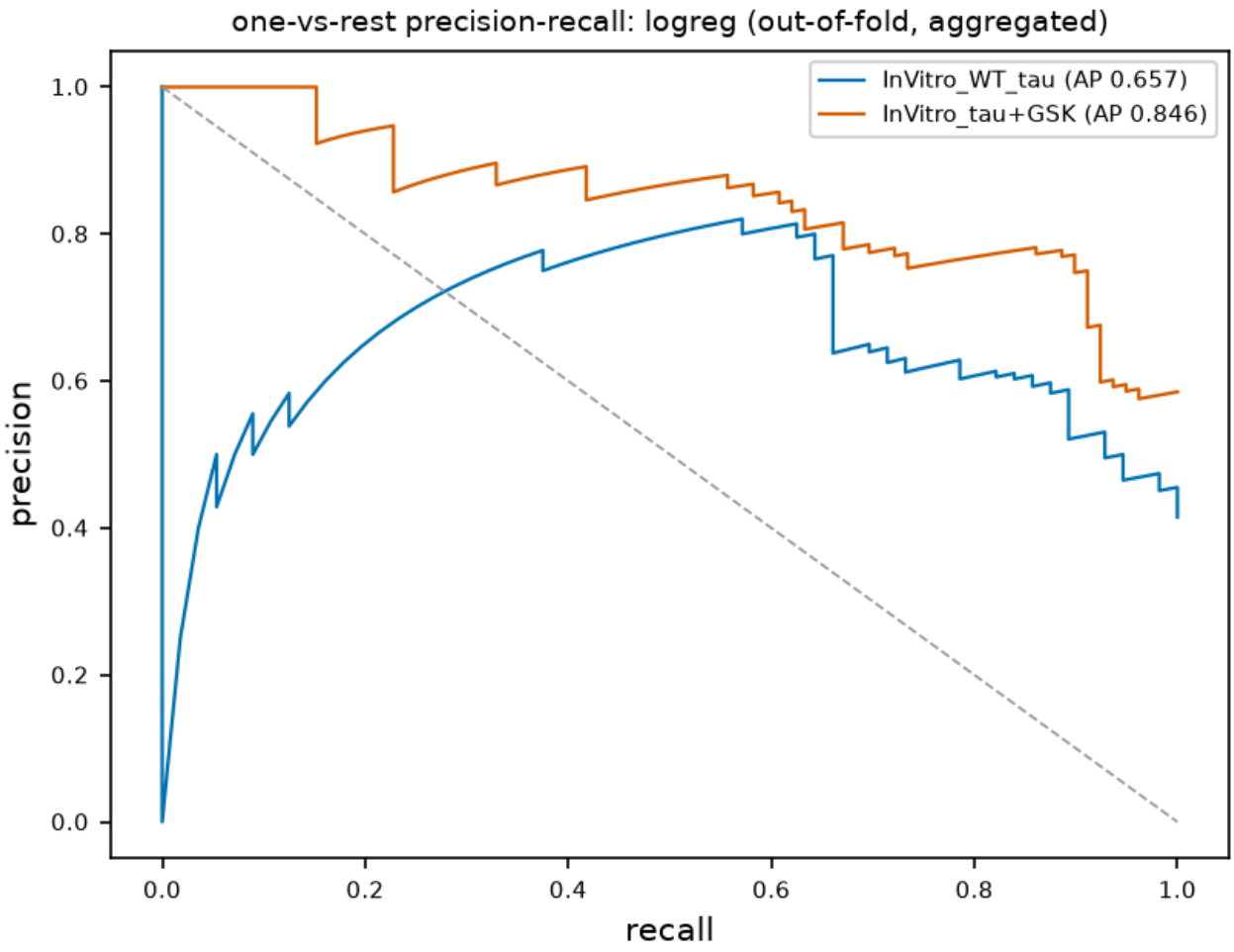
roc logreg



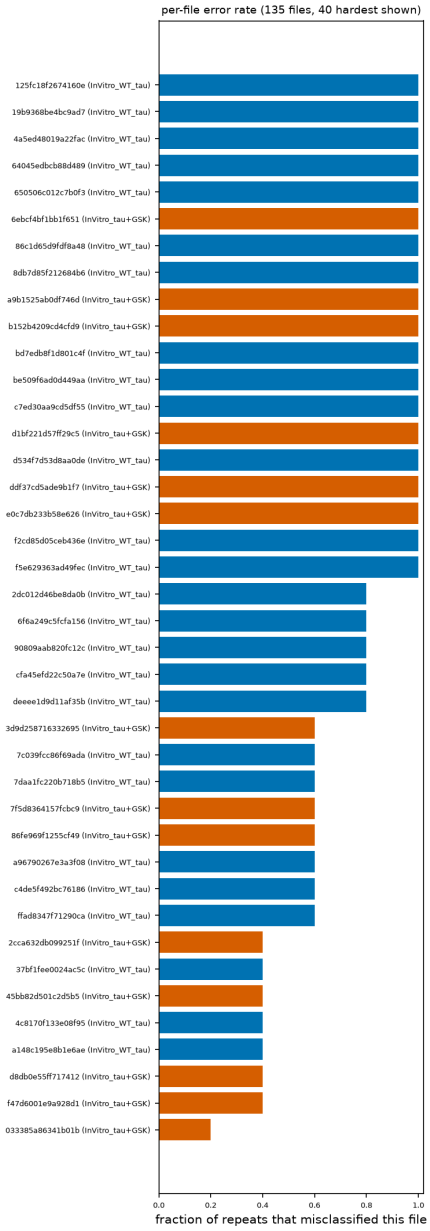
calibration logreg

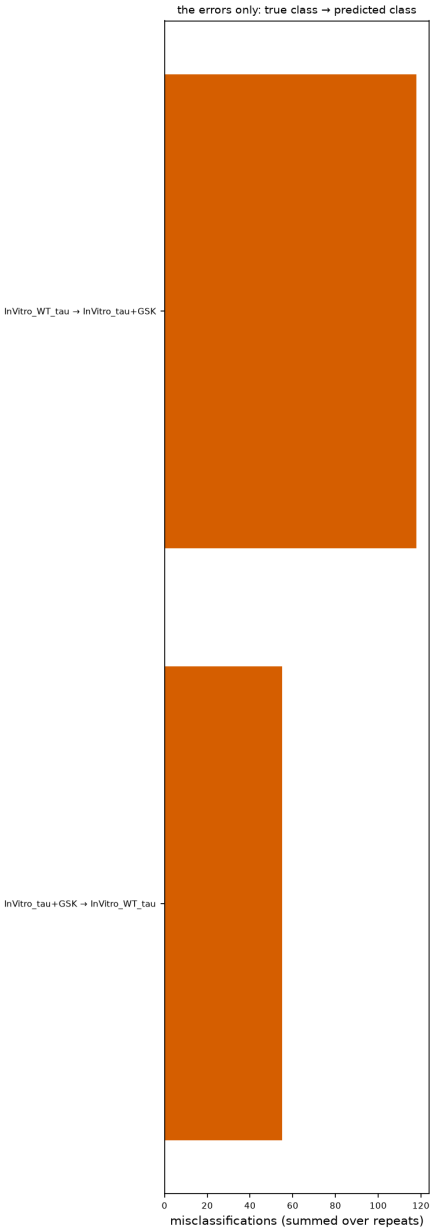

logreg
bin counts: [42, 34, 28, 14, 17]
observed accuracy
mean predicted confidence
0.0
0.2
0.4
0.6
0.8
1.0

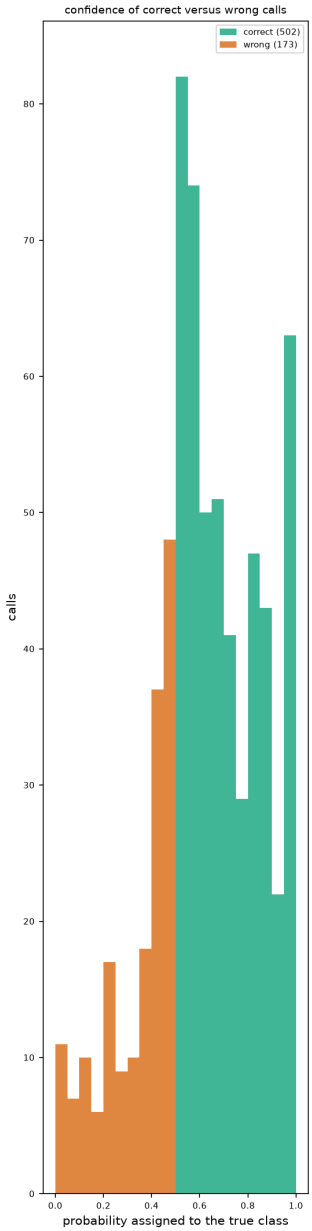
pr logreg



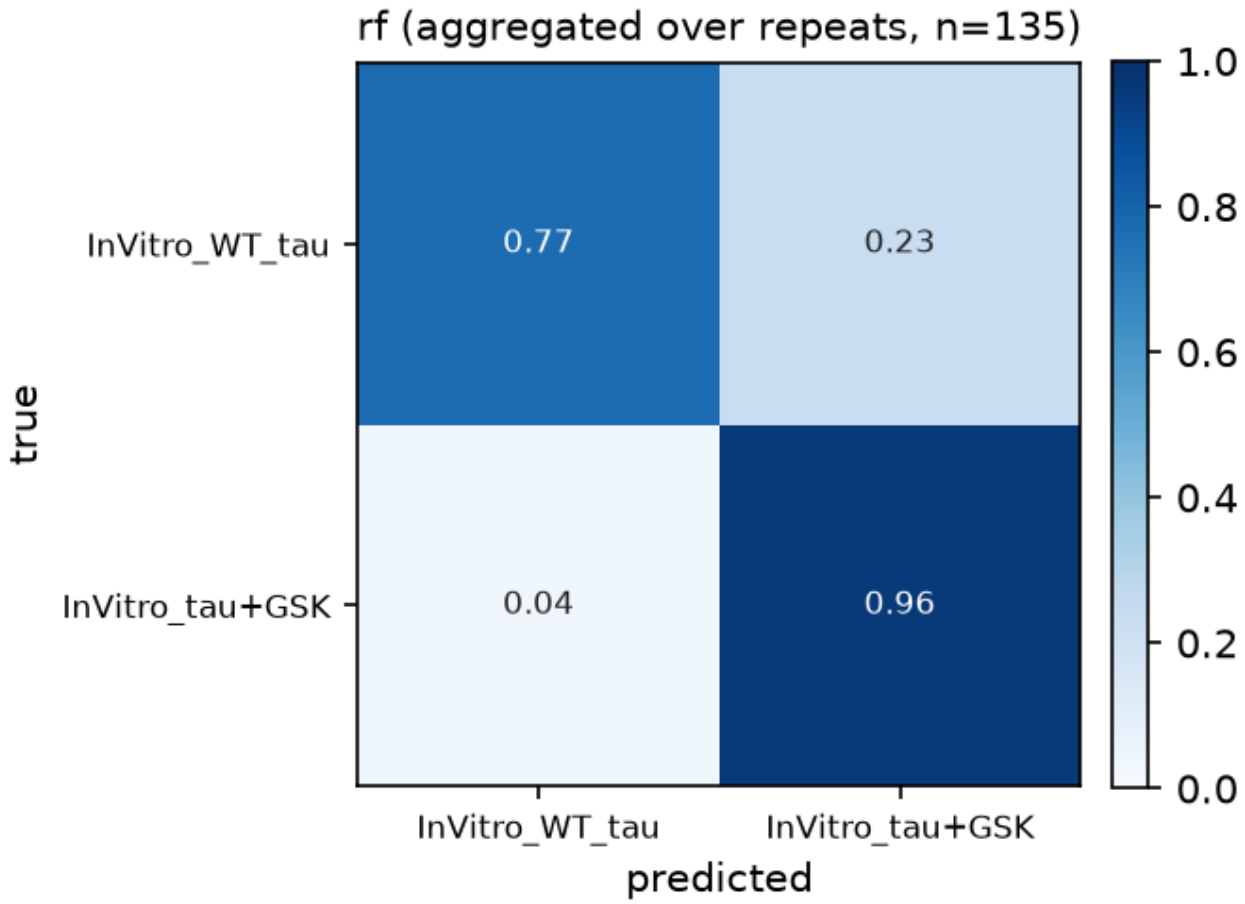
error analysis logreg



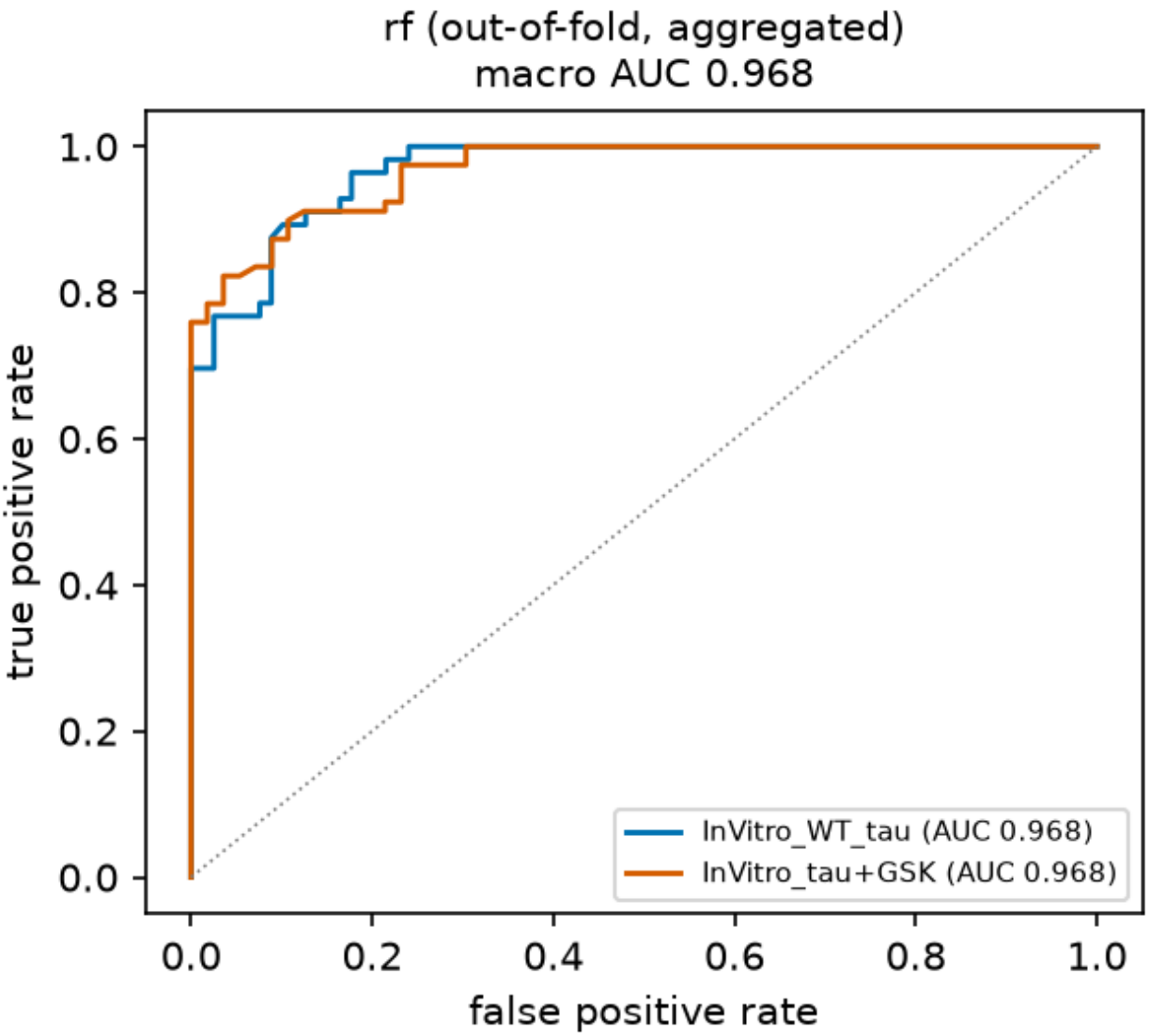




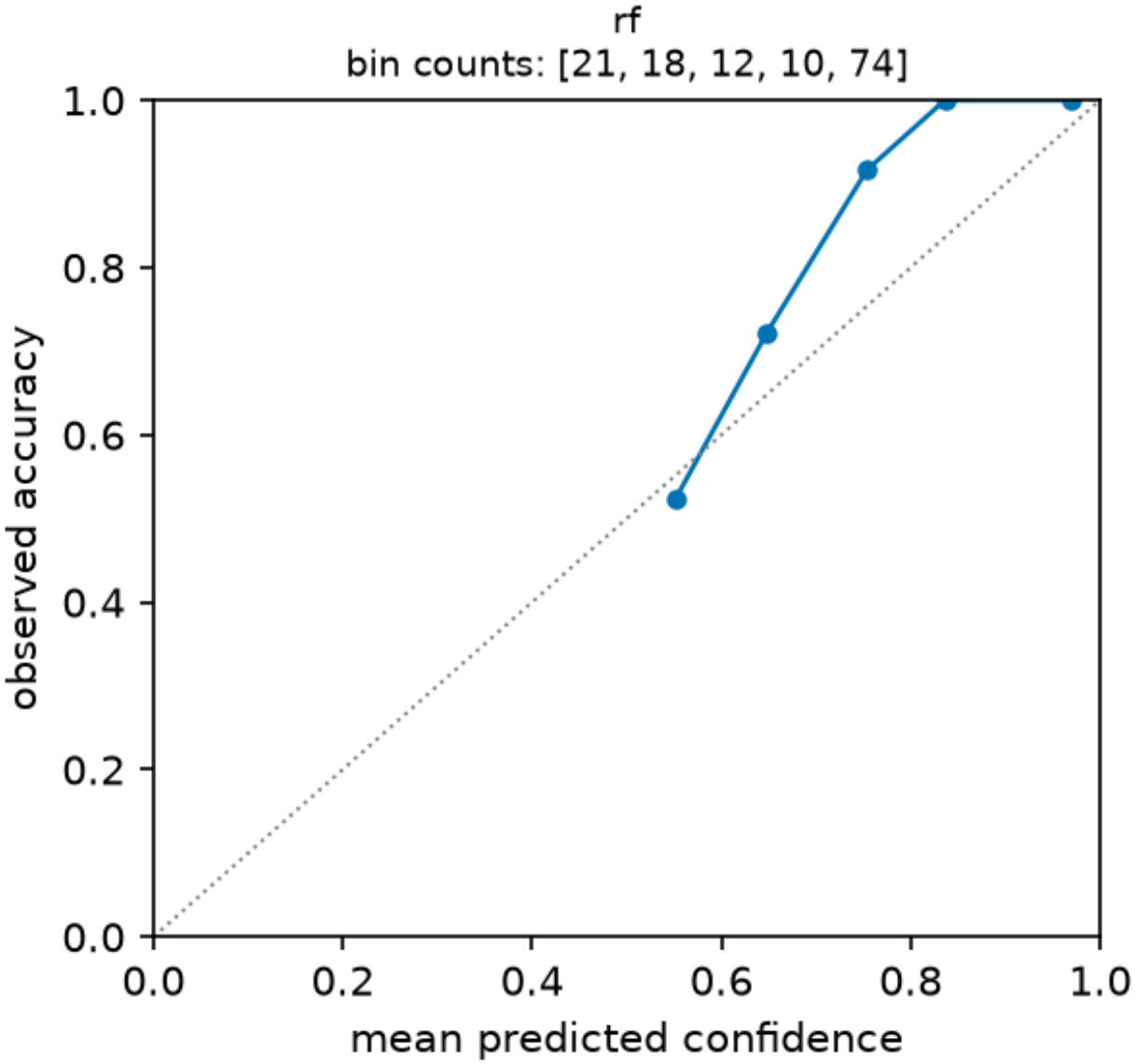
confusion rf


rf (aggregated over repeats, n=135)
InVitro_WT_tau
InVitro_tau+GSK
true
0.77
0.23
0.04
0.96
InVitro_WT_tau
InVitro_tau+GSK
predicted
1.0
0.8
0.6
0.4
0.2
0.0

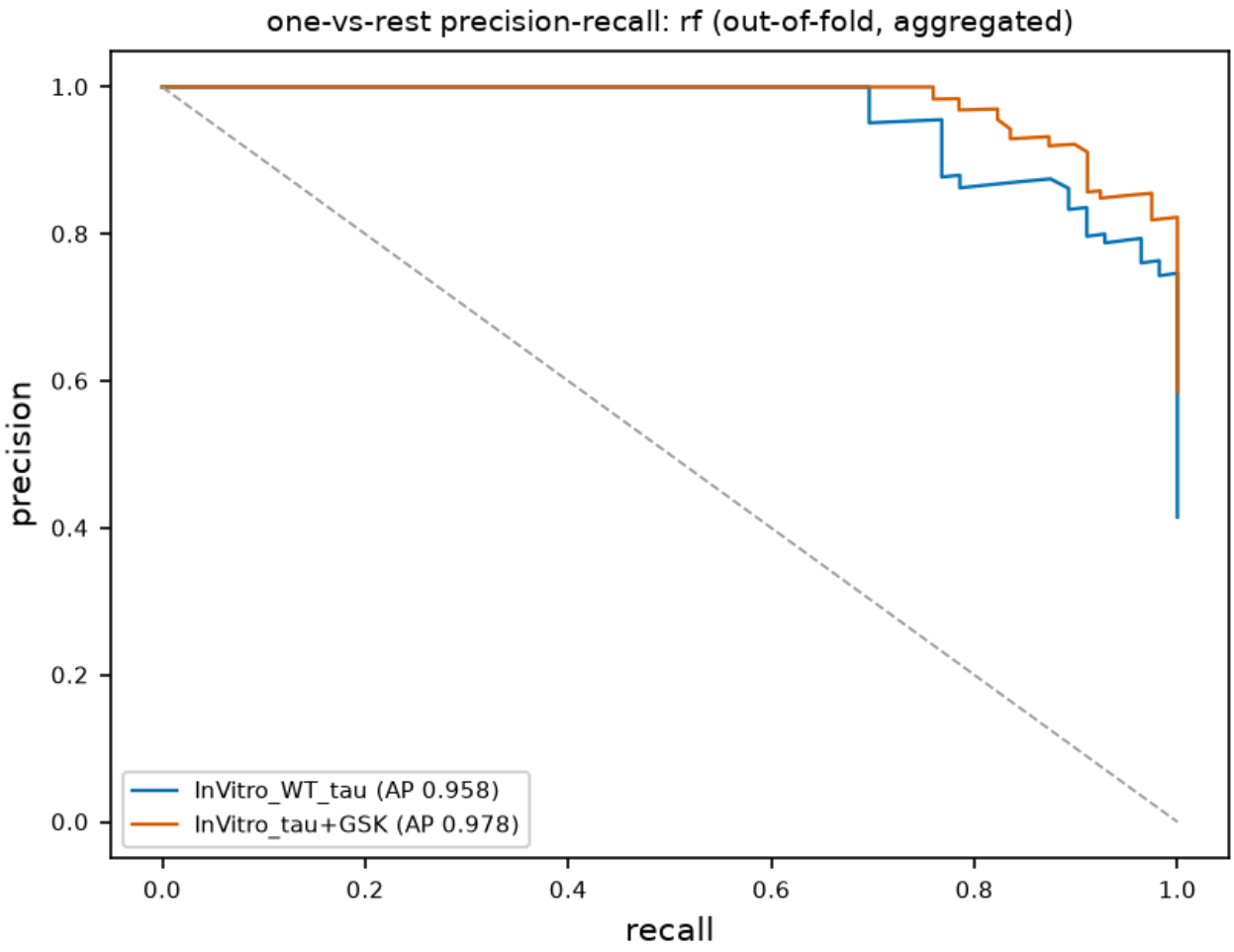
roc rf



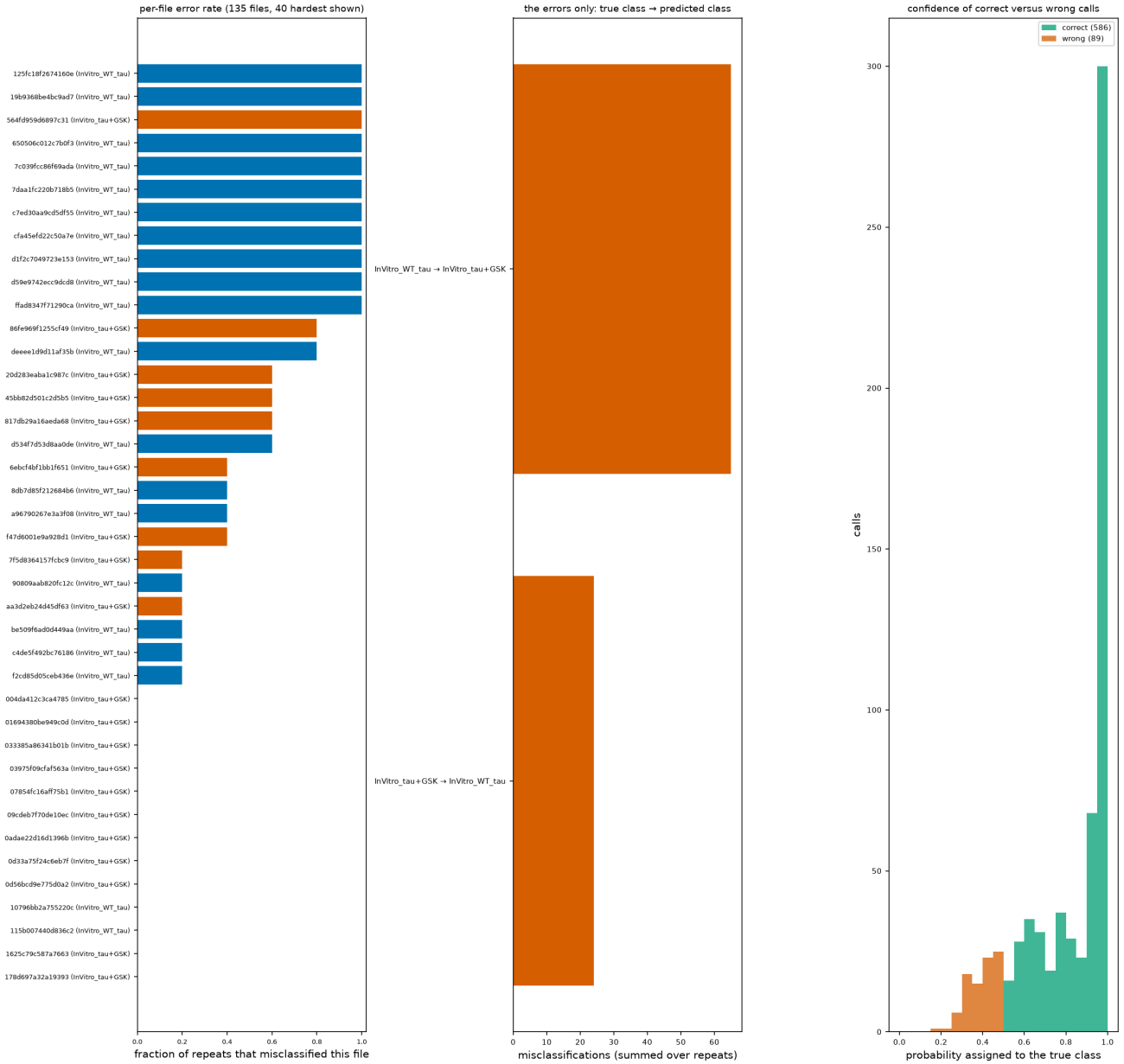
calibration rf


rf
bin counts: [21, 18, 12, 10, 74]
observed accuracy
mean predicted confidence
0.0
0.2
0.4
0.6
0.8
1.0

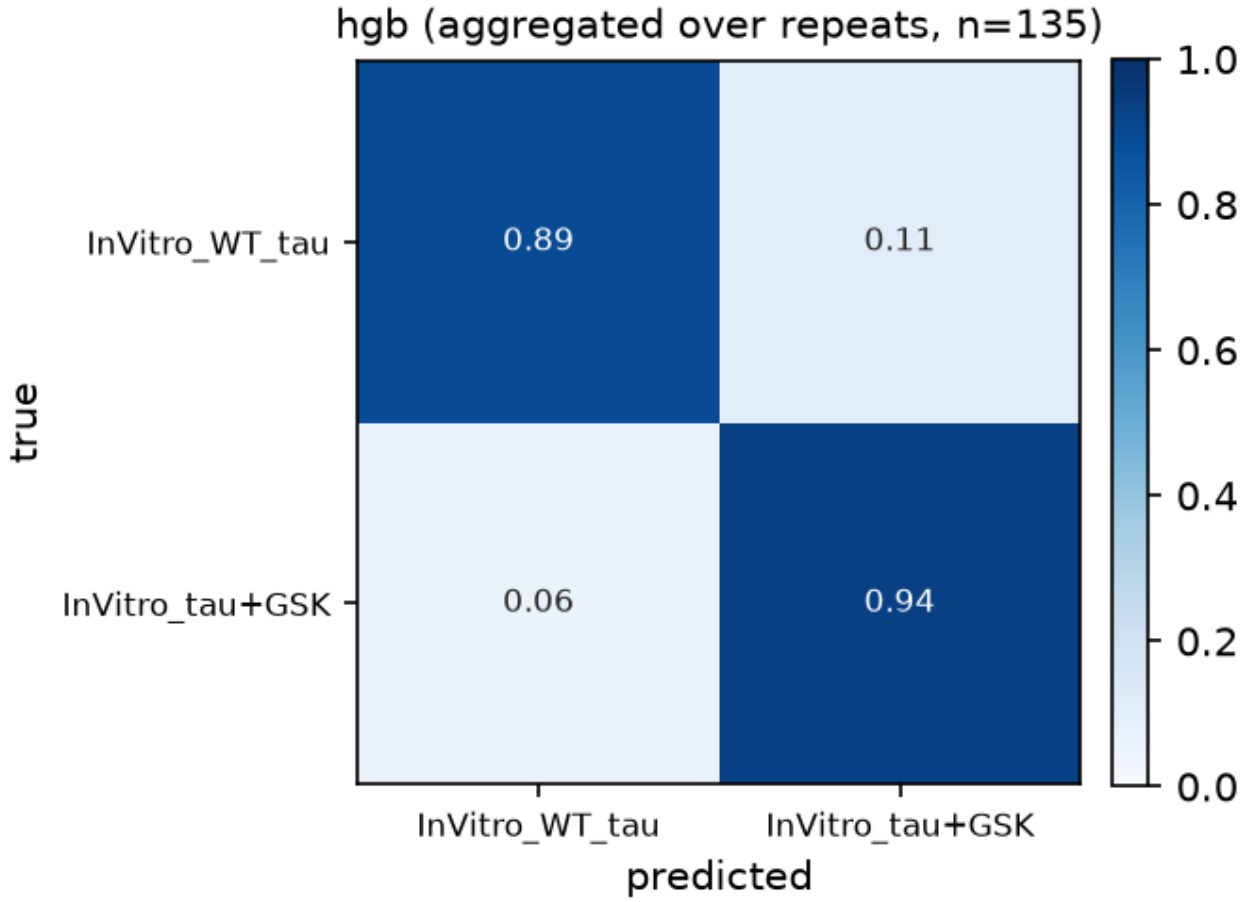
pr rf



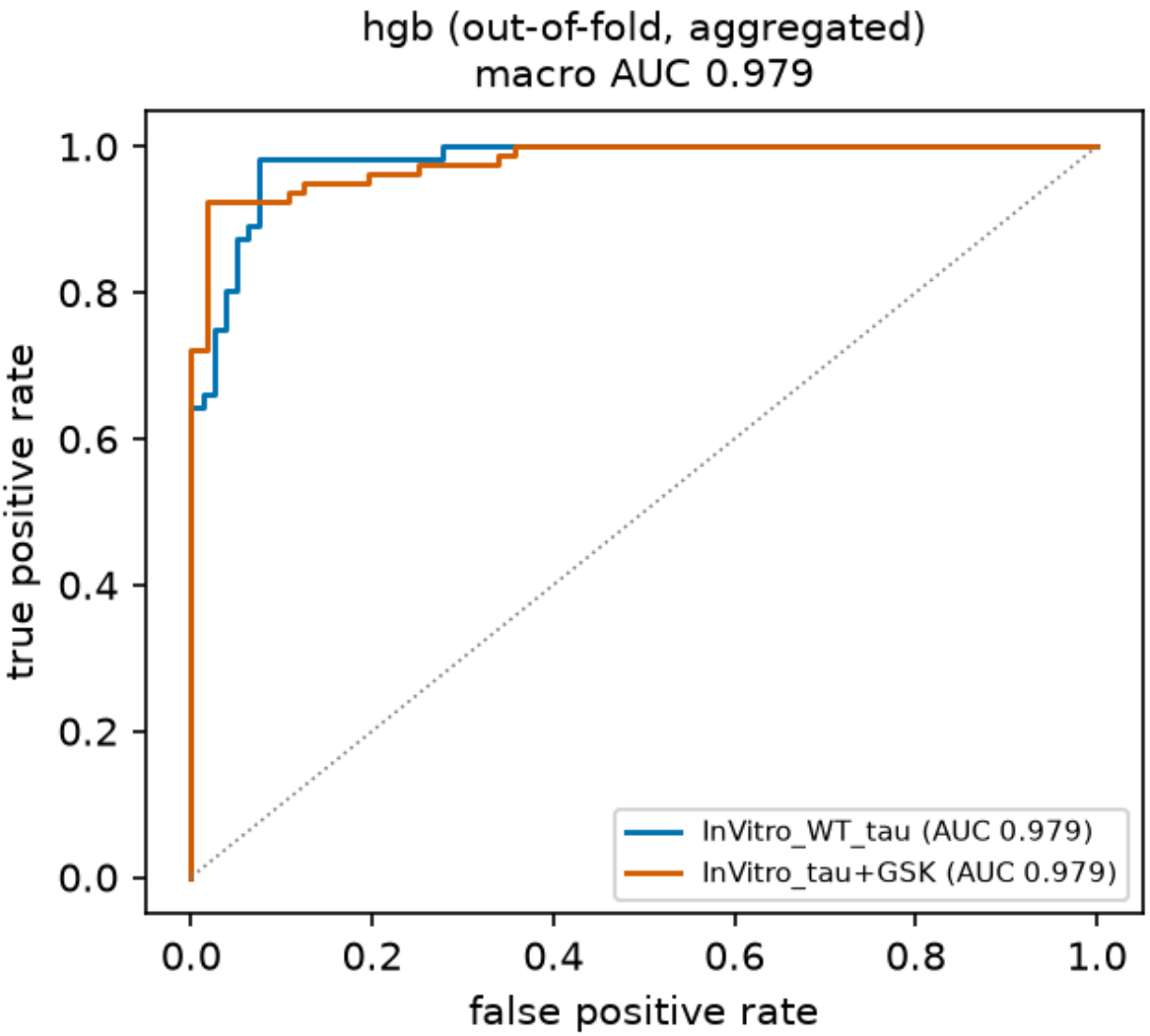
error analysis rf



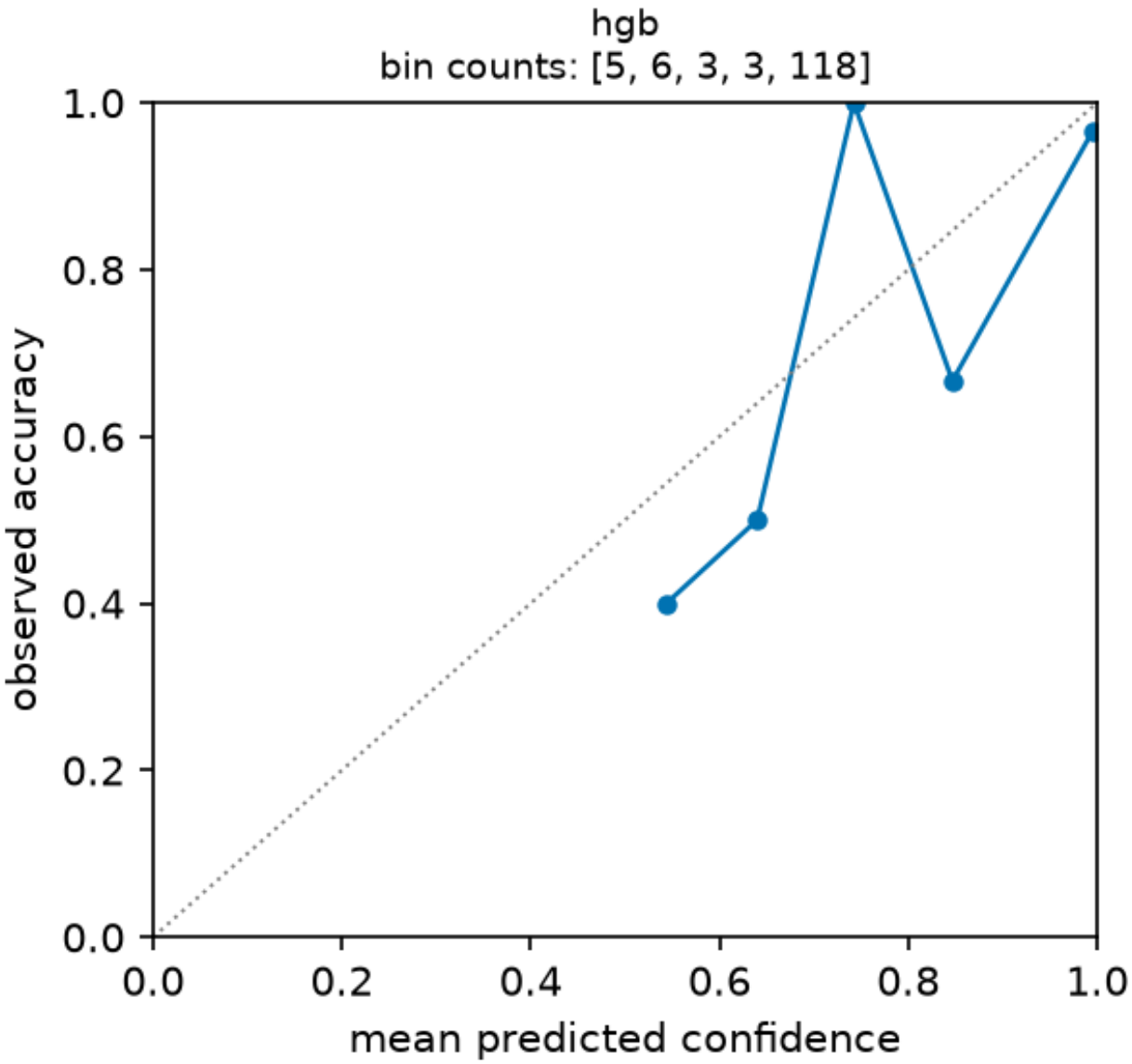
confusion hgb


hgb (aggregated over repeats, n=135)
InVitro_WT_tau
InVitro_tau+GSK
true
0.89
0.11
0.06
0.94
InVitro_WT_tau
InVitro_tau+GSK
predicted
1.0
0.8
0.6
0.4
0.2
0.0

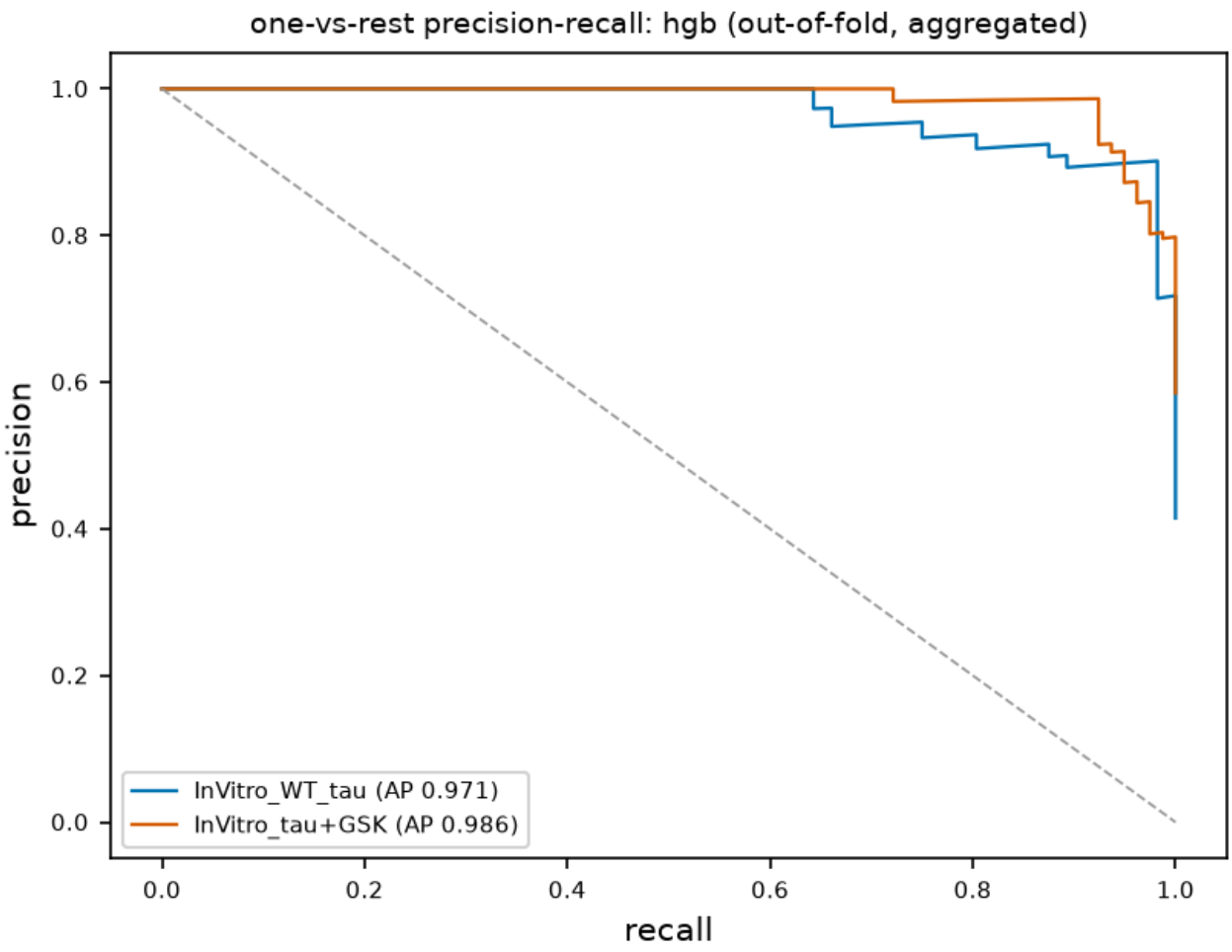
roc hgb



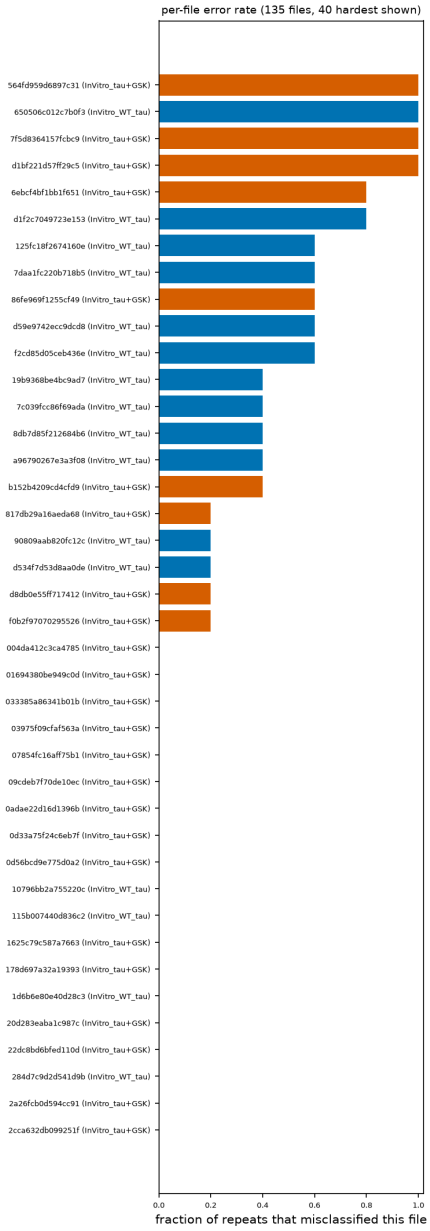
calibration hgb

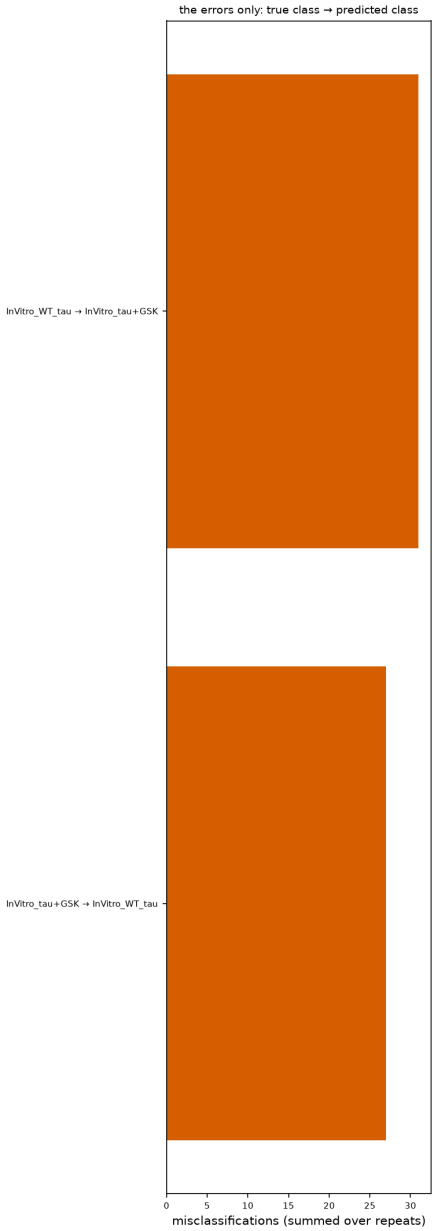

hgb
bin counts: [5, 6, 3, 3, 118]
observed accuracy
mean predicted confidence
0.0
0.2
0.4
0.6
0.8
1.0

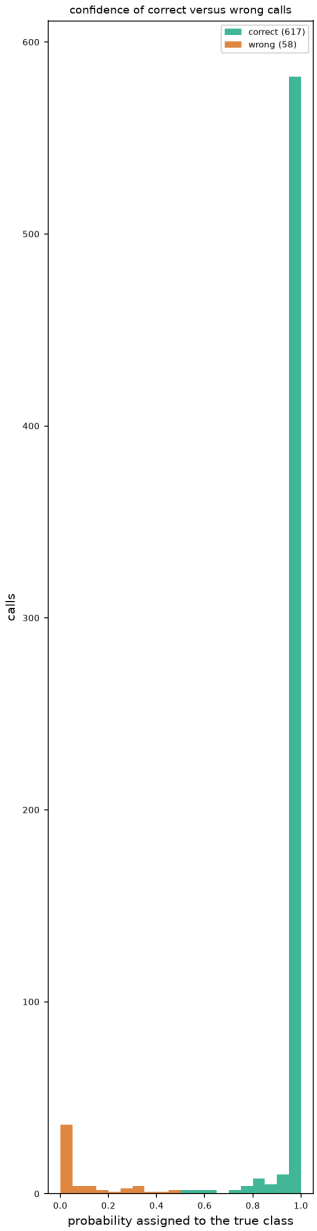
pr hgb



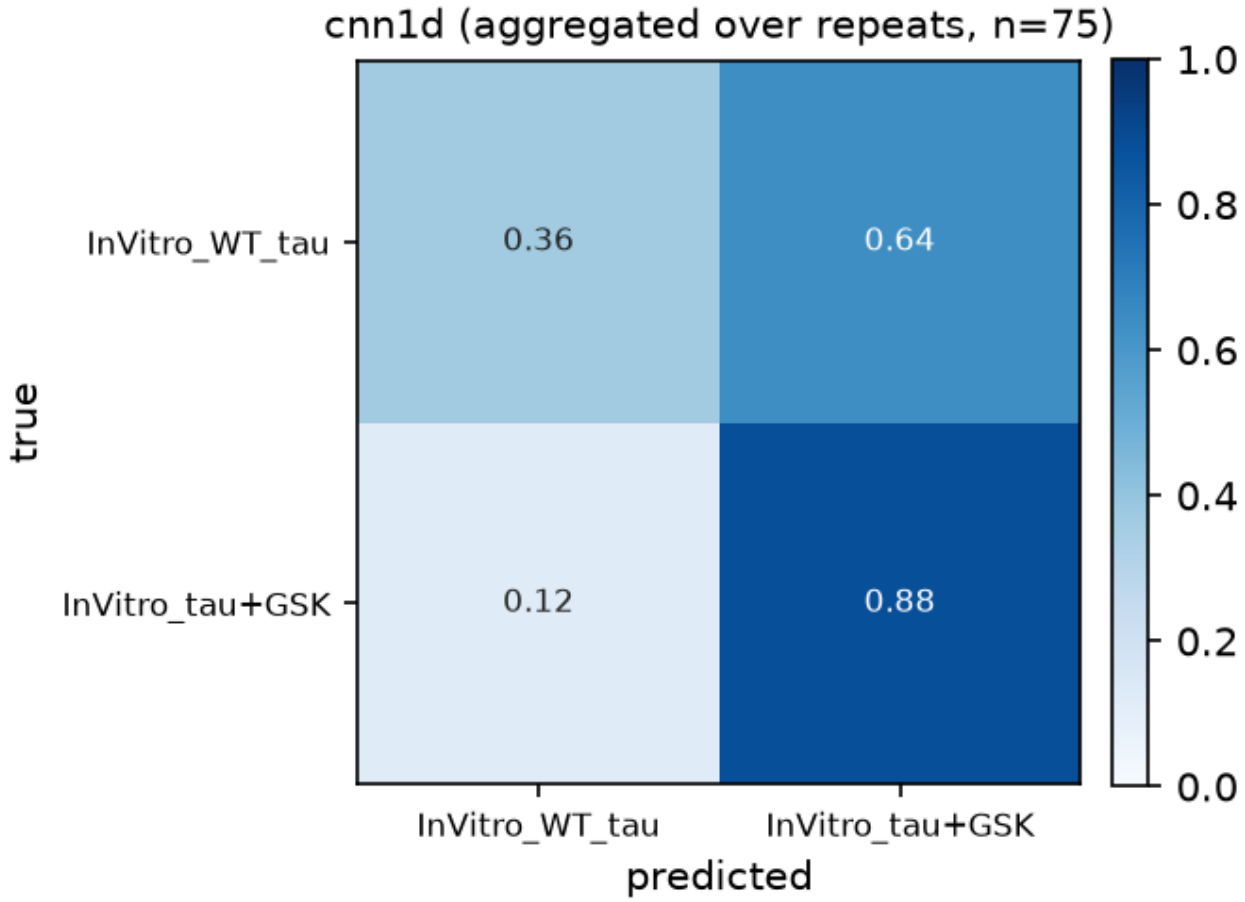
error analysis hgb



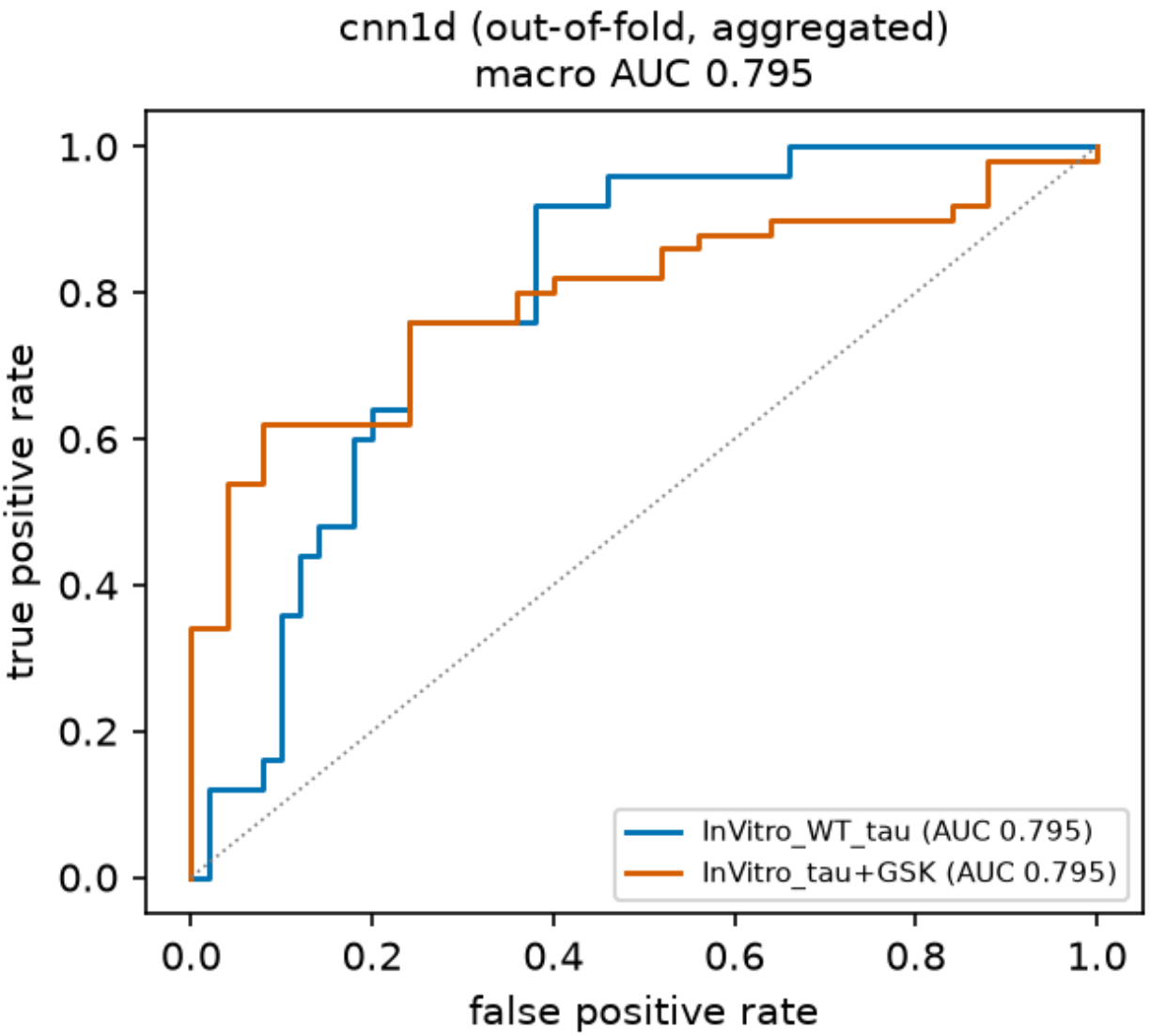




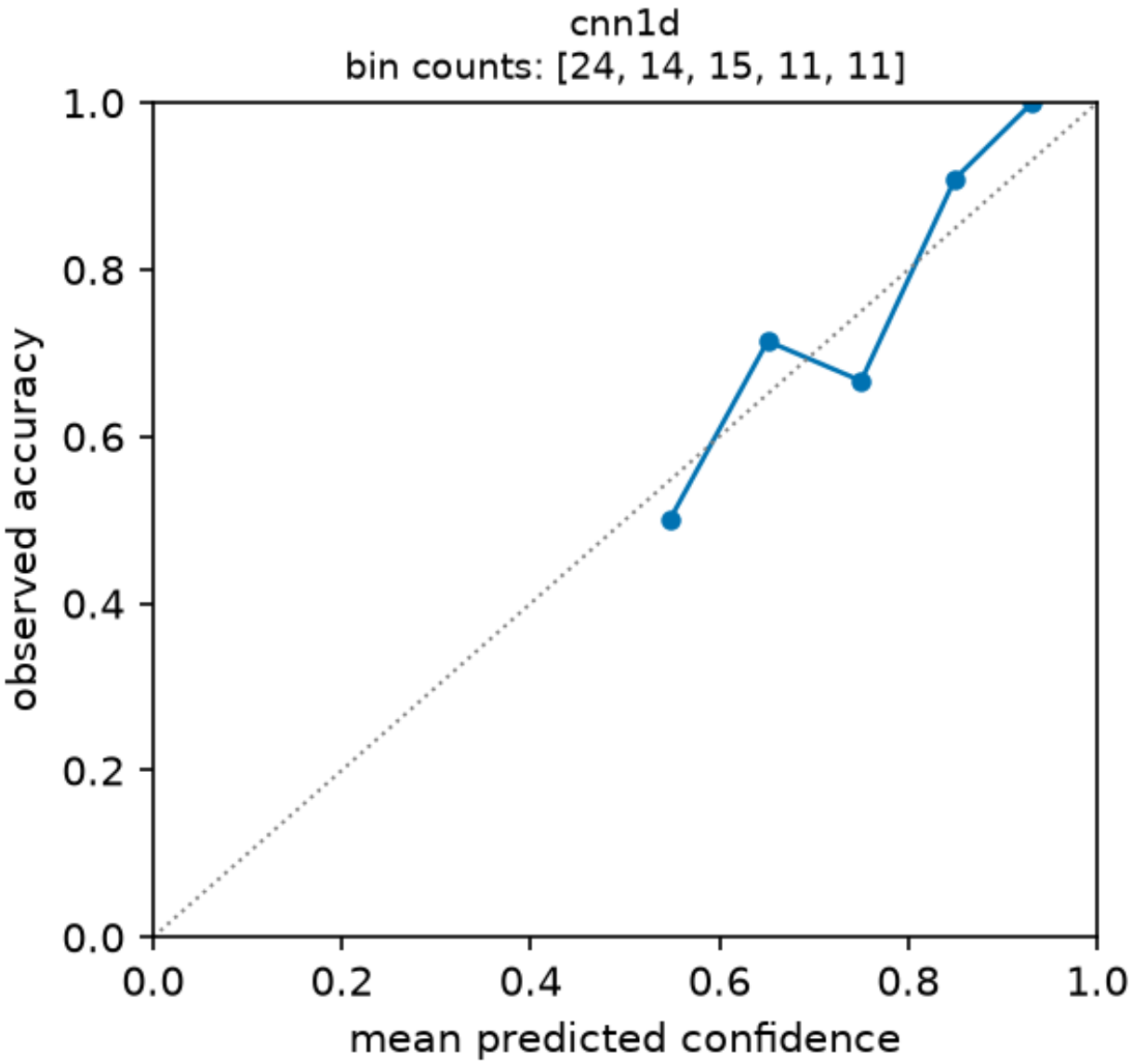
confusion cnn1d


cnn1d (aggregated over repeats, n=75)
InVitro_WT_tau
InVitro_tau+GSK
true
0.36
0.64
0.12
0.88
InVitro_WT_tau
InVitro_tau+GSK
predicted
1.0
0.8
0.6
0.4
0.2
0.0

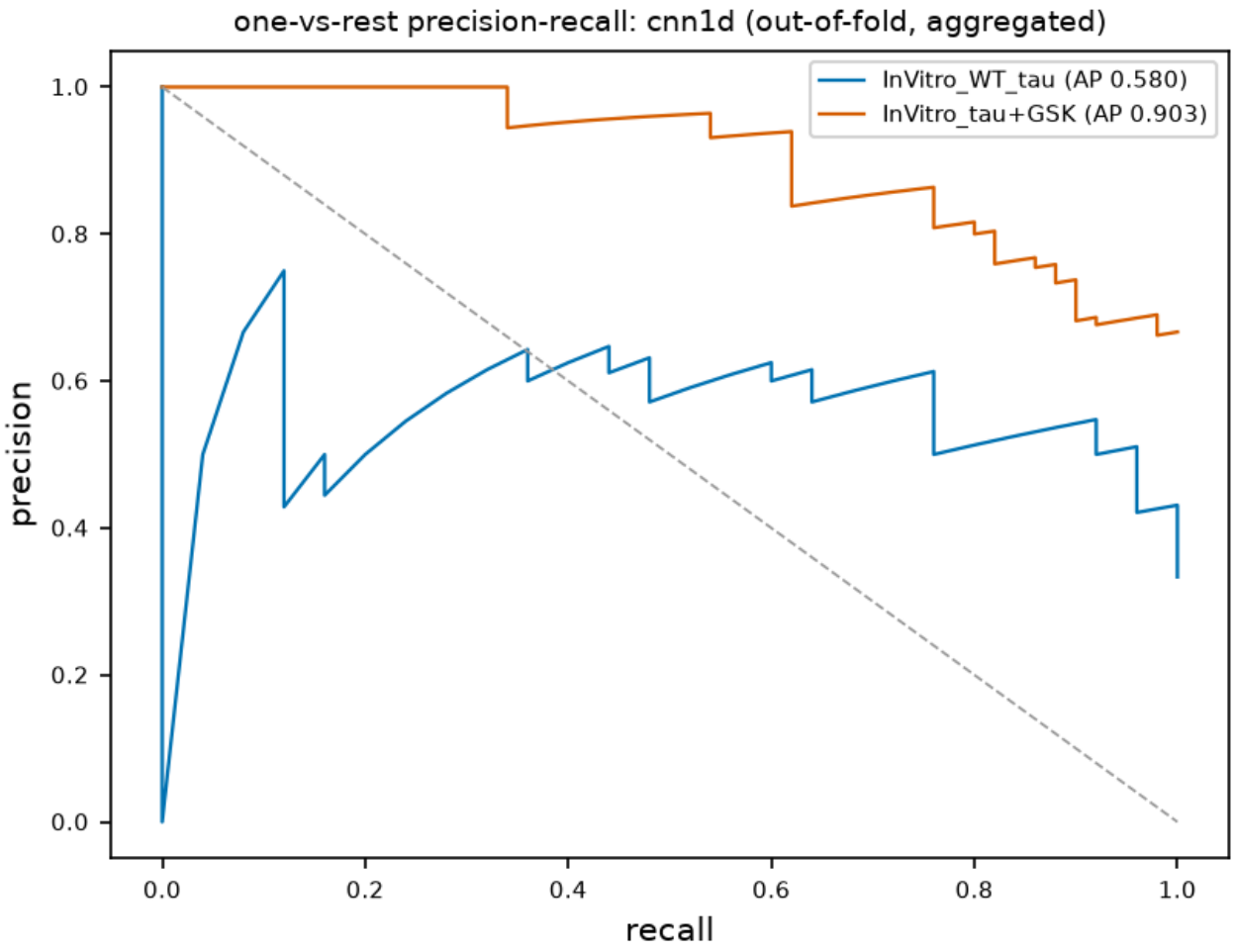
roc cnn1d



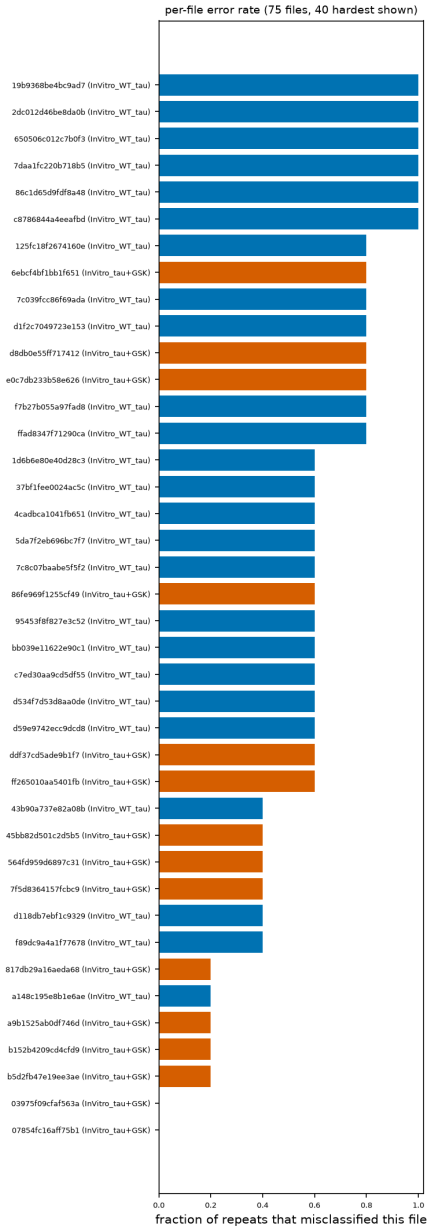
calibration cnn1d

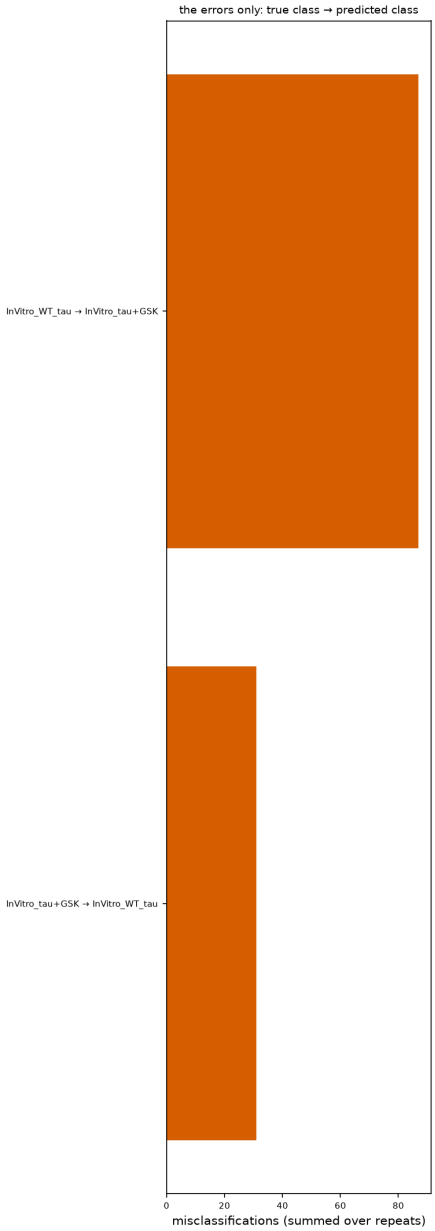

cnn1d
bin counts: [24, 14, 15, 11, 11]
observed accuracy
mean predicted confidence
0.0
0.2
0.4
0.6
0.8
1.0

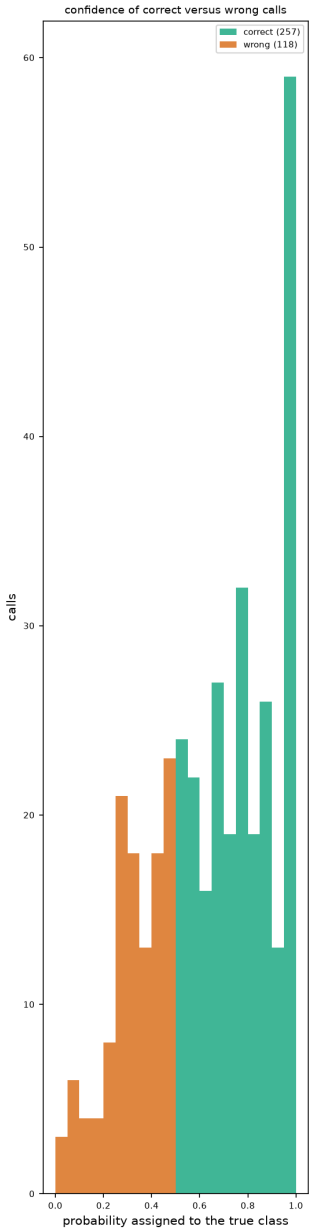
pr cnn1d



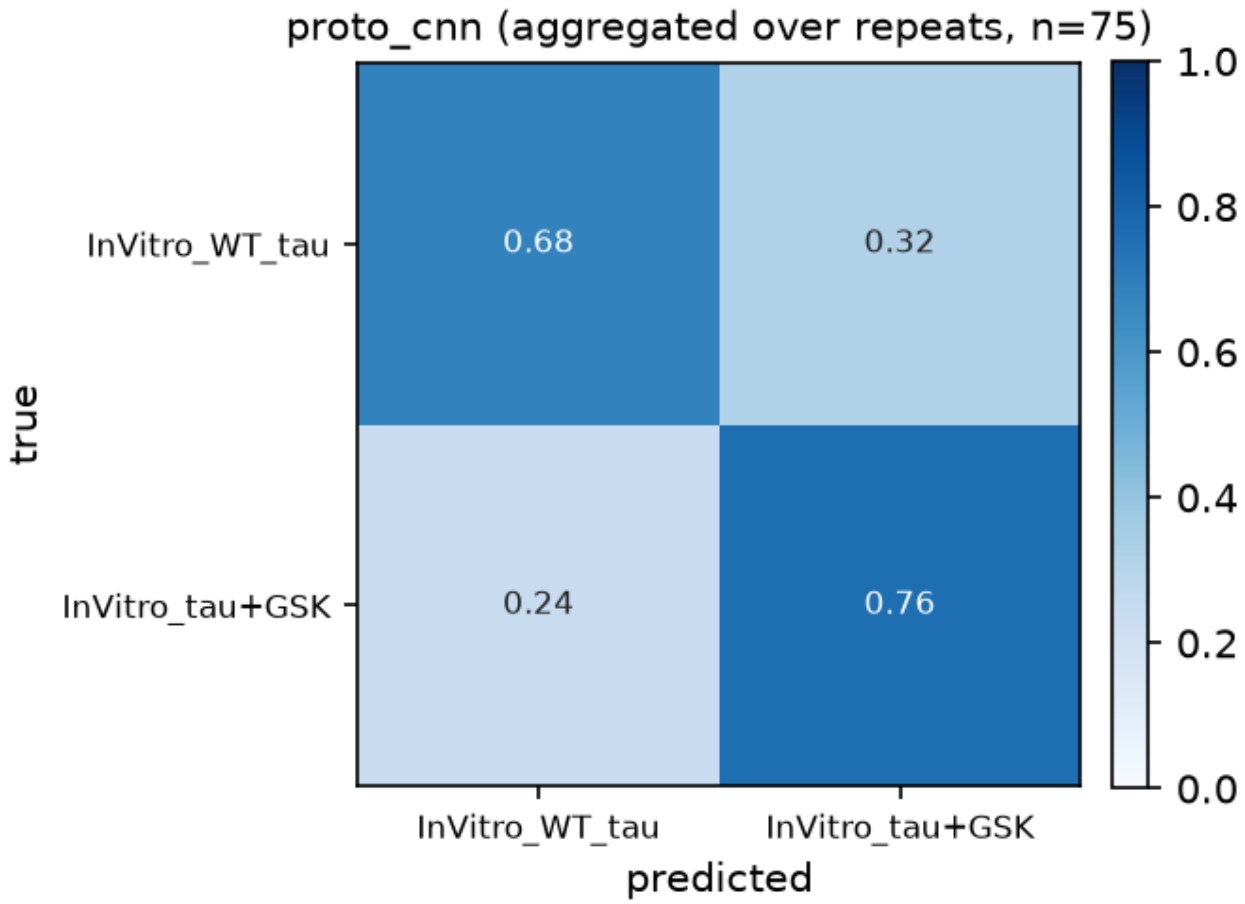
error analysis cnn1d



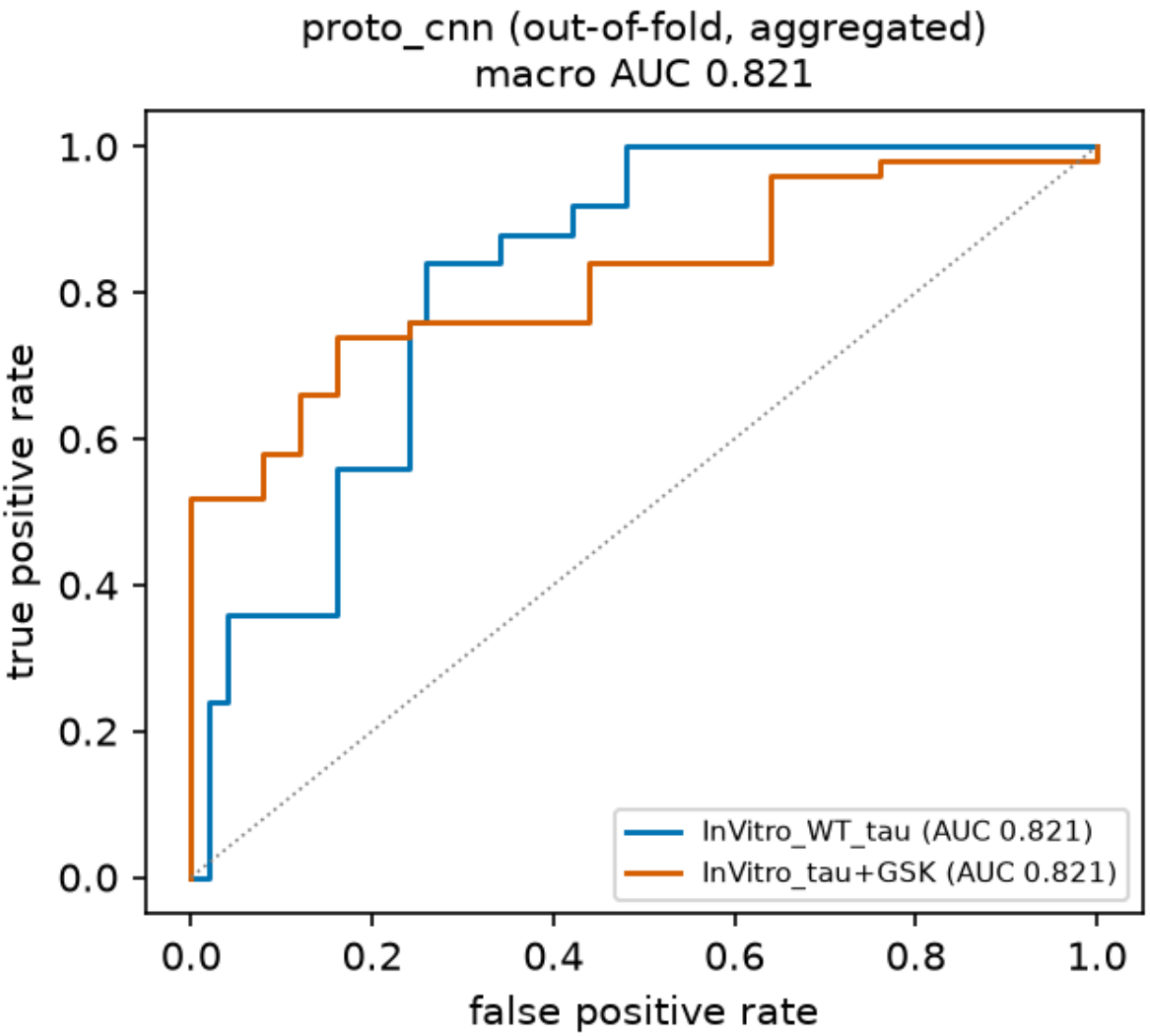




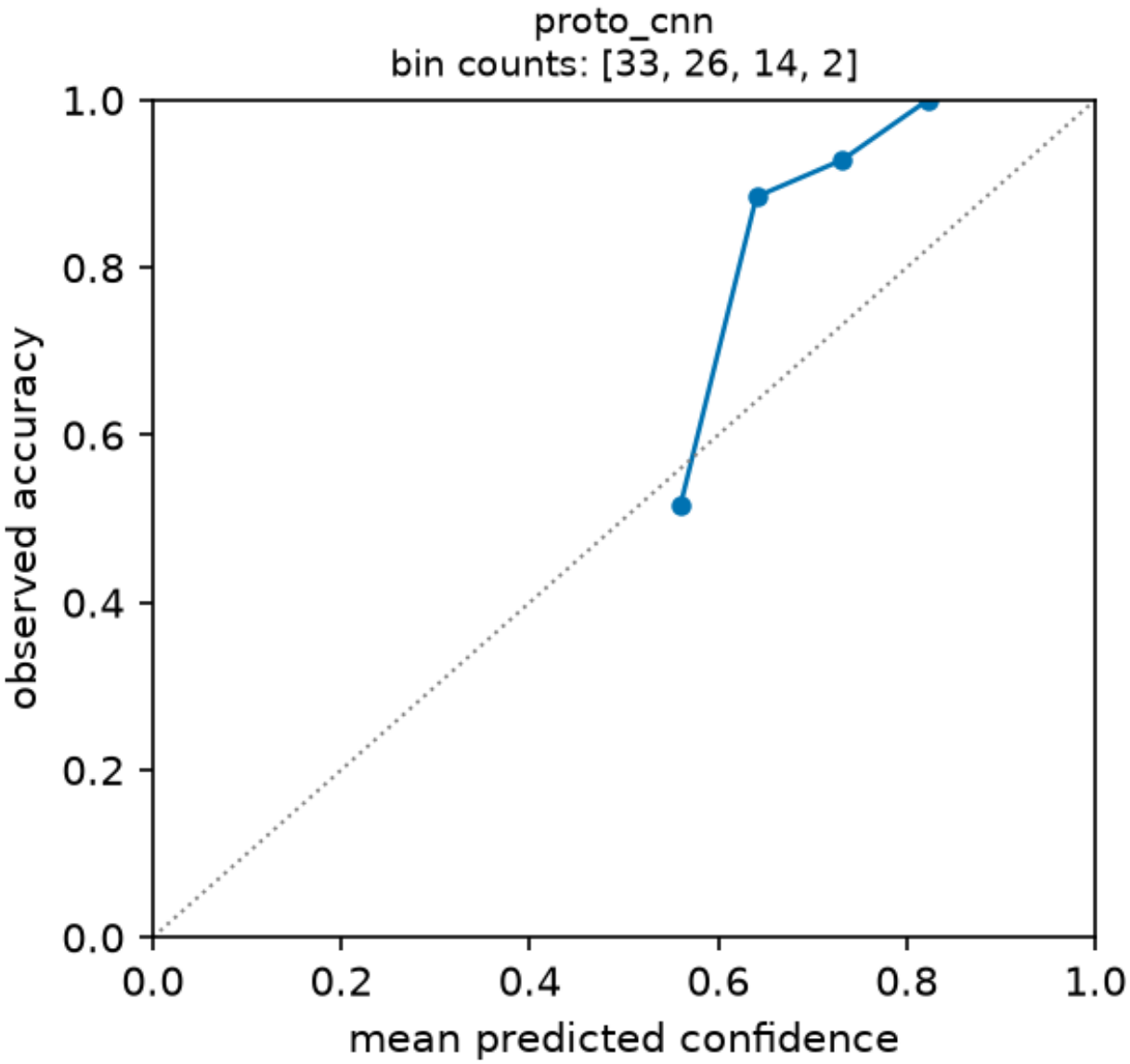
confusion proto cnn



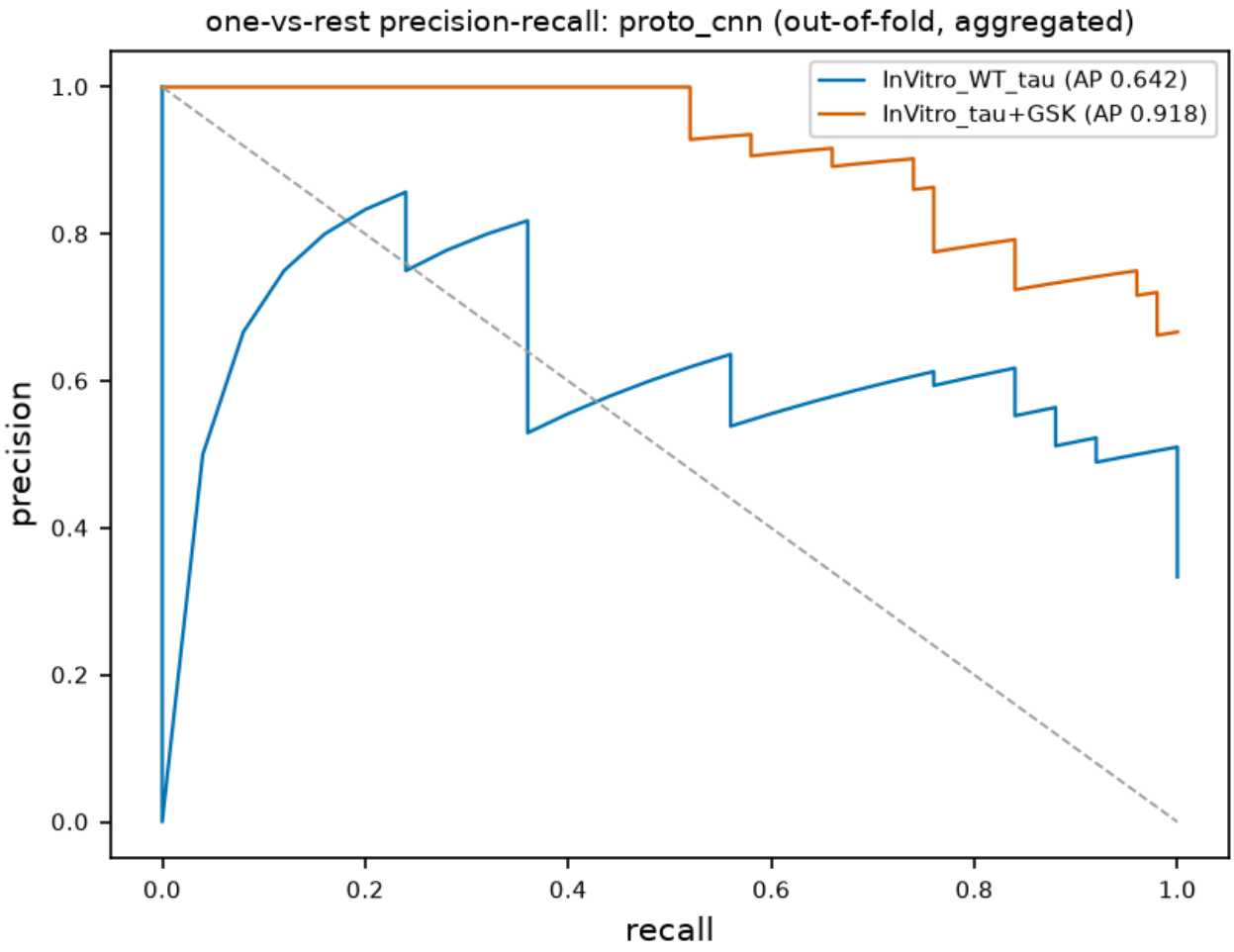
roc proto cnn



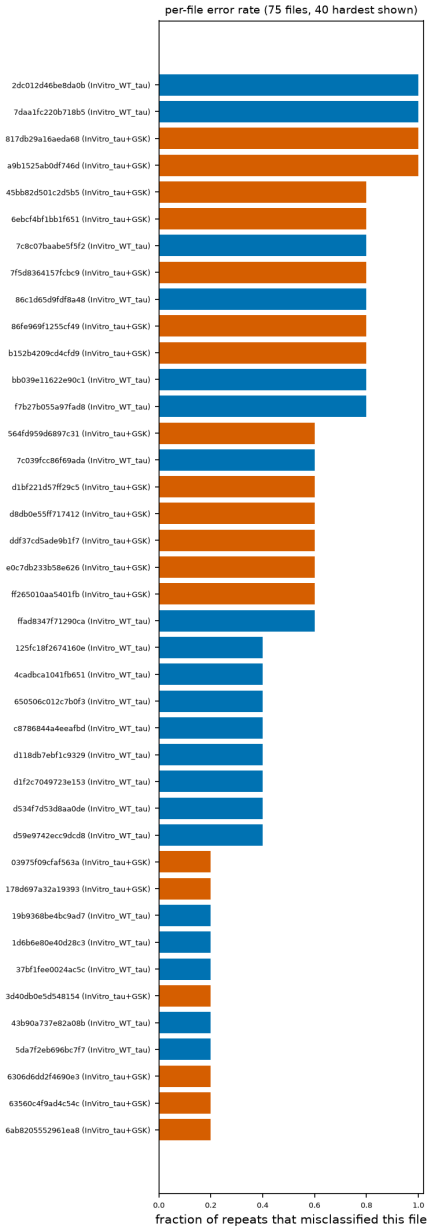
calibration proto cnn

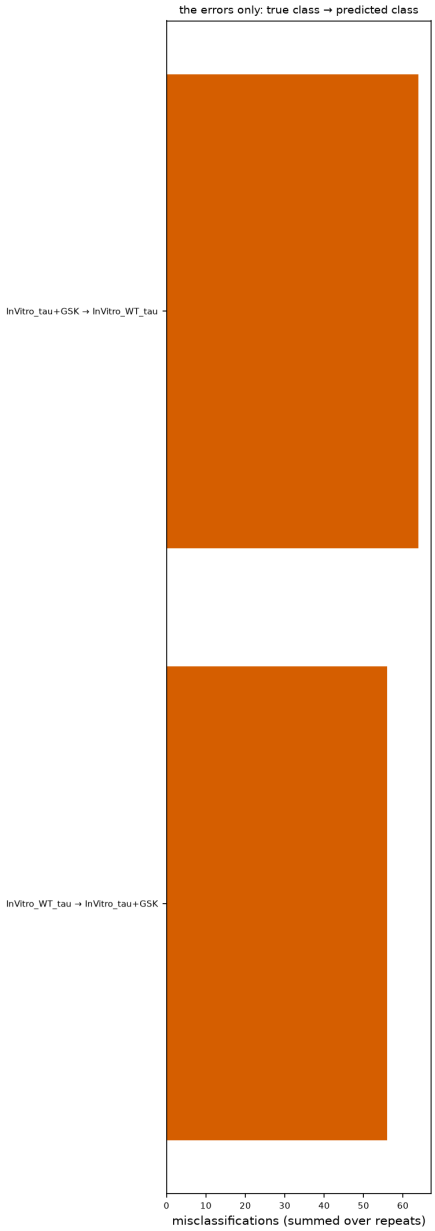

proto_cnn
bin counts: [33, 26, 14, 2]
observed accuracy
mean predicted confidence
0.0
0.2
0.4
0.6
0.8
1.0

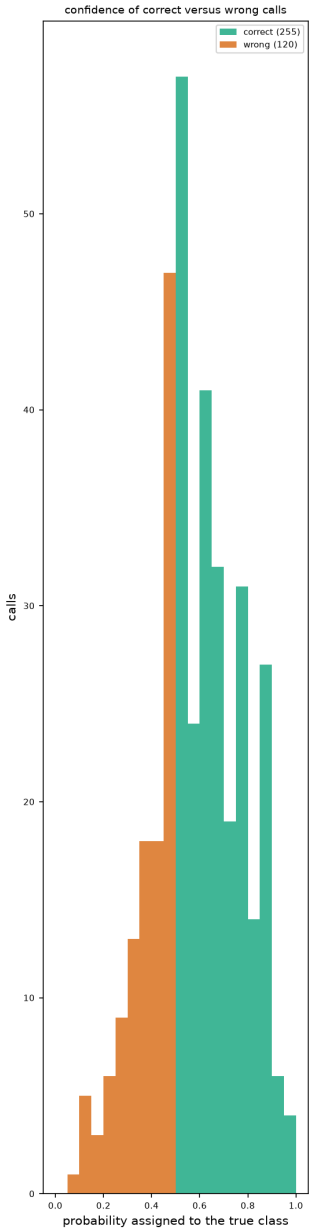
pr proto cnn



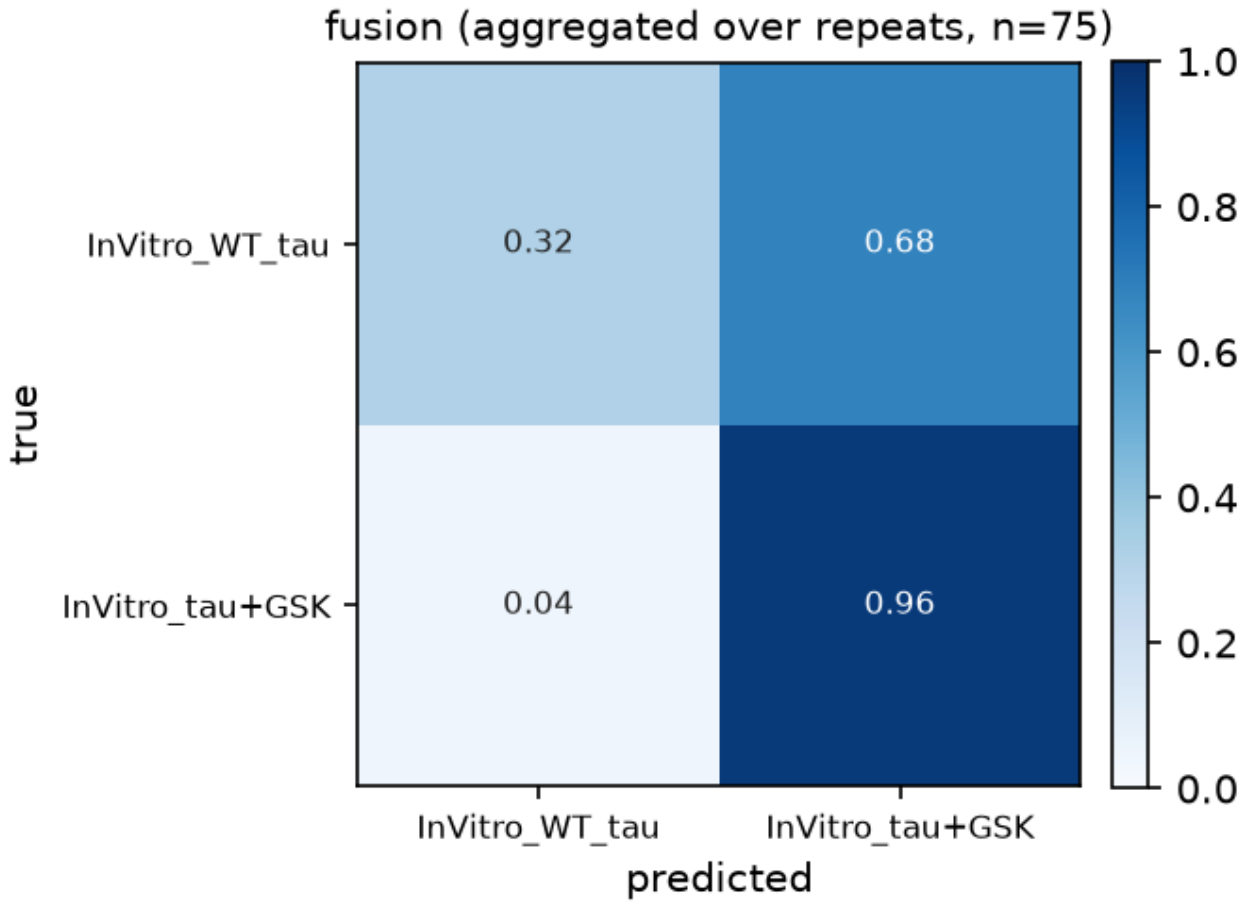
error analysis proto cnn



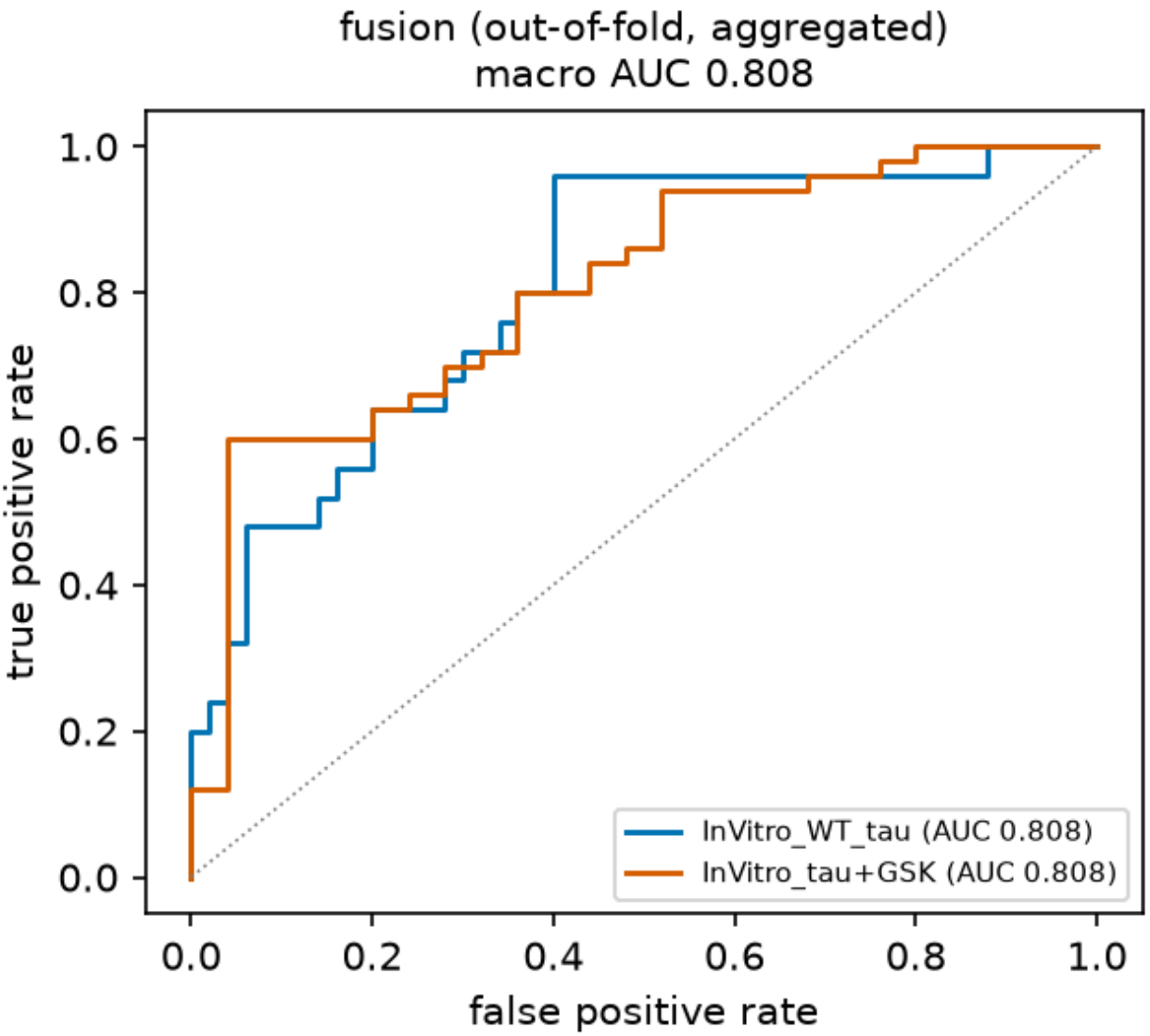




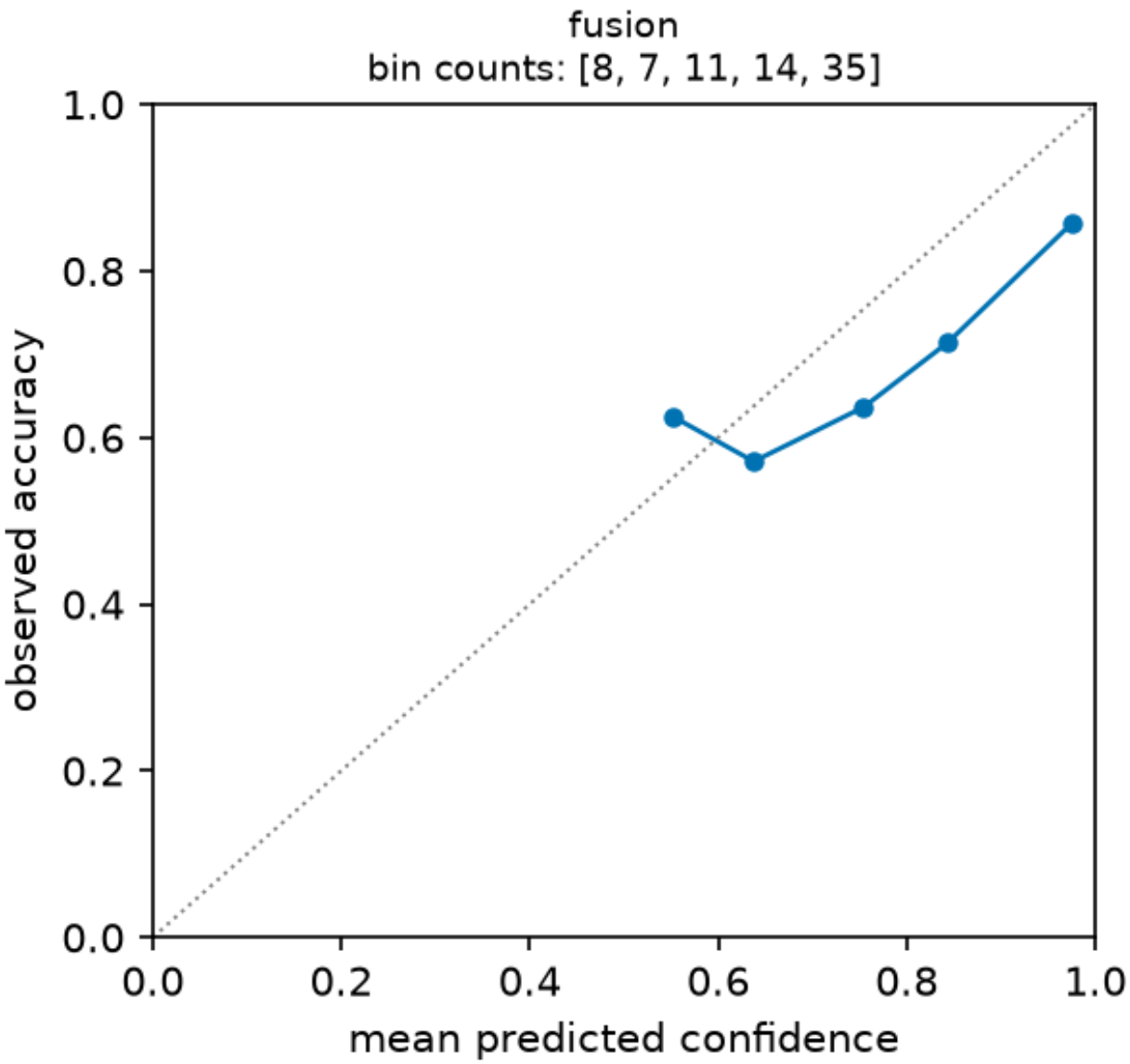
confusion fusion


fusion (aggregated over repeats, n=75)
InVitro_WT_tau
InVitro_tau+GSK
true
0.32
0.68
0.04
0.96
InVitro_WT_tau
InVitro_tau+GSK
predicted
1.0
0.8
0.6
0.4
0.2
0.0

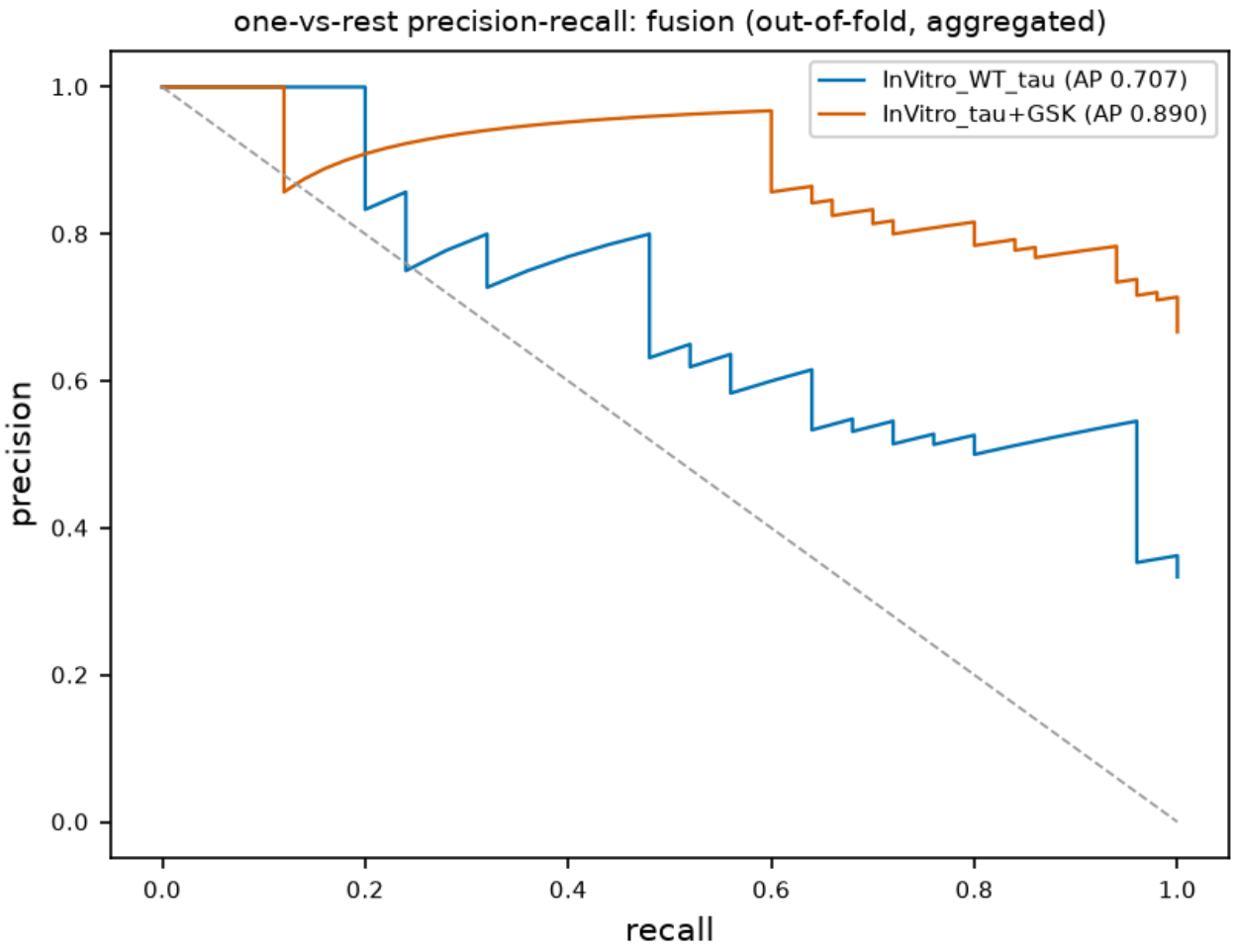
roc fusion



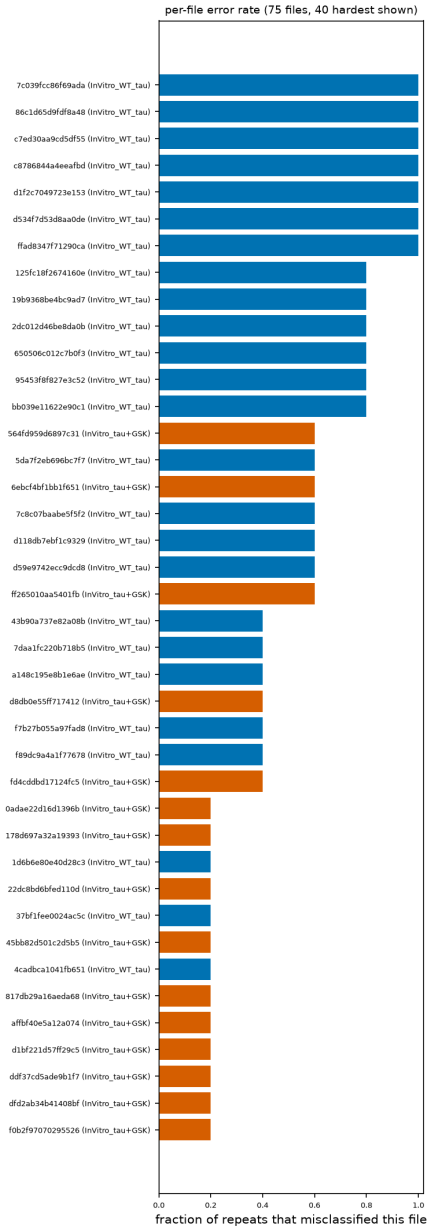
calibration fusion

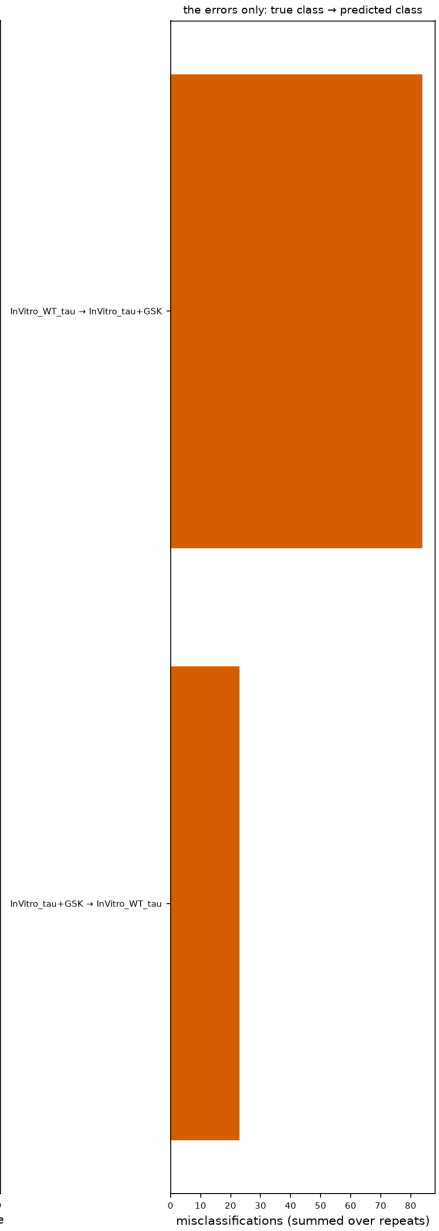

fusion
bin counts: [8, 7, 11, 14, 35]
observed accuracy
mean predicted confidence
0.0
0.2
0.4
0.6
0.8
1.0

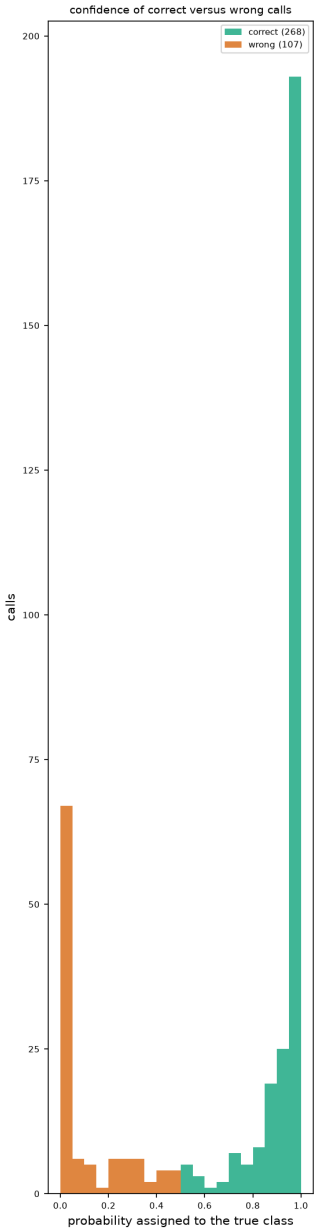
pr fusion



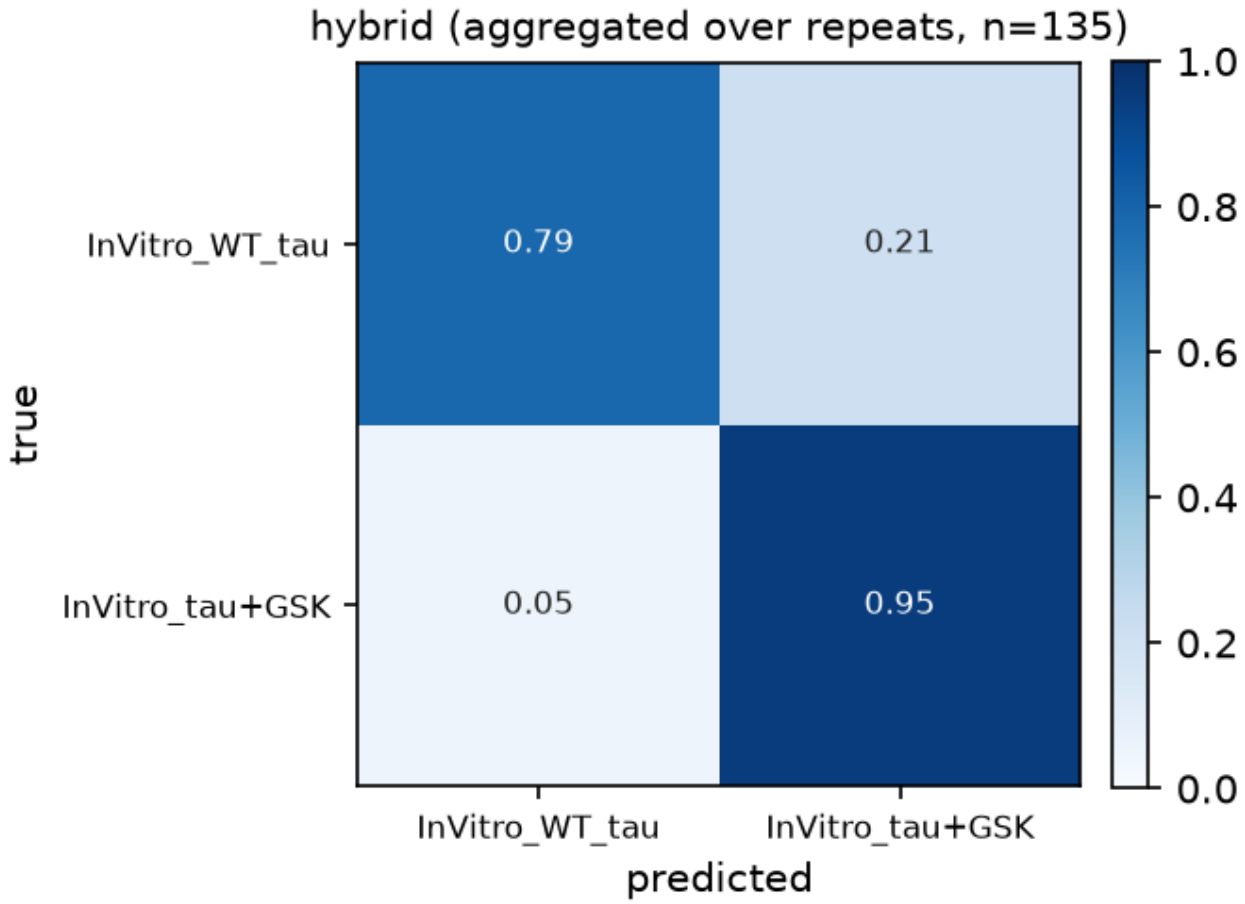
error analysis fusion



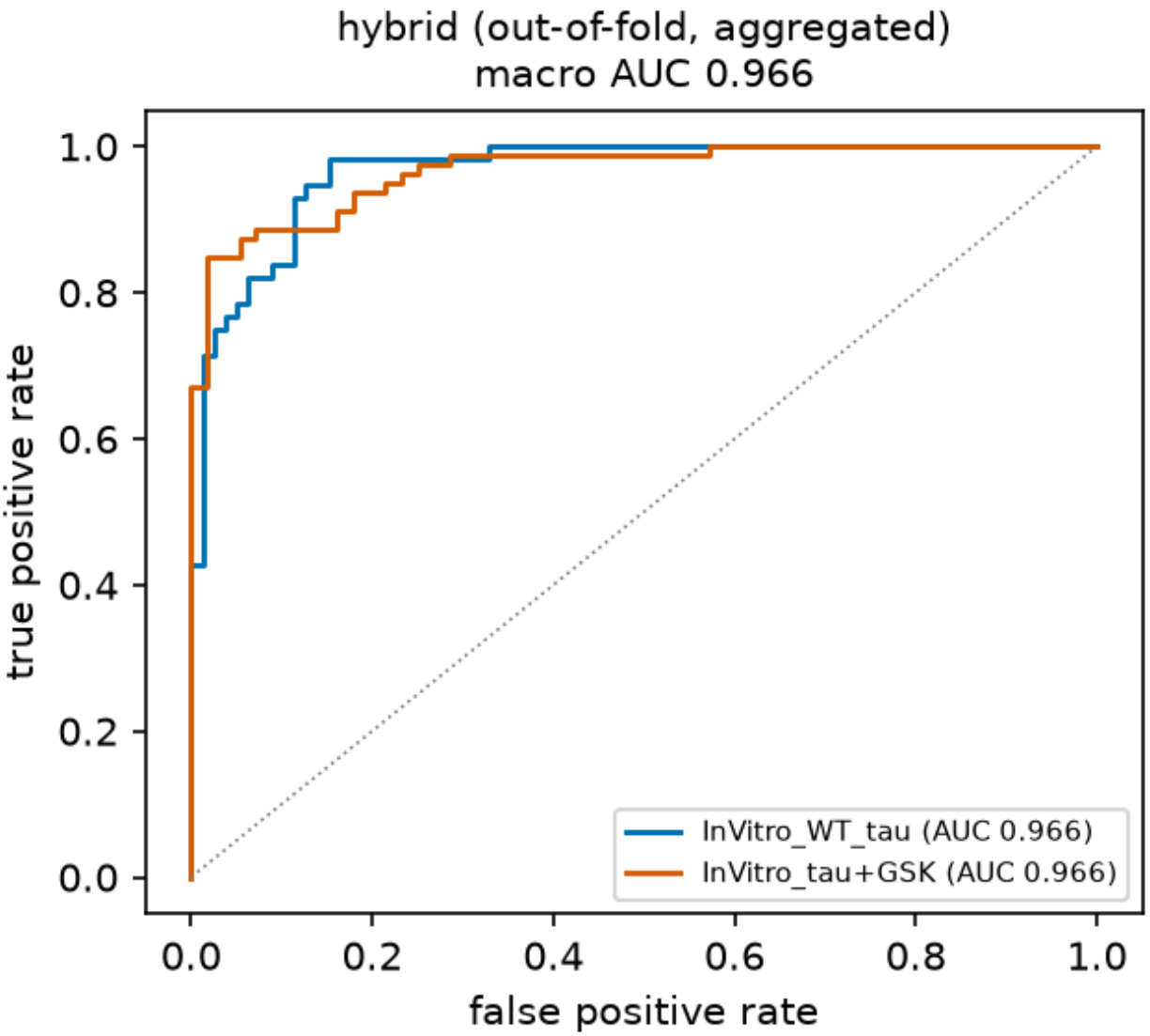




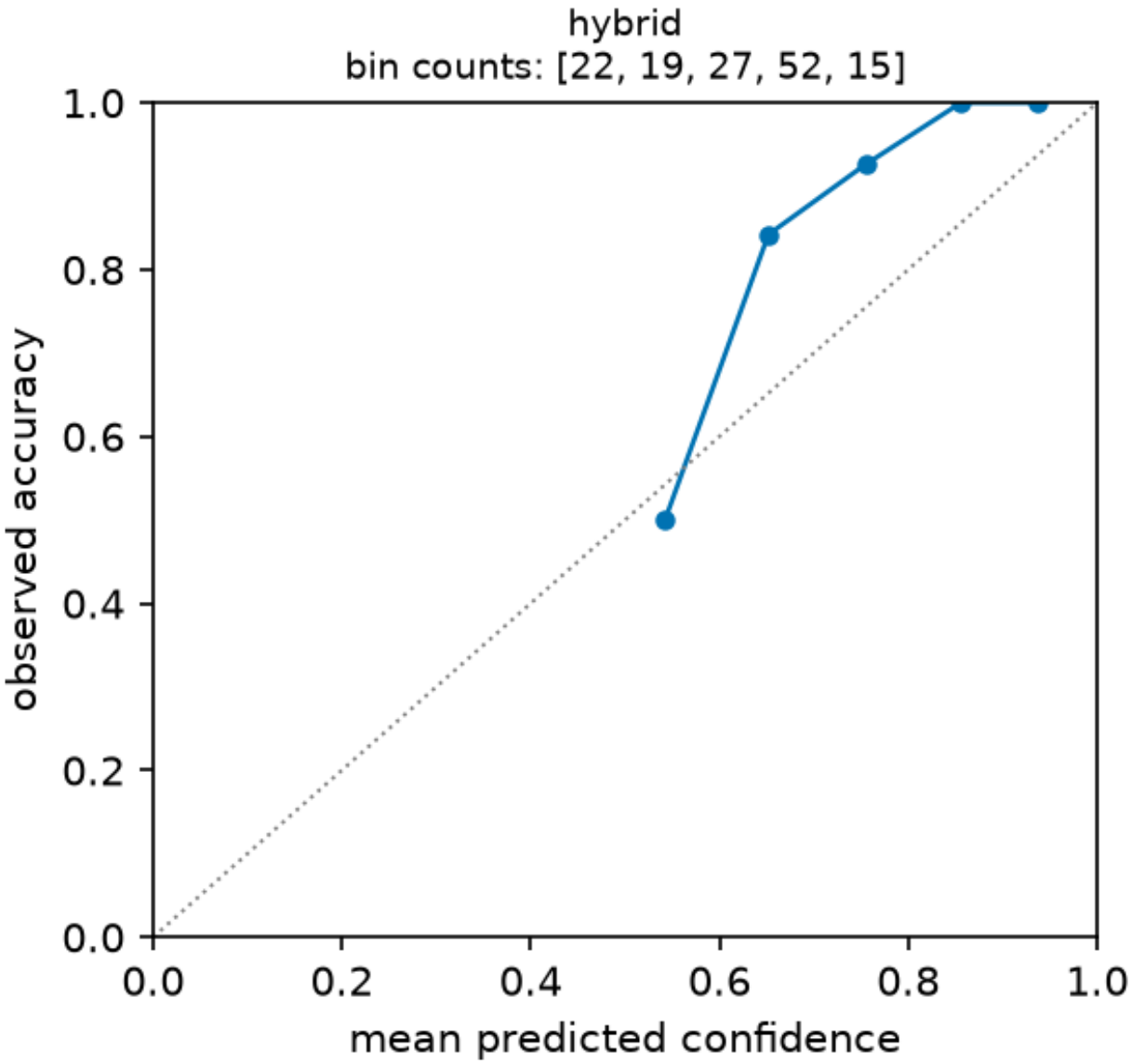
confusion hybrid


hybrid (aggregated over repeats, n=135)
InVitro_WT_tau
InVitro_tau+GSK
true
0.79
0.21
0.05
0.95
InVitro_WT_tau
InVitro_tau+GSK
predicted
1.0
0.8
0.6
0.4
0.2
0.0

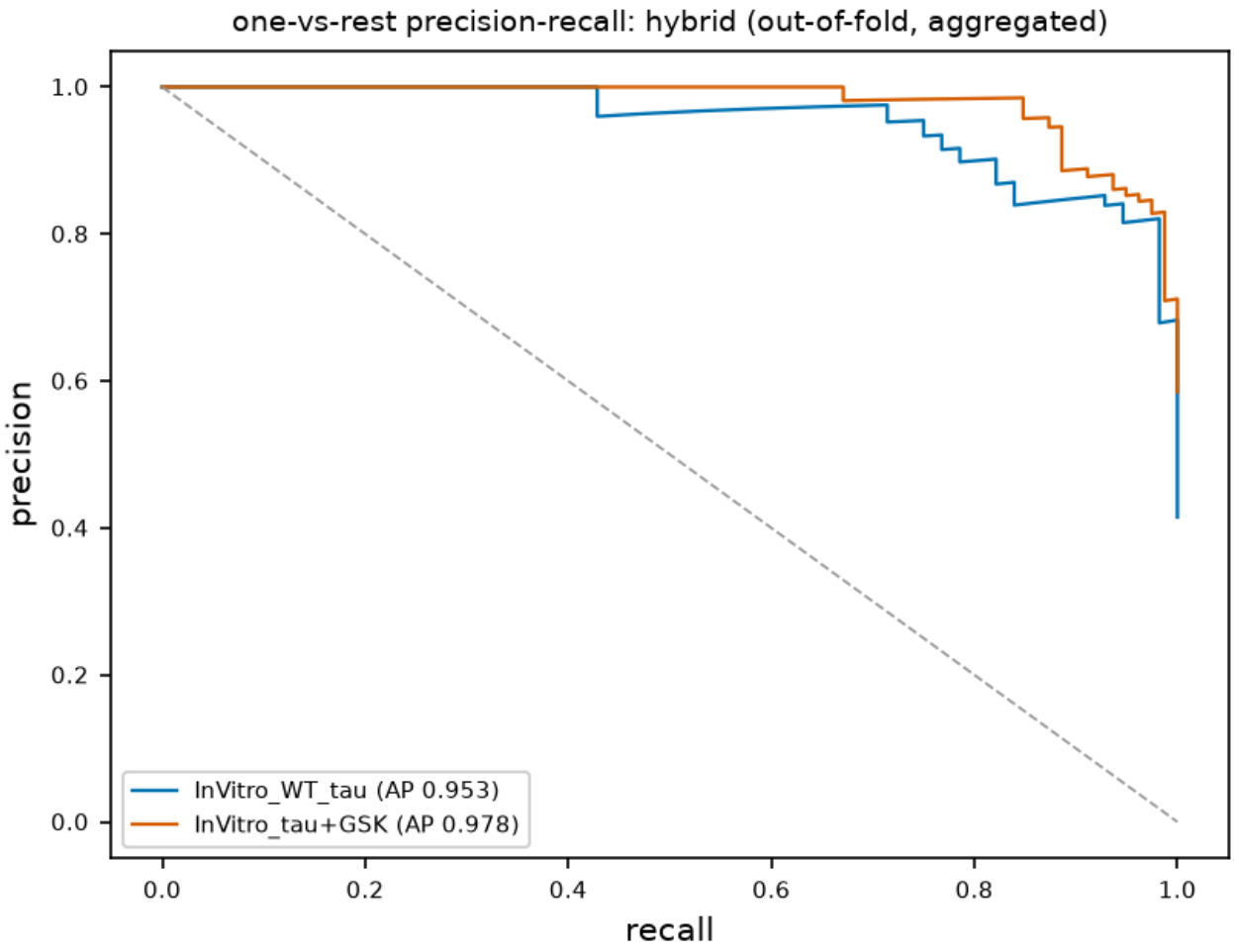
roc hybrid



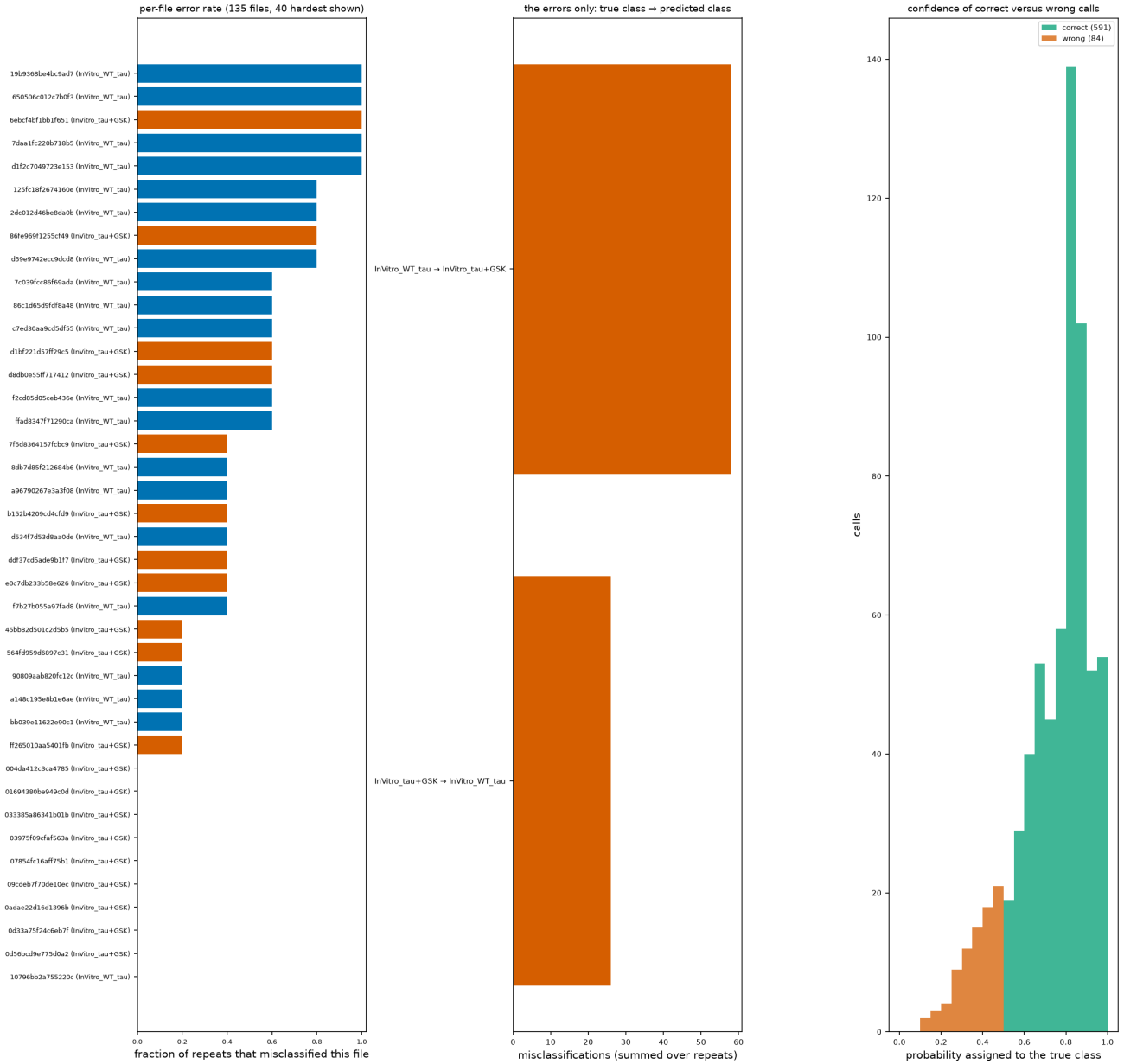
calibration hybrid


hybrid
bin counts: [22, 19, 27, 52, 15]
observed accuracy
mean predicted confidence
0.0
0.2
0.4
0.6
0.8
1.0

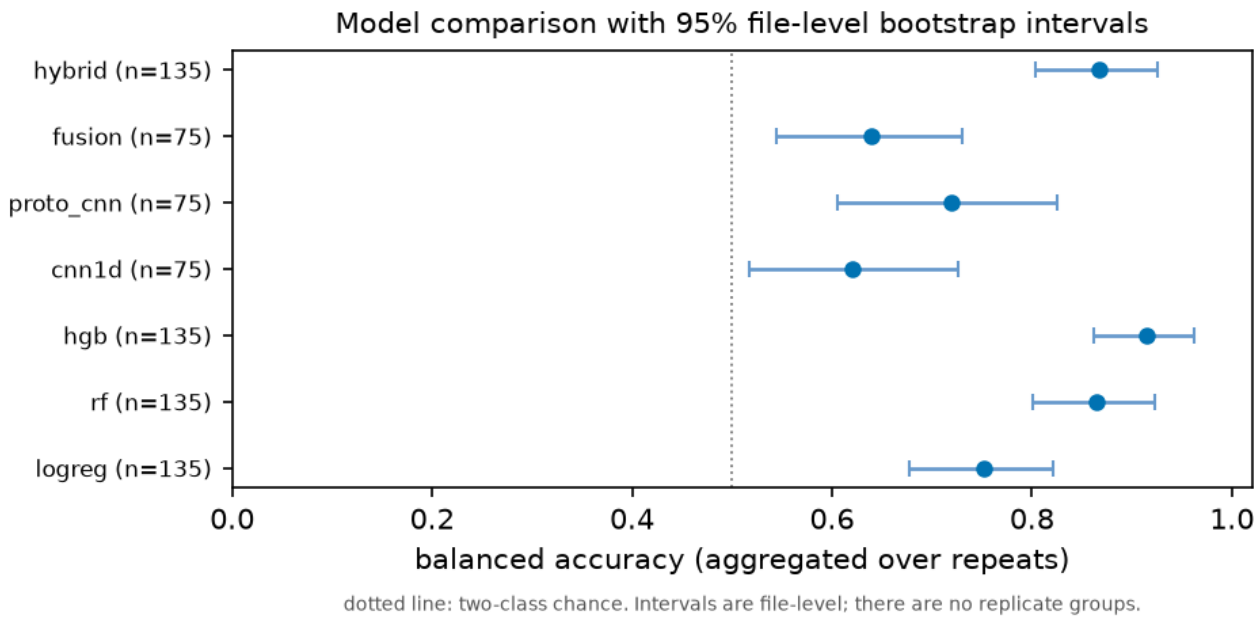
pr hybrid



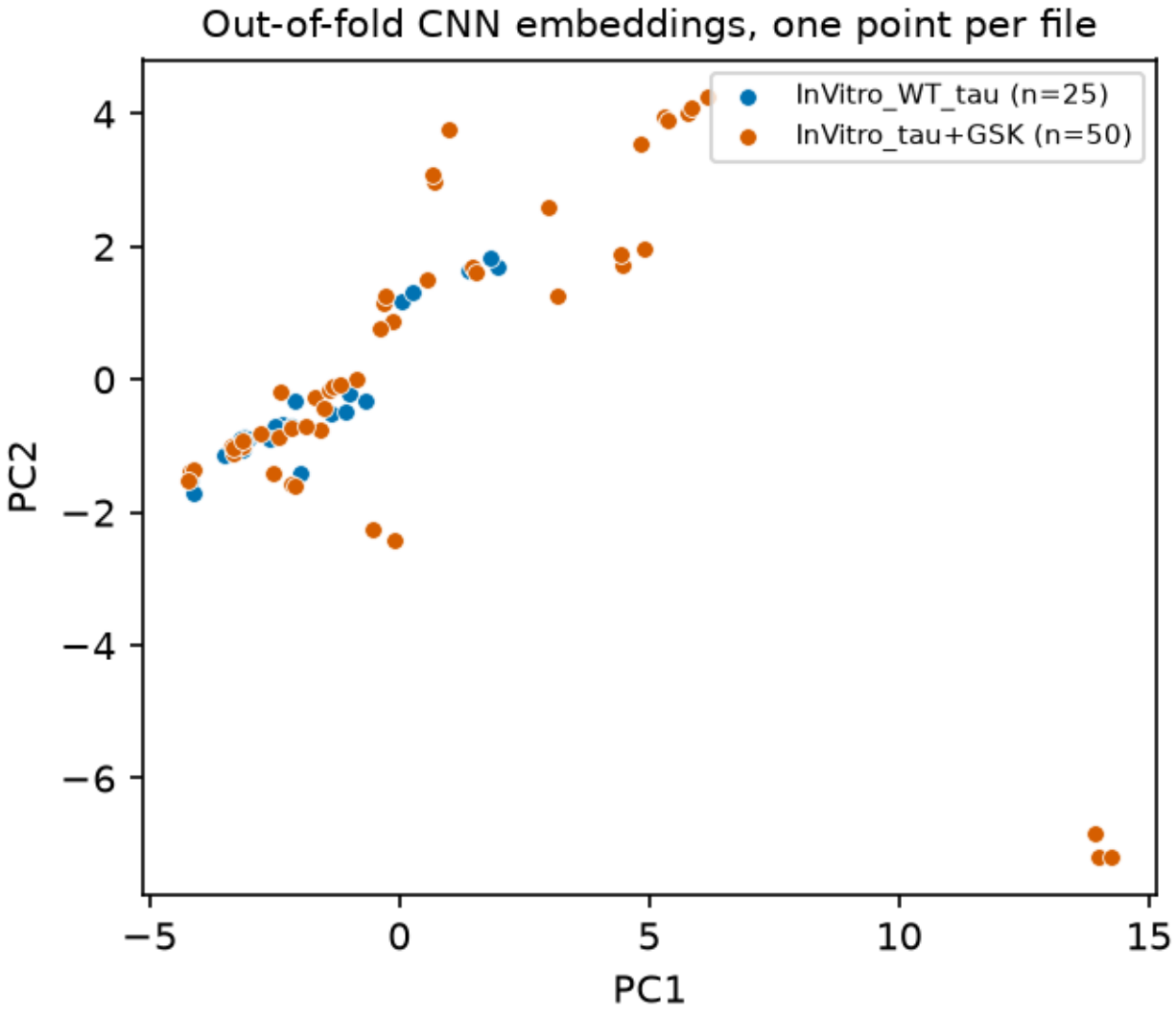
error analysis hybrid



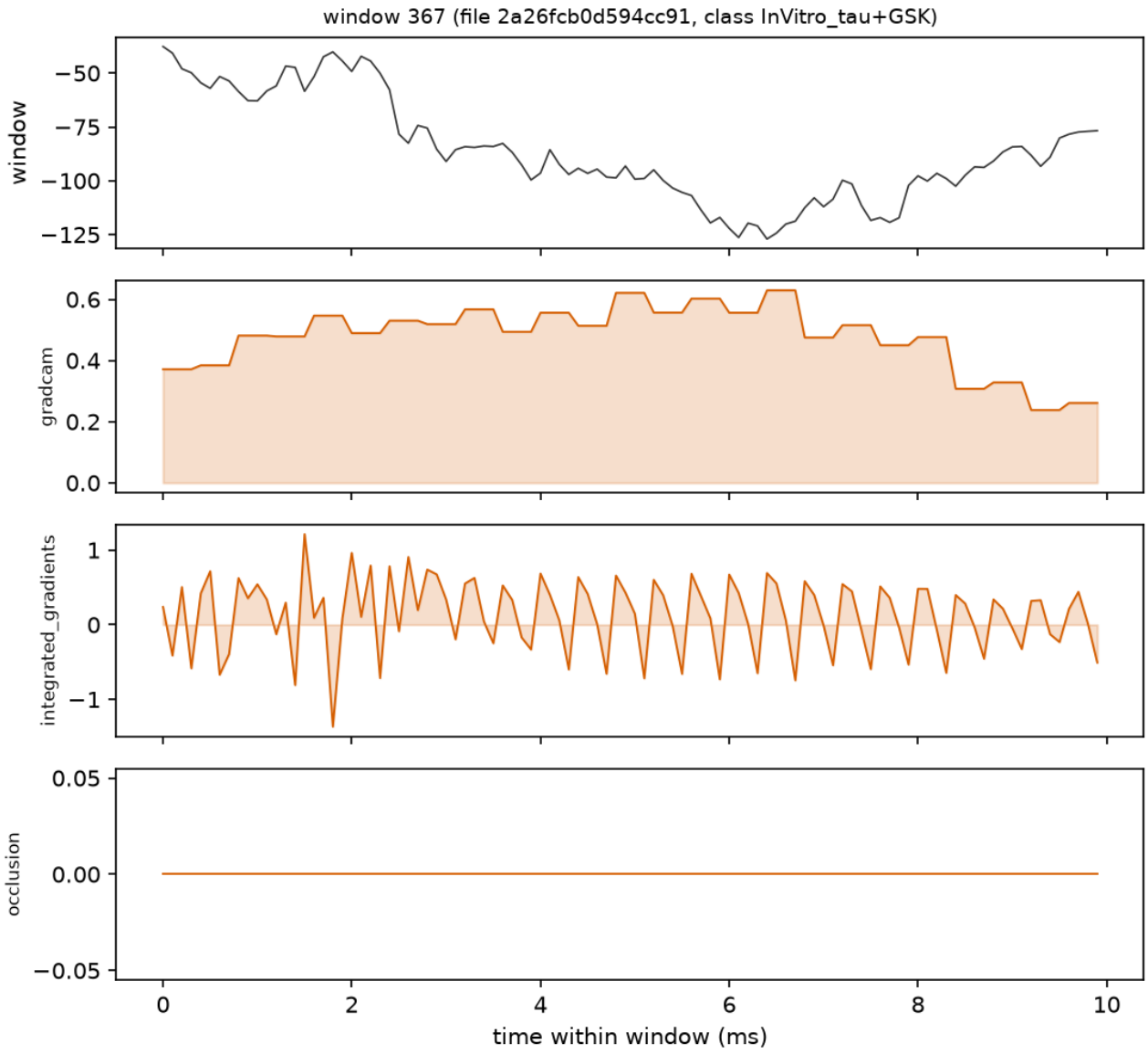
model comparison with bootstrap intervals



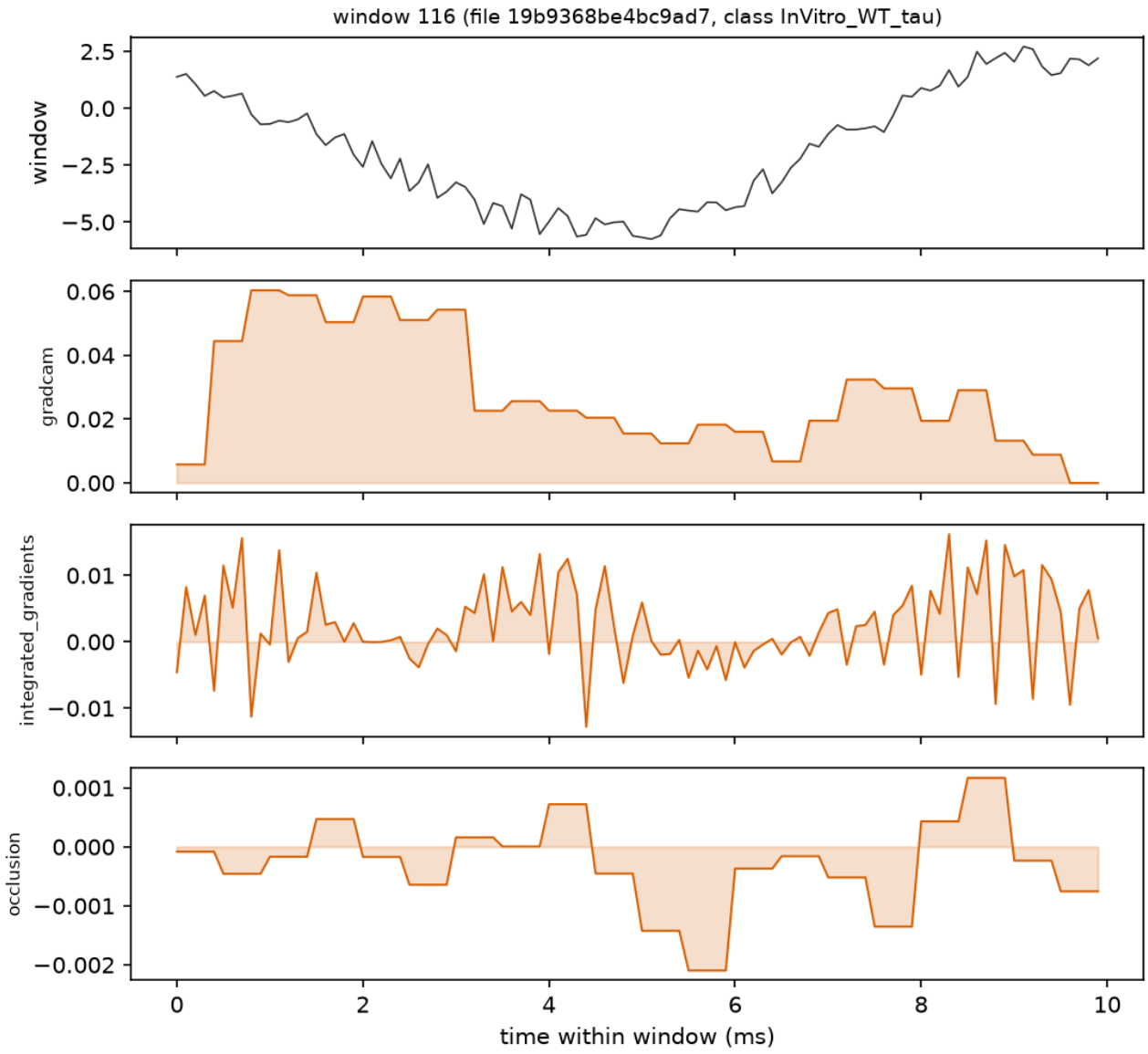
out of fold embedding pca



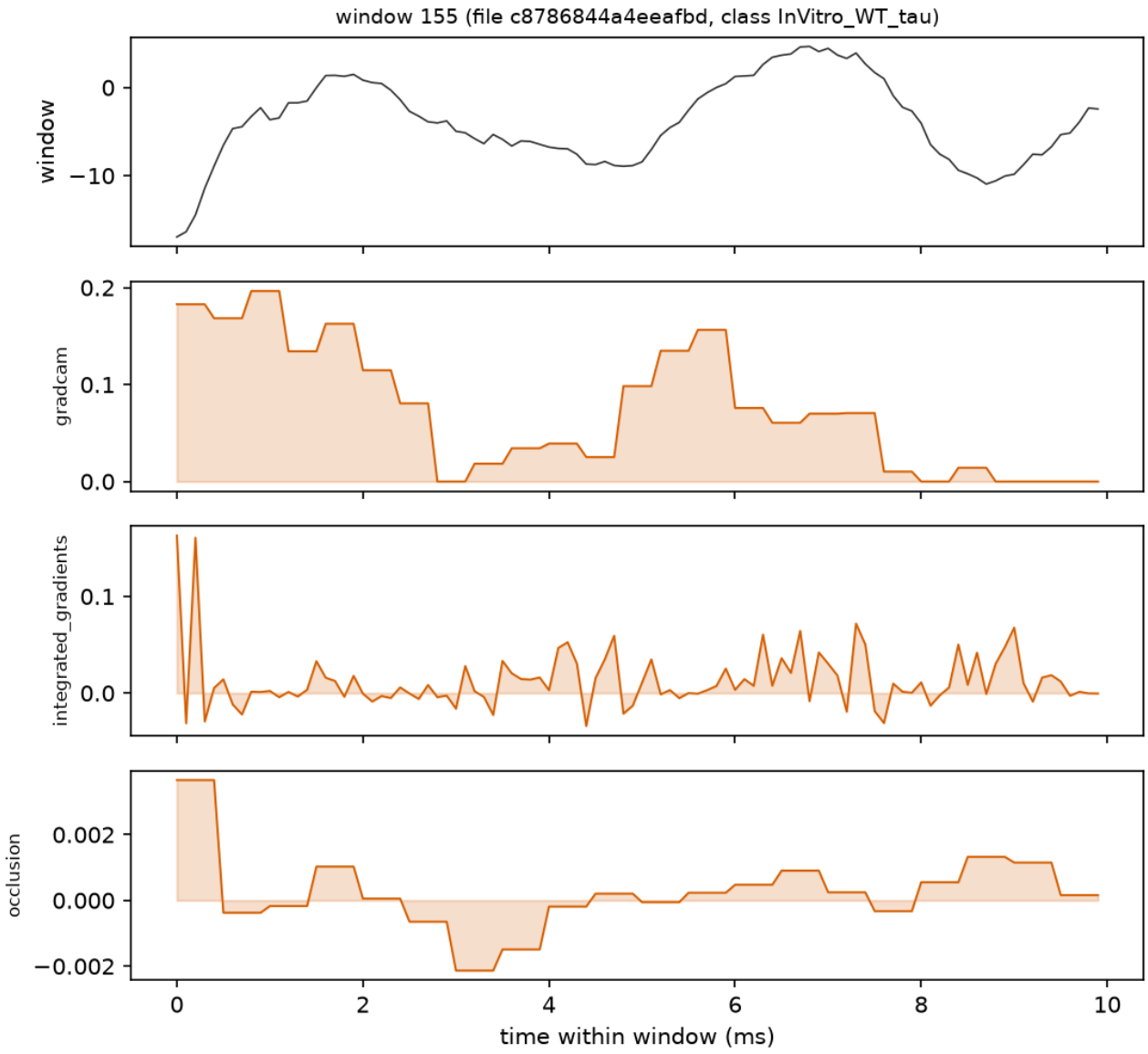
attribution window 367



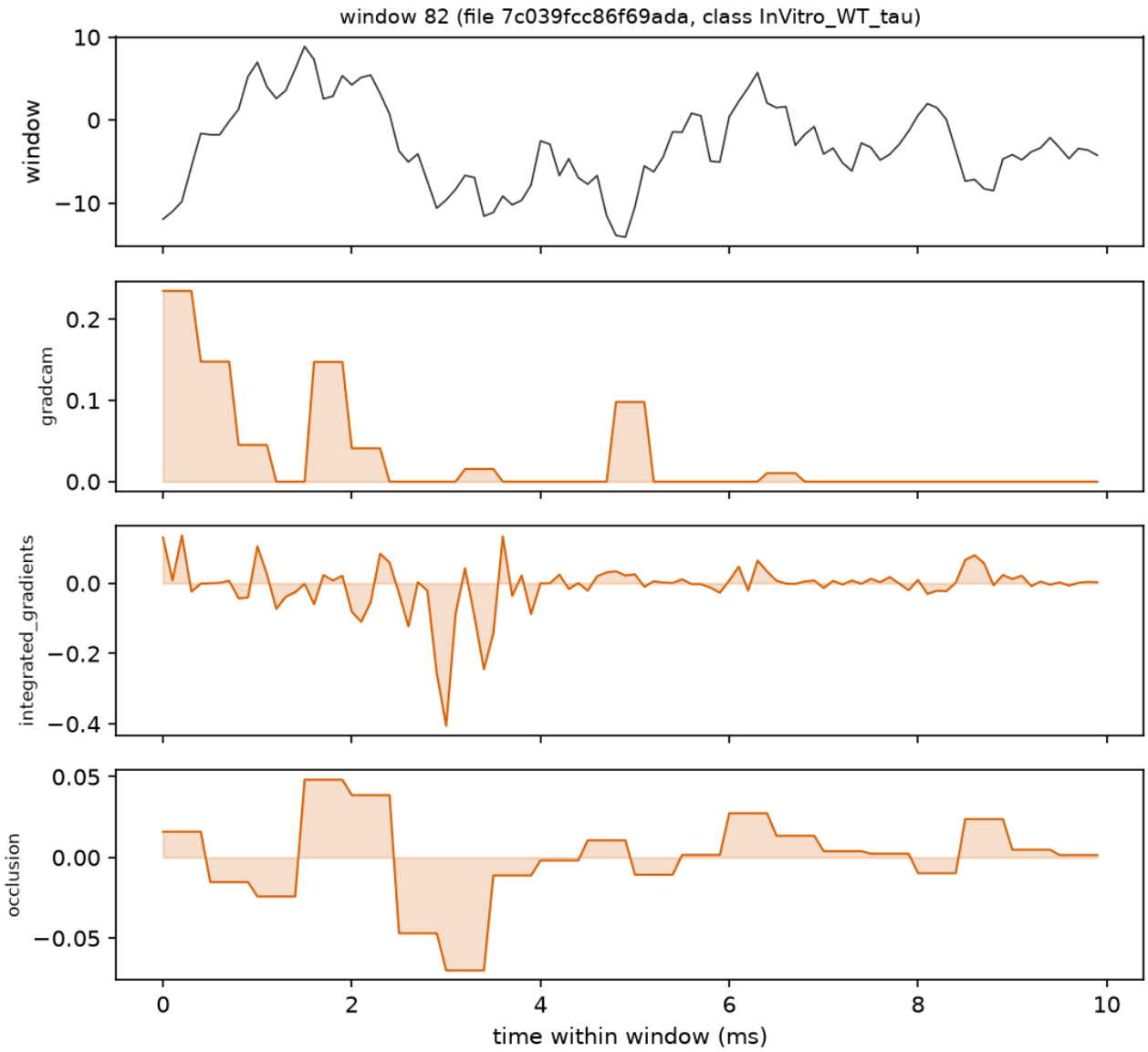
attribution window 116



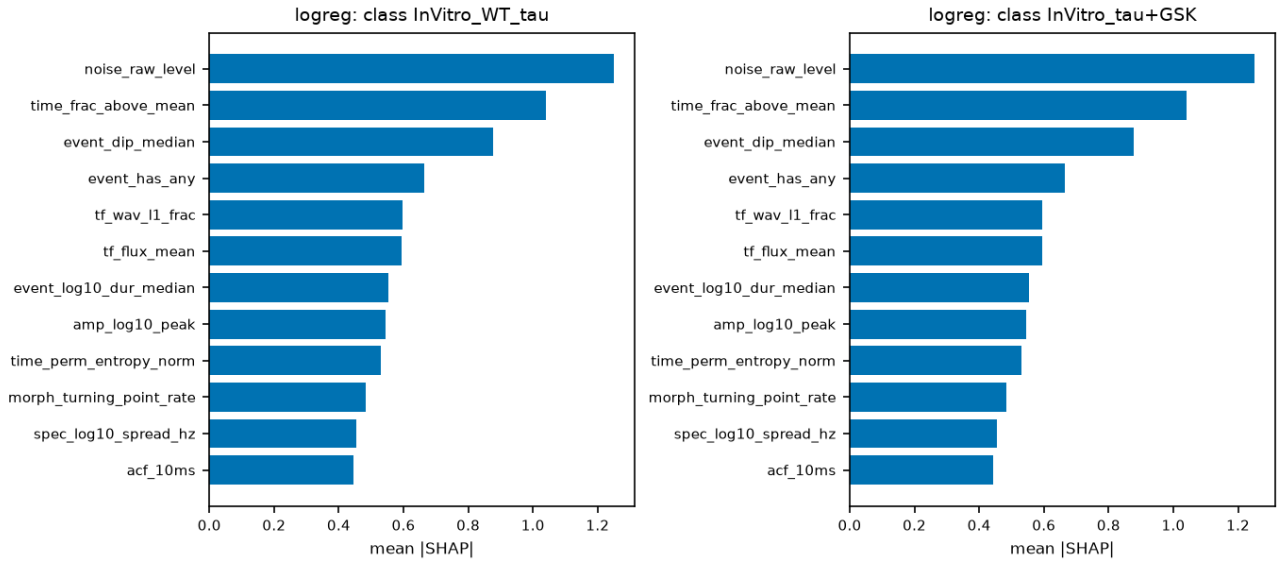
attribution window 155



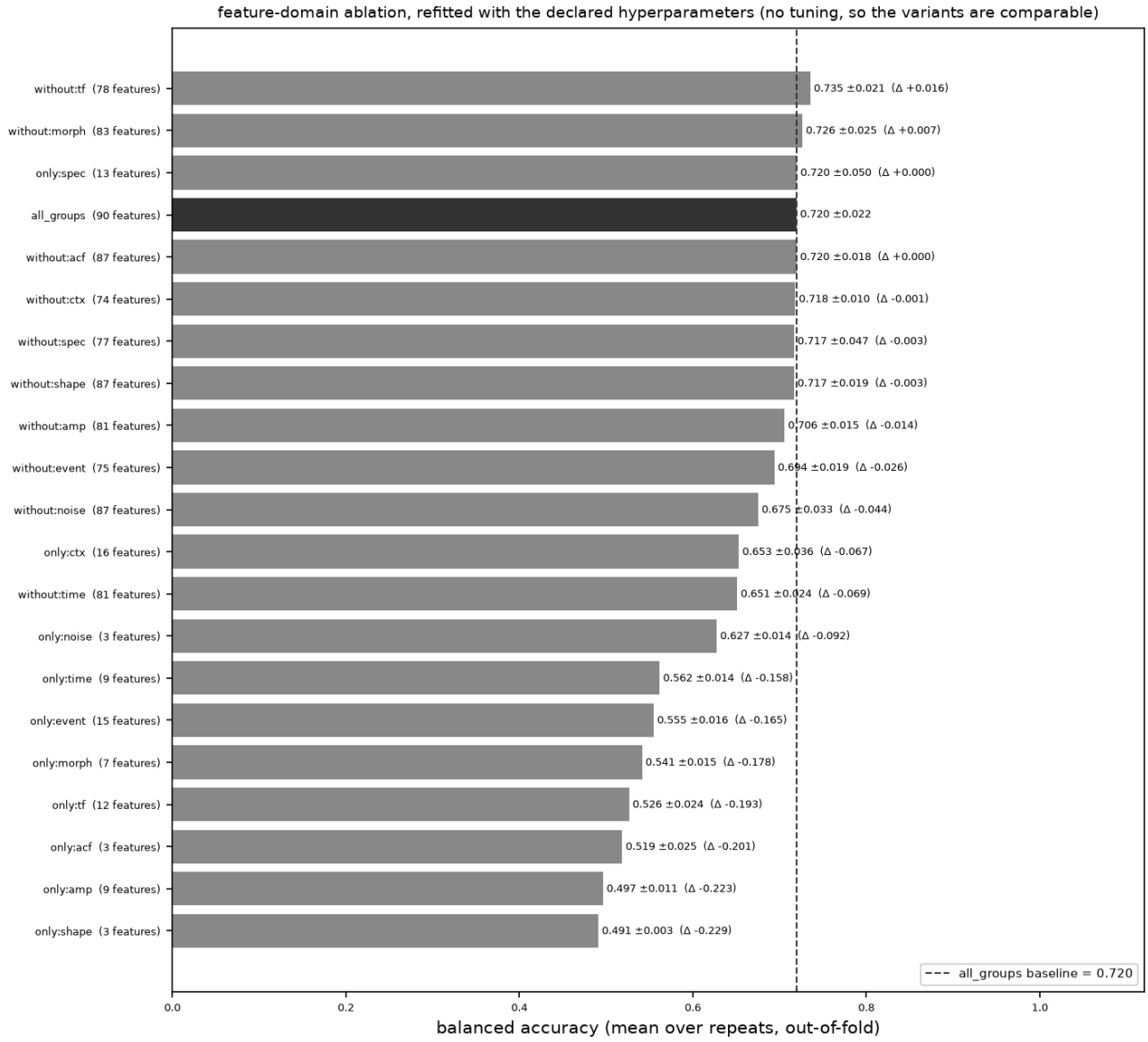
attribution window 82



model a feature importance



ablation feature domains


feature-domain ablation, refitted with the declared hyperparameters (no tuning, so the variants are comparable)
without:tf (78 features)
without:morph (83 features)
only:spec (13 features)
all_groups (90 features)
without:acf (87 features)
without:ctx (74 features)
without:spec (77 features)
without:shape (87 features)
without:amp (81 features)
without:event (75 features)
without:noise (87 features)
only:ctx (16 features)
without:time (81 features)
only:noise (3 features)
only:time (9 features)
only:event (15 features)
only:morph (7 features)
only:tf (12 features)
only:acf (3 features)
only:amp (9 features)
only:shape (3 features)
0.735 ±0.021 (Δ +0.016)
0.726 ±0.025 (Δ +0.007)
0.720 ±0.050 (Δ +0.000)
0.720 ±0.022
0.720 ±0.018 (Δ +0.000)
0.718 ±0.010 (Δ -0.001)
0.717 ±0.047 (Δ -0.003)
0.717 ±0.019 (Δ -0.003)
0.706 ±0.015 (Δ -0.014)
0.694 ±0.019 (Δ -0.026)
0.675 ±0.033 (Δ -0.044)
0.653 ±0.036 (Δ -0.067)
0.651 ±0.024 (Δ -0.069)
0.627 ±0.014 (Δ -0.092)
0.562 ±0.014 (Δ -0.158)
0.555 ±0.016 (Δ -0.165)
0.541 ±0.015 (Δ -0.178)
0.526 ±0.024 (Δ -0.193)
0.519 ±0.025 (Δ -0.201)
0.497 ±0.011 (Δ -0.223)
0.491 ±0.003 (Δ -0.229)
all_groups baseline = 0.720
0.0
0.2
0.4
0.6
0.8
1.0
balanced accuracy (mean over repeats, out-of-fold)

5. Explainability

CSV twin: `report_explainability.csv` carries the same content.

conv -> global average pooling -> single linear head, so Grad-CAM is mathematically valid (design R22); attributions are labelled gradcam, not CAM

Temporal attribution on 20 windows using gradcam, integrated_gradients, occlusion. fitted on the full pool for attribution only; no performance number in this report comes from it

Shortcut probe (mandatory): a baseline-only classifier reached balanced accuracy 0.542 against a chance level of 0.500 on 75 files. baseline alone does not predict above chance.

Time-shift null: the attribution peak moved with the shifted event in 0.400 of probed windows (shift 12 samples). attribution does not reliably track the shifted event: the model may be keyed to window position

logreg, class InVitro_WT_tau: noise_raw_level (1.25), time_frac_above_mean (1.04), event_dip_median (0.877), event_has_any (0.664), tf_wav_l1_frac (0.596), tf_flux_mean (0.595)

logreg, class InVitro_tau+GSK: noise_raw_level (1.25), time_frac_above_mean (1.04), event_dip_median (0.877), event_has_any (0.664), tf_wav_l1_frac (0.596), tf_flux_mean (0.595)

rf, class InVitro_WT_tau: noise_log10_sigma (0.00542), noise_log10_raw_std (0.00542)

rf, class InVitro_tau+GSK: noise_log10_raw_std (0.00291), noise_log10_sigma (0.00291)

hgb, class InVitro_WT_tau: noise_log10_raw_std (0.0573), noise_log10_sigma (0.0573)

hgb, class InVitro_tau+GSK: noise_log10_raw_std (0.168), noise_log10_sigma (0.168)

7. Limitations and scientific safeguards

CSV twin: `report_limitations.csv` carries the same content.

Limitation	Consequence	Safeguard applied
Small sample: 135 files across 2 classes	Confidence intervals are wide; a high point estimate is not evidence of a large effect	Every metric is reported with a file-level bootstrap interval and a label-shuffle null
No replicate or session metadata exists	Batch or day effects cannot be separated from the class label, and no hierarchical bootstrap or session-level holdout is possible	Files are the split unit; the bootstrap is labelled file-level; the baseline-only shortcut probe tests whether pre-event baseline alone predicts the class
No held-out test set	Every number is cross-validated, so it describes this pool of files, not new recordings	Repeated grouped cross-validation with tuning confined to inner folds
File-level labels over event-level signals	A file whose events do not match its folder label injects label noise that no model can fix	The label-noise rate (files with zero usable events) is reported; those files are kept and listed rather than dropped
Window length was derived from this pool's event-duration distribution	The same configuration on a different experiment yields a different window length, so numbers are not directly comparable across datasets	The resolved window length and its source are recorded in summary.json
No label-shuffle null for cnn1d	There is no permutation-based check that this model's accuracy exceeds chance	Reason recorded: deep.n_permutations is 0; each CNN permutation costs one full repeated cross-validation on CPU. Raise it if a Model B null is required.
No label-shuffle null for hybrid	There is no permutation-based check that this model's accuracy exceeds chance	Reason recorded: the ensemble has no free parameters, so its null is implied by the nulls of Model A and Model B; running it would double the cost without adding evidence

8. Information gaps

CSV twin: `report_information_gaps.csv` carries the same content.

Missing	Why it matters	What would close it
Replicate or session identifiers	Without them, leakage between files of one recording session cannot be ruled out and variance cannot be partitioned	A metadata CSV giving session_id and replicate_id per file
Ground-truth event annotations	Detected events are threshold crossings, not verified trapping events; sensitivity and specificity of detection are unknown	A manually annotated subset of files, even 10 per class
An unknown or out-of-class set	Novelty detection cannot be evaluated, only diagnosed by leave-one-class-out	Recordings of a third condition, or buffer-only controls
Instrument and acquisition metadata	Batch effects cannot be modelled or removed, and units cannot be converted to physical current	Per-file acquisition date, amplifier gain, and a calibration factor
Independent validation data	Cross-validated numbers on this pool cannot confirm performance on new recordings	A second, separately acquired dataset

9. Reproducibility

CSV twin: `report_reproducibility.csv` carries the same content.

trapsig_version: 0.1.0

config_fingerprint: 7da5ffdaffc35258976d4a0af199792ece7d2b2cfe3701f764724491f941c176

seed: 42

model_set: hybrid

n_repeats: 5

n_splits: 5

analysis_rate_hz: 10000.0

filter_mode: none

deep: {"window_ms": 10.0, "window_ms_source": "auto", "window_rate_hz": 10000.0, "window_length_samples": 100, "aggregation": "mean_proba", "n_windows": 868, "architecture": "stem(strided conv) -> residual dilated conv blocks -> GAP -> linear", "n_parameters": 31682, "select_metric": "macro_f1", "temperature_mean": 2.3001795325297745, "total_train_seconds": 62.6}

python: 3.11.17

platform: Linux-5.15.0-173-generic-x86_64-with-glibc2.39

packages: {"numpy": "2.4.6", "scipy": "1.17.1", "scikit-learn": "1.9.1", "pandas": "3.0.6", "PyYAML": "6.0.3", "matplotlib": "3.11.2", "trapsig": "0.1.0"}

10. Feature ablation and redundancy

CSV twin: `report_ablation.csv` carries the same content, row for row.

This run computed 90 handcrafted features in 10 declared groups, then measured what each group contributes by refitting the same model on the same folds with a different feature subset. Near-duplicate features are reported, never dropped: choosing columns on the pooled matrix would use information from the test folds.

10.1 Feature groups

Group	Features	What it measures
noise	3	robust residual sigma, raw level and spread
time	9	temporal pattern: zero crossings, Hjorth, line length, Teager energy, permutation entropy, segment variability, drift
amp	9	amplitude domain: RMS, peak, peak-to-peak, crest factor, percentiles, dynamic range
event	15	event statistics: count, rate, duration, dip depth, amplitude
ctx	16	context: inter-event intervals, burstiness, onset distribution, rate drift, amplitude-duration coupling, baseline stability, reference level
spec	13	frequency domain: Welch band powers, slope, peak, centroid, spread, entropy, roll-off, harmonicity
tf	12	time-frequency: STFT flux, per-band energy drift, frame entropy, db2 wavelet levels
acf	3	normalised autocorrelation of the residual at the configured lags
shape	3	distribution shape: skewness, excess kurtosis, tail fraction
morph	7	morphology: turning points, event rise/fall, area-to-peak, sharpness, self-similarity to this recording's mean event waveform

10.2 Ablation

Mode both, model logreg. same folds, same files, same model settings as the main cross-validation; only the feature columns change

Feature subset	Balanced accuracy (mean +/- sd)	Change against all groups
all_groups (90 features)	0.720 +/- 0.022	baseline
only:noise (3 features)	0.627 +/- 0.014	delta vs all groups: -0.092
only:time (9 features)	0.562 +/- 0.014	delta vs all groups: -0.158
only:amp (9 features)	0.497 +/- 0.011	delta vs all groups: -0.223
only:event (15 features)	0.555 +/- 0.016	delta vs all groups: -0.165
only:ctx (16 features)	0.653 +/- 0.036	delta vs all groups: -0.067
only:spec (13 features)	0.720 +/- 0.050	delta vs all groups: +0.000
only:tf (12 features)	0.526 +/- 0.024	delta vs all groups: -0.193
only:acf (3 features)	0.519 +/- 0.025	delta vs all groups: -0.201
only:shape (3 features)	0.491 +/- 0.003	delta vs all groups: -0.229
only:morph (7 features)	0.541 +/- 0.015	delta vs all groups: -0.178
without:noise (87 features)	0.675 +/- 0.033	delta vs all groups: -0.044
without:time (81 features)	0.651 +/- 0.024	delta vs all groups: -0.069
without:amp (81 features)	0.706 +/- 0.015	delta vs all groups: -0.014
without:event (75 features)	0.694 +/- 0.019	delta vs all groups: -0.026
without:ctx (74 features)	0.718 +/- 0.010	delta vs all groups: -0.001
without:spec (77 features)	0.717 +/- 0.047	delta vs all groups: -0.003
without:tf (78 features)	0.735 +/- 0.021	delta vs all groups: +0.016
without:acf (87 features)	0.720 +/- 0.018	delta vs all groups: +0.000
without:shape (87 features)	0.717 +/- 0.019	delta vs all groups: -0.003
without:morph (83 features)	0.726 +/- 0.025	delta vs all groups: +0.007

a group whose 'only:' score is near the all-groups score carries most of the signal; a group whose 'without:' score is near the all-groups score adds little. Differences are on the same folds, so they are paired comparisons, but they are not independent tests: no p-value is claimed here

10.3 Redundant feature pairs

Threshold |Spearman| >= 0.98; 31 pair(s) flagged. flagged pairs are reported in feature_redundancy.csv and never dropped: dropping columns on the pooled matrix would use test-fold information

Feature A	Feature B	Rank correlation
event_n	event_rate_per_min	|rho| = 1.0000 on 135 files
time_hjorth_mobility	acf_0p1ms	|rho| = -1.0000 on 135 files
amp_abs_p99_over_sigma	amp_dynamic_range_db	|rho| = 1.0000 on 135 files
amp_log10_rms	spec_log10_total	|rho| = 0.9996 on 135 files
spec_log10_band_1000_5000Hz	tf_wav_log10_l2	|rho| = 0.9984 on 135 files
spec_log10_band_100_1000Hz	tf_wav_log10_l4	|rho| = 0.9968 on 135 files
time_hjorth_complexity	spec_log10_centroid_hz	|rho| = -0.9966 on 135 files
spec_log10_band_1000_5000Hz	tf_wav_log10_l1	|rho| = 0.9961 on 135 files
spec_log10_centroid_hz	spec_log10_rolloff85_hz	|rho| = 0.9940 on 135 files
amp_abs_p99_over_sigma	shape_frac_beyond_3sigma	|rho| = 0.9937 on 135 files
amp_dynamic_range_db	shape_frac_beyond_3sigma	|rho| = 0.9937 on 135 files
amp_log10_peak	amp_log10_p2p	|rho| = 0.9931 on 135 files
tf_wav_log10_l1	tf_wav_log10_l2	|rho| = 0.9930 on 135 files
amp_log10_rms	tf_wav_log10_l3	|rho| = 0.9899 on 135 files

17 further pair(s) are listed in `feature_redundancy.csv`.

Attribution and performance are different claims. A region that drives a prediction is not evidence of a biological mechanism; only an independent experiment can supply that.

11. Sampling-rate design

CSV twin: `report_sampling.csv`.

Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'Rate (Hz)' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Rate (Hz)', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] ' style': 'bulletText': None ' debug': 0) #Paragraph	Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'Nyquist limit (Hz)' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Nyquist limit (Hz)', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Typical role' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Typical role', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '1e+06' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='1e+06', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '500000' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='500000', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'acquisition rate' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='acquisition rate', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '50000' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='50000', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '25000' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='25000', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'analysis rate' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='analysis rate', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '25000' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='25000', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '12500' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='12500', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'analysis rate' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='analysis rate', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '10000' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='10000', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '5000' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='5000', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'analysis rate' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='analysis rate', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '1000' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='1000', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '500' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='500', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'analysis rate' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='analysis rate', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '100' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='100', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': '50' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='50', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'analysis rate' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='analysis rate', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Rate (Hz)' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Rate (Hz)', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Nyquist limit (Hz)' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Nyquist limit (Hz)', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Typical role' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Typical role', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '10' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='10', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '5' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='5', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'analysis rate' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='analysis rate', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '2' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='2', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '1' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='1', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'analysis rate' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='analysis rate', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

Analysis rate used by this run: **10000.0 Hz**. Acquisition rate and analysis rate are different things: the file was recorded at its own rate and everything downstream is computed at the analysis rate. Down-conversion uses staged decimation with an anti-alias filter, so no spectral content above the new Nyquist limit folds back into the band. Up-conversion cannot create information that was never sampled and is offered only as a display convenience.

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Segmentation parameter' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Segmentation parameter', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Value' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Value', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'window_s' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='window_s', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'None' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='None', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'overlap' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='overlap', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '0.5' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.5', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'stride_s' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='stride_s', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'None' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='None', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'normalize' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='normalize', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'robust' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='robust', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'detrend' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='detrend', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'none' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='none', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': ' Segmentation parameter' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Segmentation parameter', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Value' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Value', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': ' pad_mode' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='pad_mode', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'reflect' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='reflect', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': ' max_windows_per_file' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='max_windows_per_file', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '64' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='64', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None ' debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': ' allow_off_menu_rates' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='allow_off_menu_rates', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'True' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='True', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph

12. Extended evaluation metrics

CSV twin: `report_extended_metrics.csv` and `metrics_extended.csv`.

Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'Model' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Model', textColor=Color(.266667,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': 'n' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='n', textColor= Color(.26666 7,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'MCC' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='MCC', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'Brier' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Brier', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'ECE' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='ECE', textColor =Color(.266667,.266 667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'MCE' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='MCE', textColor =Color(.266667,.266 667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': ' dummy_majority' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='dummy_majori ty', textColor=Color(.266667,.2666 67,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': '135' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='135', textColor= Color(.26666 7,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.000' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.000', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.486' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.486', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.000' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.000', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.000' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.000', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'logreg' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='logreg', textColor=Color(.266667,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': '135' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='135', textColor= Color(.26666 7,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.521' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.521', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.369' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.369', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.116' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.116', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.202' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.202', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph

Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'Model' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Model', textColor=Color(.266667,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': 'n' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='n', textColor= Color(.26666 7,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'MCC' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='MCC', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'Brier' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Brier', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'ECE' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='ECE', textColor =Color(.266667,.266 667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'MCE' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='MCE', textColor =Color(.266667,.266 667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'rf' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='rf', textColor=Color(.266667,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': '135' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='135', textColor= Color(.26666 7,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.759' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.759', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.158' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.158', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.057' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.057', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.164' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.164', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'hgb' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='hgb', textColor=Color(.266667,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': '135' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='135', textColor= Color(.26666 7,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.832' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.832', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.118' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.118', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.047' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.047', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.258' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.258', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph

Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'Model' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Model', textColor=Color(.266667,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': 'n' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='n', textColor= Color(.2666 67,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'MCC' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='MCC', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'Brier' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Brier', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'ECE' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='ECE', textColor =Color(.266667,.266 667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'MCE' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='MCE', textColor =Color(.266667,.266 667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'cnn1d' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='cnn1d', textColor=Color(.266667,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': '75' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='75', textColor= Color(.2666 67,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.283' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.283', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.350' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.350', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.062' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.062', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.083' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.083', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'proto_cnn' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='proto_cnn', textColor=Color(.266667,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': '75' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='75', textColor= Color(.2666 67,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.426' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.426', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.373' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.373', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.146' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.146', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.244' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.244', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph

Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'Model' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Model', textColor=Color(.266667,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': 'n' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='n', textColor= Color(.2666 67,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'MCC' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='MCC', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'Brier' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Brier', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'ECE' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='ECE', textColor =Color(.266667,.266 667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'MCE' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='MCE', textColor =Color(.266667,.266 667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'fusion' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='fusion', textColor=Color(.266667,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': '75' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='75', textColor= Color(.2666 67,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.388' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.388', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.373' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.373', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.110' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.110', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.130' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.130', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 ' encoding': 'utf8' 'text': 'hybrid' ' frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='hybrid', textColor=Color(.266667,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensiti ve': 1 'enco ding': 'utf8' ' text': '135' ' frags': [ParaFrag(__tag__='p ara', bold=0, fontName=' Helvetica', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text='135', textColor= Color(.2666 67,.266667,. 266667,1), us_lines=[])] 'style': ' bulletText': None 'debu g': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.757' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.757', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.186' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.186', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.130' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.130', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '0.190' ' frags': [ParaFrag(__t ag__='para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='0.190', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph

cohens_kappa: chance-corrected agreement (sklearn cohen_kappa_score); empty when the prediction set is degenerate

precision_micro/recall_micro/f1_micro: multiclass micro averages; identical to accuracy

precision_macro/recall_macro/f1_macro: unweighted mean of the per-class scores

precision_weighted/recall_weighted/f1_weighted: per-class scores weighted by support

specificity: TN/(TN+FP) one-vs-rest; None when the subset holds no negative file

mcc_multiclass: Matthews correlation from the multiclass confusion matrix

brier_multiclass: mean over files of sum_c (p_c - 1[y=c])^2

ece: sum_b (n_b/n)|acc_b - conf_b| over equal-width bins of the max probability

mce: max_b |acc_b - conf_b| over the same bins

pr_auc_ovr: one-vs-rest average precision; None when the class is absent or universal

topk_accuracy: fraction of files whose true label is among the k largest probabilities

Per-class precision, recall, specificity, F1, support and error counts, top-k accuracy and PR-AUC are in `metrics_extended.csv`; undefined values are recorded as empty cells with the reason in the note column rather than being silently imputed.

13. Four-branch deep architecture

CSV twin: `report_quad.csv`.

quad.enabled is false on this run, so the four-branch architecture (learned temporal branch, learned spectral branch, self-supervised pretraining, engineered window features, cross-branch attention fusion, supervised softmax head and metric-learning head) was not trained. This section deliberately shows no numbers instead of illustrative ones. Enable it with `quad.enabled: true`; everything else has safe defaults.

14. Visualization registry

CSV twin: `report_viz_registry.csv`; the per-plot audit trail is `viz_registry_index.csv`.

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': ' Status' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Status', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' ' text': 'Entries' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Entries', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': ' generated' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='generated', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' ' text': '124' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='124', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': ' not_available' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='not_available', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' ' text': '4' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='4', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': ' skipped' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='skipped', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' ' text': '45' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='45', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': ' total' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='total', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' ' text': '173' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='173', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Category' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Category', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Entries' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Entries', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Generated' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Generated', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Skipped' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Skipped', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Not available' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Not available', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Failed' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Failed', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'A' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='A', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '15' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='15', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '15' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='15', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'B' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='B', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '12' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='12', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '12' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='12', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'C' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='C', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '10' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='10', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '8' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='8', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '2' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='2', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Category' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Category', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Entries' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Entries', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Generated' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Generated', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Skipped' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Skipped', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Not available' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Not available', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Failed' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Failed', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'D' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='D', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '31' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='31', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '25' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='25', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '6' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='6', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'E' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='E', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '28' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='28', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '16' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='16', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '12' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='12', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'F' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='F', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '12' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='12', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '5' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='5', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '7' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='7', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Category' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Category', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Entries' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Entries', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Generated' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Generated', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Skipped' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Skipped', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Not available' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Not available', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Failed' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Failed', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'K' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='K', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '15' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='15', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '15' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='15', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Q' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Q', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '12' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='12', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '12' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='12', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'S' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='S', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '6' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='6', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '5' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='5', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '1' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='1', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph

Paragraph('caseSensitive': 1 'encoding': 'utf8' ' text': 'Category' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Category', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': ' Entries' 'frags': [ParaFrag(__tag __='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Entries', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] ' style': 'bulletTex t': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 ' encoding': 'utf8' ' text': 'Generated' ' frags': [ParaFrag(__tag__='para', bold=0, fontName ='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Generated', textColor=Color(.2 66667,.266667,.2 66667,1), us_line s=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': ' Skipped' 'frags': [ParaFrag(__tag __='para', bold=0, fontNa me='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Skipped', textColor=Color(. 266667,.266667, .266667,1), us_lines=[])] ' style': 'bulletTex t': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': 'Not available' 'frags': [ParaFrag(__tag__=' para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='Not available', textColor=Color(.266 667,.266667,.26666 7,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': ' Failed' 'frags': [ParaFrag(__t ag__='para', bold=0, fontN ame='Helvetic a', fontSize=7. 2, greek=0, italic=0, link=[], rise=0, text=' Failed', textCo lor=Color(.266 667,.266667,. 266667,1), us_lines=[])] ' style': 'bulletT ext': None ' debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' ' text': 'M' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='M', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': '8' ' frags': [ParaFra g(__tag__='para' , bold=0, fontName='Helv etica', fontSize= 7.2, greek=0, italic=0, link=[], rise=0, text='8', textColor=Color(. 266667,.266667, .266667,1), us_lines=[])] ' style': 'bulletTex t': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 ' encoding': 'utf8' ' text': '2' 'frags': [ParaFrag(__tag_ _='para', bold=0, fontName='Helveti ca', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='2', textColor =Color(.266667,.2 66667,.266667,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': '4' ' frags': [ParaFra g(__tag__='para' , bold=0, fontName='Helv etica', fontSize= 7.2, greek=0, italic=0, link=[], rise=0, text='4', textColor=Color(. 266667,.266667, .266667,1), us_lines=[])] ' style': 'bulletTex t': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '2' 'frags': [ParaFrag(__tag__=' para', bold=0, fontName='Helvetica' , fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='2', textColor=C olor(.266667,.26666 7,.266667,1), us_lines=[])] 'style': ' bulletText': None ' debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': '' ' frags': [] 'style': 'bulletText': None 'debug': 0) #Paragrap h
Paragraph('caseSensitive': 1 'encoding': 'utf8' ' text': 'T' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='T', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': '22' ' frags': [ParaFra g(__tag__='para' , bold=0, fontName='Helv etica', fontSize= 7.2, greek=0, italic=0, link=[], rise=0, text='22', textColor=Color (.266667,.26666 7,.266667,1), us_lines=[])] ' style': 'bulletTex t': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 ' encoding': 'utf8' ' text': '7' 'frags': [ParaFrag(__tag_ _='para', bold=0, fontName='Helveti ca', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='7', textColor =Color(.266667,.2 66667,.266667,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': '15' ' frags': [ParaFra g(__tag__='para' , bold=0, fontName='Helv etica', fontSize= 7.2, greek=0, italic=0, link=[], rise=0, text='15', textColor=Color (.266667,.26666 7,.266667,1), us_lines=[])] ' style': 'bulletTex t': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '' 'frags': [] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': '' ' frags': [] 'style': 'bulletText': None 'debug': 0) #Paragrap h
Paragraph('caseSensitive': 1 'encoding': 'utf8' ' text': 'G' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Helvetica', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='G', textColor=Color(.266667,.266667,.266667,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': '2' ' frags': [ParaFra g(__tag__='para' , bold=0, fontName='Helv etica', fontSize= 7.2, greek=0, italic=0, link=[], rise=0, text='2', textColor=Color(. 266667,.266667, .266667,1), us_lines=[])] ' style': 'bulletTex t': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 ' encoding': 'utf8' ' text': '2' 'frags': [ParaFrag(__tag_ _='para', bold=0, fontName='Helveti ca', fontSize=7.2, greek=0, italic=0, link=[], rise=0, text='2', textColor =Color(.266667,.2 66667,.266667,1), us_lines=[])] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': '' ' frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSen sitive': 1 'encoding': ' utf8' 'text': '' 'frags': [] ' style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph(' caseSensitive': 1 'encoding': ' utf8' 'text': '' ' frags': [] 'style': 'bulletText': None 'debug': 0) #Paragrap h

Every registry entry declares its category (A time domain, B frequency, C time-frequency, D nonlinear/complexity, E dataset/class, F deep diagnostics), its prerequisites and parameters in `viz_registry_index.csv`. Entries whose prerequisites the run did not produce are recorded as skipped with the reason; optional methods that are not shipped (EMD, VMD) are recorded as not_available. No plot in this report is illustrative: each one was drawn from arrays this run computed, and plots carrying numbers also wrote a CSV twin.